On the Complexity of Computation and Graph Homomorphisms

by

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Abstract

Here goes the abstract.

List of Publications

The contents of this thesis have been published in the following manner:

- **Section 2.2: A concise proof of the L_0 dichotomy**
  Presentation: Fields Institute Set Theory Seminar, Toronto (March 28, 2025).
  Note: Contains original proofs first disseminated at the Fields Institute.

- **Section 2.4: No countable basis for Borel directed graphs of dichromatic number at least three**
  Presentation: Caltech Logic Seminar (April 29, 2026).
  Note: This new result completes the picture for finite Borel chromatic and dichromatic thresholds.

- **Section 2.5: Uncountable sets and an infinite linear order game**
  Source: Adapted from [MWS25].
  Presentation: 2023 Summer Meeting of the Canadian Mathematical Society, Ottawa.
  Note: This work provides a solution to a problem originally posed by Brian and Clontz about games.

- **Chapter 3: Deep Computations**
  Source: Primarily based on [DIMW$^+$26].
  Presentation: A multi-part series at the Fields Institute Set Theory Seminar (2024–2026).
  Note: This chapter focuses on the author's contributions to the separation of learnability classes with computational structures and their connections to other fields.

- **Section 4.2: Border tracing in oriented adjacency graphs of polygonal tilings**
  Source: Based on my contributions to [WMW26a].

- **Section 4.3: Recursive constructions**
  Source: Extends joint work found in [EHMWV25] and [CGH$^+$23].
  Presentation: 2022 Summer Meeting of the Canadian Mathematical Society, St John's.
  Note: This section presents original generalizations and recursive constructions that extend beyond the currently published literature.

- **Section 4.4: Amoeba trees**
  Source: Based on [GGHMW].

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Chapter 1

Introduction

This thesis is the culmination of mainly three independent research projects I worked on during my Ph.D. studies at the University of Toronto. Before reading my work, there are two important remarks I want to make to any interested reader:

- The chapters focus on my own main contributions to the fields of research and my perspectives on existing questions, results and concepts. For this reason, there are aspects of the joint work that are alluded in the text but not comprehensively covered, even if I was very much a part of said research: only those results and ideas in which I felt I played a major role are mentioned and explained in detail. This comment matters in particular to readers already familiar with the field of research, as they may notice relevant results that first appeared in my own papers that I did not include in this thesis.

- Each chapter is self-contained and assumes no prior deep knowledge of the research presented. Concretely, I do not expect readers to be familiar or have read any of the cited papers that I (co-)authored. In addition, each chapter offers a comprehensive introduction covering, or, in the case of more elementary material, containing detailed references of, all of the necessary preliminaries. The chapters have several overarching themes that are often not explicitly present in the published research, and so I went to great efforts to highlight these connections: from including new results and generalizations, to reformulating notions in a lingua franca of sorts.

Chapter 2

Descriptive Combinatorics

This is the descriptive combinatorics chapter of the thesis.

2.1 Background

Throughout this entire chapter, G denotes an analytic graph on a Polish space $V(G) = X$. That is, X is a separable completely metrizable topological space, and G is a symmetric and irreflexive relation on X which, as a subspace of $X \times X$, is the continuous image of a Polish space.

I employ the symbol $H \to_c G$ (respectively, $H \to_B G$) to abbreviate the fact that there is a continuous (respectively Borel) homomorphism from the analytic graph H to the analytic graph G.

2.1.1 The Borel chromatic number

The *Borel chromatic number* $\chi_B(G)$ of a Borel graph G was introduced by Kechris, Solecki, and Todorčević [KST99], who proved the celebrated G_0 *dichotomy*: there exists a single Borel graph $\mathbb{G}_0$ such that

$$\chi_B(G) > \aleph_0 \quad \iff \quad \mathbb{G}_0 \to_B G. \tag{2.1}$$

In other words, the graph $\mathbb{G}_0$ forms a one-element *basis* for the class of Borel graphs with uncountable Borel chromatic number.

This dichotomy breaks down when one moves from uncountable to merely infinite Borel chromatic number. In [TV21], Todorčević and Vidnyánszky recently proved that the set of Borel graphs with infinite Borel chromatic number (equivalently, Borel chromatic number ≥ 4 for closed subgraphs of the shift graph) is $\mathbf{\Sigma}^1_2$-complete. In particular, no countable family of Borel graphs can serve as a basis for this class. Moreover, their argument rules out the existence of a simple G_0-type dichotomy substituting $\aleph_0$ in (2.1) for any value in $3, 4, 5 \ldots$. Interestingly, when substituting the value 2, there again is a single Borel graph that makes a dichotomy like (2.1) hold. This is known as the L_0 dichotomy and is the main result of [CMSV21]. This thesis contains a more concise proof of said result in Section 2.2.

Fact 2.1. *If $H \to_c G$ or, more generally if $H \to_B G$, then $\chi_B(H) \leq \chi_B(G)$.*

Using a standard greedy algorithm argument, it is clear that any finite graph G of maximum degree Δ satisfies $\chi(G) \leq \Delta + 1$. Interestingly, a similar fact holds for Borel graphs of uniformly bounded degree.

Theorem 2.2 (Proposition 4.6 in [KST99])**.** *If G is a Borel graph on a Polish space all of whose degrees are bounded by the natural number k, then $\chi_B(G) \leq k + 1$.*

A common situation is having an analytic set and the desire to apply to it a statement that holds for Borel sets instead. One way to circumvent this problem is by using Borel separation to "inflate" the analytic set into a Borel set with the same property. This technique is crucial, for instance, for the proof of the G_0 dichotomy found in [Ber24]. Concretely, since the dichotomy is a statement about independent sets of vertices, the proof makes use of the following fact.

Proposition 2.3. *In the graph G, any analytic independent set of vertices is contained in a Borel independent set.*

We present two proofs of this fact, original to this exposition, for several reasons. First, because [Ber24] omits it [1]; it is an illustrative example of how different notions separation get used in an argument; it motivates the introduction of a more powerful technique that we need in Section 2.2. Specifically, Claim 2.10.1 states an analog result for odd-walks, instead of merely independent sets. The added complexity is that odd-walks can have different lengths, and so proving the aforementioned claim is not as straightforward without invoking Lemma 2.5.

Proof. Suppose that $I \subseteq V(G)$ is an analytic independent set. We view the edge relation of G as a subset of $V(G) \times V(G)$, which is analytic by assumption.

First, consider the set

$$A_0 := \{x_0 \in V(G) : \exists x_1 \in I \ (x_0 x_1 \in G)\},$$

which consists of all vertices with at least one neighbour in I. To see that A_0 is analytic, write it as the projection onto the first coordinate of the set

$$(V(G) \times I) \cap G \ = \ \{(x_0, x_1) : x_1 \in I \text{ and } x_0 x_1 \in G\}.$$

This is the intersection of the analytic set $V(G) \times I$ with the analytic set G, hence analytic. Since I is independent, $I \cap A_0 = \emptyset$. As I and A_0 are disjoint analytic sets, the Lusin Separation Theorem [Kec95, Theorem 14.7] yields a Borel set $B_0 \subseteq V(G)$ with

$$I \subseteq B_0 \quad \text{and} \quad B_0 \cap A_0 = \emptyset.$$

The second condition means that no vertex of B_0 has any neighbour in I.

Next, let

$$A_1 := \{x_1 \in V(G) : \exists x_0 \in B_0 \ (x_0 x_1 \in G)\}$$

be the set of all vertices with at least one neighbour in B_0. Analogously to before, A_1 is the projection

[1] The fact is indeed left as an exercise to the reader.

onto the second coordinate of

$$(B_0 \times V(G)) \cap G \;=\; \{(x_0, x_1) : x_0 \in B_0 \text{ and } x_0x_1 \in G\},$$

which is the intersection of the Borel set $B_0 \times V(G)$ with the analytic set G, and is therefore analytic. We claim that $I \cap A_1 = \emptyset$. Indeed, if some $x_1 \in I$ had a neighbour $x_0 \in B_0$, then x_0 would be a vertex of B_0 with a neighbour $x_1 \in I$, that is $x_0 \in A_0$, contradicting $B_0 \cap A_0 = \emptyset$. So again we have two disjoint analytic sets, and separation yields a Borel set $B_1 \subseteq V(G)$ with

$$I \subseteq B_1 \quad \text{and} \quad B_1 \cap A_1 = \emptyset.$$

The second condition means that no vertex of B_1 has any neighbour in B_0.

Set $B := B_0 \cap B_1$, which is Borel and contains I. It remains to check that B is independent. Suppose for contradiction that $x_0, x_1 \in B$ with $x_0x_1 \in G$. Then $x_0 \in B_0$, so x_1 is a vertex with a neighbour in B_0, giving $x_1 \in A_1$. But $x_1 \in B_1$ and $B_1 \cap A_1 = \emptyset$, a contradiction. □

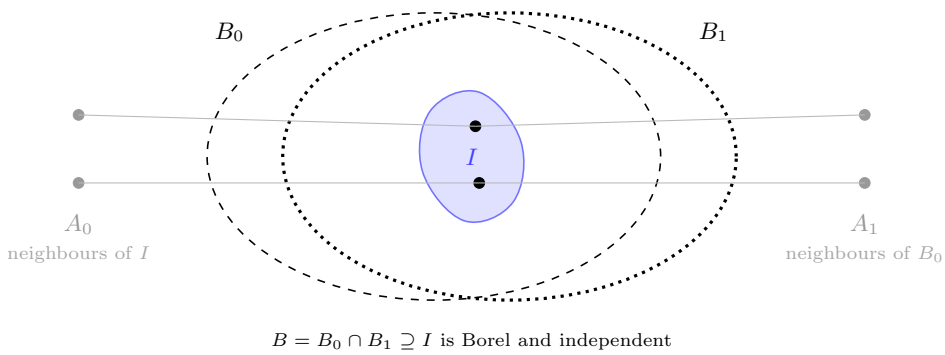

Figure 2.1: An illustration of Borel Separation in vertex sets.

The following notion, taken from [Kec95, Theorem 35.10, p. 285], is a well-known fact about analytic sets.

Definition 2.4. A collection of subsets Φ of a Polish space X is said to be Π^1_1 *on* Σ^1_1 if for any Polish Y and any Σ^1_1 set $A \subseteq Y \times X$, the set $A_\Phi := \{y \in Y : A_y := \{x \in X : (y, x) \in A\} \in \Phi\}$ is Π^1_1.

Lemma 2.5 (First Reflection Theorem)**.** *If Φ is as in the preceding definition, then for any Σ^1_1 set $A \in \Phi$, there is a Borel set $B \subseteq X$ such that $A \subseteq B$ and $B \in \Phi$.*

For example, the property of being an independent set of vertices is, as expected, Π^1_1 on Σ^1_1. Indeed, let Φ denote the collection of all independent subsets of $V(G)$, let Y be a Polish space, and let $A \subseteq Y \times V(G)$ be Σ^1_1. We must show that $A_\Phi = \{y \in Y : A_y \in \Phi\}$ is Π^1_1. Observe that its complement consists precisely of those $y \in Y$ for which A_y contains an edge:

$$Y \setminus A_\Phi = \{y \in Y : \exists x_0, x_1 \in V(G)\ (x_0x_1 \in E(G) \wedge (y, x_0) \in A \wedge (y, x_1) \in A)\}.$$

The set $\{(y, x_0, x_1) : (y, x_0) \in A \wedge (y, x_1) \in A\}$ is Σ^1_1, being the preimage of $A \times A$ under the continuous map $(y, x_0, x_1) \mapsto ((y, x_0), (y, x_1))$; intersecting with the Borel set $\{(y, x_0, x_1) : x_0x_1 \in E(G)\}$ keeps it Σ^1_1. Projecting onto Y therefore yields a Σ^1_1 set, so A_Φ is Π^1_1, as required.

Consequently, the Proposition above is an immediate corollary of Lemma 2.5: given an analytic independent set $I \in \Phi$, the First Reflection Theorem directly produces a Borel set $B \supseteq I$ with $B \in \Phi$. Note that this second proof, unlike the elementary one given above, generalises readily: one only needs to verify the Π^1_1 on Σ^1_1 property for the relevant family Φ, which, for more complex combinatorial properties, is typically a non-trivial step.

2.1.2 The Borel dichromatic number

The *dichromatic number* of a directed graph was introduced by Neumann-Lara [NL82] as a natural directed analogue of the chromatic number. A *dicolouring* of a directed graph D is a map $c\colon V(D) \to k$ such that every colour class induces a subgraph without any directed cycles; the *dichromatic number* $\vec{\chi}(D)$ is the least such k. Equivalently, $\vec{\chi}(D)$ is the minimum number of acyclic sets into which the vertex set can be partitioned.

This notion has attracted considerable attention in finite combinatorics. Neumann-Lara [NL82] established several foundational results, including analogues of classical lower bounds for the chromatic number. Bokal, Fijavž, Juvan, Kayll, and Mohar [BFJ+04] later introduced the circular dichromatic number of a directed graph and proved that the decision problem of whether a directed graph D satisfies $\vec{\chi}(D) \leq 2$ or not is NP-complete; this is in contrast with the usual chromatic number for (undirected) graphs, where one gets NP-completeness only after posing the question for three colours or more.

Very recently, Raghavan and Xiao [RX24] established a dichotomy for the Borel dichromatic number that generalizes the G_0 dichotomy to directed graphs: they showed that a continuum-size family of directed graphs characterizes when a Borel directed graph has uncountable Borel dichromatic number. This continuum-size basis consists of pairwise Borel-incomparable directed graphs, but it is unknown whether any smaller basis exists; in particular, the question of whether a *countable* basis suffices is open.

One can interpret the existence of a small (i.e. singleton, or finite, or countable) basis as a sort of measure of complexity of the associated decision problem. For instance, the existence of $\mathbb{G}_0$ says that deciding whether a Borel graph has uncountable Borel chromatic number is "straightforward:" one only has to verify one possible ill-behaved property, namely containing a homomorphic copy of $\mathbb{G}_0$, to give a positive answer; and in some sense, finding this homomorphic copy is the *only* way of proving that a Borel graph has uncountable Borel chromatic number. In contrast, deciding whether a Borel graph has infinite Borel chromatic number is a much more complex task, as no easily describable method exists for deciding this in terms of graph homomorphisms, and so different graphs probably require wildly different proving methods, if any.

The Borel dichromatic number can recover the Borel chromatic number in the following sense.

Remark 2.6. Given an undirected graph G, let $\vec{G}$ denote the directed graph obtained by replacing each edge $\{u, v\}$ with two opposing arcs (u, v) and (v, u). Then $\vec{\chi}_B(\vec{G}) = \chi_B(G)$. Indeed, a subset of vertices is acyclic in $\vec{G}$ if and only if it is independent in G: any directed cycle in $\vec{G}$ would in particular contain an arc (u, v) with $\{u, v\} \in E(G)$, so a set containing both u and v is neither acyclic in $\vec{G}$ nor independent in G; conversely, an independent set in G induces a directed graph in $\vec{G}$ with no arcs at all, hence trivially acyclic. Moreover, the map $\mathrm{code}(G) \mapsto \mathrm{code}(\vec{G})$ is Δ^1_1, since it is built from the edge set using products and unions (Lemma 2.18(1),(3)). It follows that the $\mathbf{\Sigma}^1_2$-completeness of Borel k-colouring of undirected graphs [TV21] implies the $\mathbf{\Sigma}^1_2$-completeness

of Borel k-dicolouring of directed graphs, for every $k \geq 4$. The case $k = 2$ is not covered by this reduction (since deciding 2-colourability of undirected graphs is trivial), and is precisely the content of Proposition 2.15.

It is natural to ask whether the Todorčević–Vidnyánszky phenomenon also occurs in the directed setting at the remaining finite threshold, or if there is a directed analog for the L_0 dichotomy:

> *Is there a countable basis for the class of Borel directed graphs with Borel dichromatic number at least* 3*?*

Section 2.4 gives a negative answer, thereby completing the picture for all finite thresholds of both the Borel chromatic and dichromatic numbers.

The following table summarizes the basis landscape, contrasting the undirected and directed cases at different thresholds.

Class	**Existence of small basis**	**Reference**
$\chi_B(G) > \aleph_0$	Yes (size 1: $\mathbb{G}_0$)	[KST99]
$\chi_B(G) > 2$	Yes (size 1: $\mathbb{L}_0$)	[CMSV21], Section 2.2
$\chi_B(G) > k$, $k \geq 3$	No ($\mathbf{\Pi}^1_2$-complete)	[TV21]
$\vec{\chi}_B(D) > \aleph_0$	Yes (size $\mathfrak{c}$)	[RX24]
$\vec{\chi}_B(D) > k$, $k \geq 2$	No ($\mathbf{\Pi}^1_2$-complete)	Prop. 2.15, Rmk. 2.6

2.2 A concise proof of the L_0 dichotomy

Carroy, Miller, Schrittesser, and Vidnyánszky established the L_0 dichotomy: there is a Borel graph of Borel chromatic number three that admits a continuous homomorphism to every analytic graph of Borel chromatic number at least three. Their proof relies on a transfinite analysis of terminal approximations over a decreasing ω_1-sequence of analytic sets.

This section exhibits a new, substantially shorter proof of this result by adapting the graph-theoretic framework recently introduced by Bernshteyn for the G_0 dichotomy. The central device is a σ-ideal of *small* sets of homomorphisms from finite path approximations into the target graph, where smallness is witnessed by a bounded odd-walk condition on vertex projections. The key lemma that largeness is preserved under the doubling operation is established via the First Reflection Theorem, replacing the original transfinite construction with a single Borel reflection argument. The continuous homomorphism from the canonical graph $\mathbb{L}_c$ into the target is then obtained as a limit of shrinking families of copies, in direct analogy with Bernshteyn's proof for G_0.

2.2.1 Finite path approximations

Fix the parameter $c \in \omega^\omega$ and a path of length ω on the vertex set $\{p_n\}_{n<\omega}$. That is, the edge set $\{(p_n, p_{n+1}) : n < \omega\}$. Naturally, one could assume that $p_n = n$, but the labeling helps distinguish these vertices from other indices.

Recall that if s and t are finite strings (of, say, natural numbers), then $s^\frown t$ is the *concatenation* of s and t. Similarly, t^n is the repeated concatenation of t with itself, and (i) denotes the string of single value i. Next, recursively construct a family of graphs $\{L^c_n : n < \omega\}$ such that for all $n < \omega$:

L_n^c is a finite path; if $n > 0$, the endpoints of L_n^c are $e_i^n := (p_0)^\frown(0)^{n-1}{}^\frown(i)$, for $i < 2$; and

$$\text{every vertex of } L_n^c \text{ is of the form } (p_k)^\frown t \text{ for some } k \text{ and } t \in 2^{\leq n}. \tag{2.2}$$

Unless $t = \emptyset$, call such vertex a *non-path vertex*, and call the vertices of the form (p_k) the *path vertices* of L_n^c.

Start with L_0^c as the graph consisting of a single vertex p_0 and no edges, and set $e_0^0 = e_1^0 = p_0$. Now suppose that L_n^c has been constructed. L_{n+1}^c will be a path obtained from joining two copies of L_n^c by a path of $c(n)+2$ edges (with $c(n)+1$ *new* path vertices), but one must update the vertex labels in order to distinguish the two copies. One accomplishes this by simply appending 0 to each vertex of the first copy, and 1 to the other. Formally, start by adding the vertex $v^\frown(i)$ for every $i < 2$ and $v \in L_n^c$. Adding the path

$$((e_1^n)^\frown(0), p_0, p_1, \ldots, p_{c(n)}, (e_1^n)^\frown(1))$$

completes the construction of L_{n+1}^c, and its endpoints are

$$e_i^{n+1} = (e_0^n)^\frown(i) = (p_0)^\frown(0)^n{}^\frown(i),$$

for $i < 2$. This finishes the recursion. Note that to construct L_{n+1}^c from L_n^c, one only requires knowledge of the value $c(n)$.

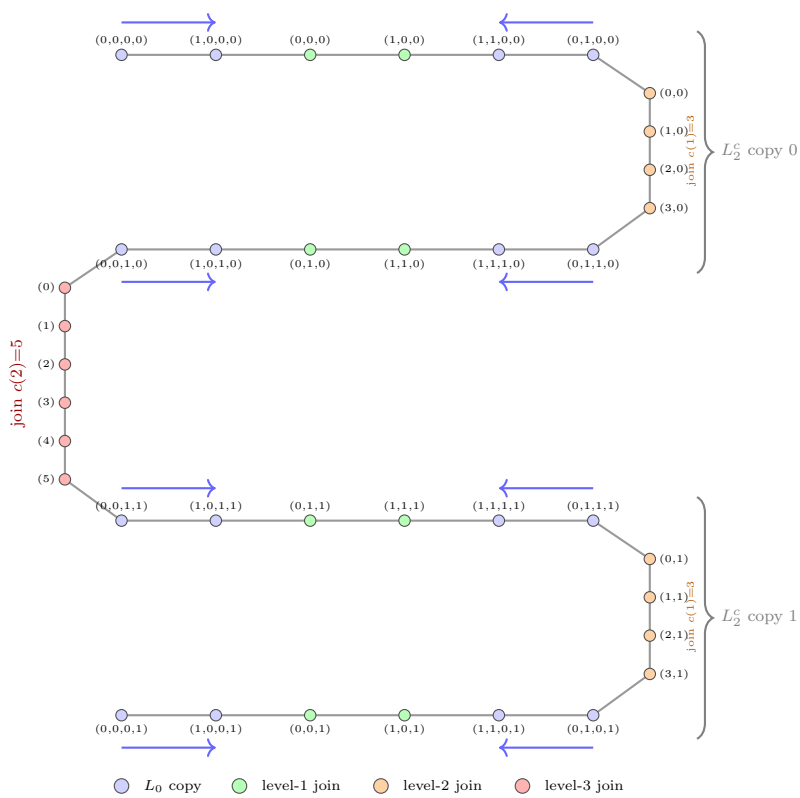

Figure 2.2: L_3^c for $c(0) = 1, c(1) = 3, c(2) = 5$.

To develop some intuition on what (2.2) says about a vertex v in L_n^c: the number $|v| - 1$ indicates

that v was added that many stages ago in the recursion, at the position of p_k in the path joining the two copies of the previous paths; the sequence t tells the story, in order, of which of the two copies of the previous path the vertex lies in. In other words, vertices of the form $(p_k)^\frown t$ for $t \in 2^n$ are precisely those that were copied from the original L_0^c in all possible positions along L_n^c, doubling the number of copies at each stage.

Based on this discussion, a vertex in some L_n^c is completely determined by the stage $n - m$ at which it was added in the position of p_k and then copied according to the binary sequence t.

2.2.2 A notion of smallness for copies

Now, define a notion of smallness analogous to the one presented in [Ber24]. By definition, the set of edges of G is the continuous image of some Polish space E, let us say under π. For a finite graph H, consider $\mathrm{Hom}(H, G)$ as the set of all maps

$$\varphi\colon V(H) \to X$$
$$\varphi\colon E(H) \to E$$

that satisfy that for all $uv \in E(H)$, $\pi(\varphi(uv)) = \varphi(u)\varphi(v)$.[2] Recall that $X = V(G)$ is the Polish space of vertices on which the analytic graph G is defined. Notice that $\mathrm{Hom}(H, G)$ is also Polish space, being a closed subset of the product space $X^{V(H)} \times E^{E(H)}$.

Given $\mathcal{H} \subseteq \mathrm{Hom}(H, G)$, define $\mathcal{H}(u) := \{\varphi(u) : \varphi \in \mathcal{H}\}$, which is analytic whenever $\mathcal{H}$ is Borel. Also, let $\mathcal{H}(uv) := \{\varphi(uv) : \varphi \in \mathcal{H}\}$.

Definition 2.7.

1. Let $k < \omega$. Say that the set A has property $\Phi(A, k)$ if $A \subseteq X$ is an analytic set of vertices of G such that all odd walks of G with endpoints in A have length at most $2k - 1$.

2. A Borel set $\mathcal{L} \subseteq \mathrm{Hom}(L_n^c, G)$ is called *tiny* if there is a vertex $u \in V(L_n^c)$ and a natural k such that $\Phi(\mathcal{L}(u), k)$.

3. A set $\mathcal{L} \subseteq \mathrm{Hom}(L_n^c, G)$ is *small* if it is the countable union of tiny sets. This notion defines a σ-ideal.

4. A set is *large* if it is not small. Notice that when H is the graph with a single vertex $H = \bullet\,(= L_0^c)$, $\mathrm{Hom}(H, G)$ can be identified with X and then for any $\mathcal{H} \subseteq \mathrm{Hom}(\bullet, G)$, $\mathcal{H}(\bullet)$ is identified with $\mathcal{H}$.

Now invoke the following auxiliary result that links the Φ property to Borel 2-colourability. The proof proceeds by induction on n using analytic separation; see [CMSV21, Claims 3.3–3.5] for details. Recall that a set of vertices of G is *invariant* if it is the union of connected components of G. In addition, if $A \subseteq X$, then $[A]_{E_G}$ is the union of all connected components of G that intersect A, that is, the set of vertices that can be reached from some element of A by a finite path. Therefore, a set A is invariant if and only if $A = [A]_{E_G}$.

[2] Formally, φ is a pair of two maps, however one can assume that $V(H) \cap E(H) = \emptyset$ and then there is no risk of confusion.

Lemma 2.8 (Claim 3.5 of [CMSV21])**.** *If $\Phi(A, n)$, then there is an invariant Borel set $B \supseteq [A]_{E_G}$ that induces a 2-Borel-colourable subgraph of G, that is, there is a Borel proper colouring $c_B \colon G\restriction_B \to 2$.*

Proposition 2.9. *If $\chi_B(G) > 2$ then X (via its identification with $\mathrm{Hom}(\bullet, G)$) is large.*

Proof. Suppose that $\mathrm{Hom}(\bullet, G) = \bigcup_{m<\omega} \mathcal{H}_m$ where each $\mathcal{H}_m$ is a tiny Borel set. For every $m < \omega$, Lemma 2.8 produces an invariant Borel set $B_m \supseteq [\mathcal{H}_m]_{E_G}$ with a Borel 2-colouring c_m. Define $c \colon X \to 2$ by letting $m(x) := \min\{m < \omega : x \in B_m\}$ and $c(x) = c_{m(x)}(x)$. Then c is a Borel 2-colouring because each B_m is invariant. □

Given that L^c_{n+1} is obtained from two copies of L^c_n joined by a path, it is convenient to have a means to refering to these distinct copies. Keeping property (2.2) in mind, for every $\varphi \in \mathrm{Hom}(L^c_{n+1}, G)$ and $i < 2$, define $\varphi^i \in \mathrm{Hom}(L^c_n, G)$ by $\varphi^i(u) = \varphi(u^\frown(i))$ for all $u \in V(L^c_n)$ and $\varphi^i(uv) = \varphi(u^\frown(i), v^\frown(i))$ for all $(u, v) \in E(L^c_n)$. Thus, given $\mathcal{L} \subseteq \mathrm{Hom}(L^c_n, G)$, define $\mathcal{L}^{+c}$ as the set of all $\varphi \in \mathrm{Hom}(L^c_{n+1}, G)$ for which $\varphi^i \in \mathcal{L}$ for both $i < 2$. Notice that if $\mathcal{L}$ is Borel, then so is $\mathcal{L}^{+c}$.

The length of the path added needs to be updated dynamically throughout the proof. Formally, one adjusts the parameter c by moving into a different branch of the Baire space when constructing the graphs L^c_n. Notice that, for two $c, d \in \omega^\omega$ and for every $n < \omega$, the condition $d\restriction_n = c\restriction_n$ implies that $L^c_k = L^d_k$ for all $k < n$.

Lemma 2.10.

(a) Suppose that $n > 0$, that c only takes odd values, that k is a natural and $t \in 2^{<n}$ is such that $(p_k)^\frown t^\frown(i)$, for $i < 2$, are vertices of L^c_n (see (2.2)). Then, they are an odd distance apart in L^c_n.

(b) For any $n < \omega$, if $\mathcal{L} \subseteq \mathrm{Hom}(L^c_n, G)$ is a large Borel set, then for any $N < \omega$ there exists $d \in \omega^\omega$ such that $d\restriction_n = c\restriction_n$, $d(n) \geq N$ is odd, and $\mathcal{L}^{+d}$ is a nonempty subset of $\mathrm{Hom}(L^d_{n+1}, G)$.

(c) For any $n < \omega$, if $\mathcal{L} \subseteq \mathrm{Hom}(L^c_n, G)$ is a large Borel set, then there exists $d \in \omega^\omega$ such that $d\restriction_n = c\restriction_n$ and $\mathcal{L}^{+d}$ is a large subset of $\mathrm{Hom}(L^d_{n+1}, G)$.

Proof. For (a), observe that both vertices come from the same vertex $(p_k)^\frown t$ in $L^c_{|t|+1}$, and so their distance in L^c_n is twice the distance from $(p_k)^\frown t$ to $e_1^{|t|}$ in $L^c_{|t|+1}$ (since the two vertices are each in distinct copies of $L^c_{|t|+1}$ inside L^c_n and the unique path between them passes through the joining path at stage $|t|$), plus the length of the joining path at that stage, which is $c(|t|) + 2$. Since $c(|t|)$ is odd, this distance is odd.

For (b), since $\mathcal{L}$ is large, it is not tiny. Hence, for every vertex $u \in V(L^c_n)$ and every natural k, $\Phi(\mathcal{L}(u), k)$ fails. In particular, fix the gluing vertex $u_0 := e^n_1 \in V(L^c_n)$ (recall that $e^0_1 = p_0$). Then for every k there exist $\varphi_0, \varphi_1 \in \mathcal{L}$ and an odd walk $(v_0, v_1, \ldots, v_m)$ in G with $v_0 = \varphi_0(u_0)$, $v_m = \varphi_1(u_0)$, and $m > 2k - 1$. Choosing k large enough, it is ensured that $m \geq N + 2$ and m is odd. Since m is odd, $d(n) := m - 2$ is also odd and satisfies $d(n) \geq N$. For each $1 \leq j \leq m$, since $(v_{j-1}, v_j) \in E(G) = \pi[E]$, pick $e_j \in E$ with $\pi(e_j) = (v_{j-1}, v_j)$. Setting $d(x) = c(x)$ for all $x \neq n$ and noting $L^c_n = L^d_n$ (since $d\restriction_n = c\restriction_n$), define $\varphi \in \mathrm{Hom}(L^d_{n+1}, G)$ by $\varphi^0 = \varphi_0$, $\varphi^1 = \varphi_1$, $\varphi(p_i) = v_{i+1}$ for $0 \leq i \leq m - 2 = d(n)$, and the edge assignments $\varphi(u_0^\frown(0), p_0) = e_1$, $\varphi(p_j, p_{j+1}) = e_{j+1}$ for $0 \leq j < m - 2$, and $\varphi(p_{m-2}, u_0^\frown(1)) = e_m$. Then $\varphi \in \mathcal{L}^{+d}$.

Next, prove (c) by contradiction. Recall that constructing L^c_{n+1} from L^c_n only requires knowing the value of $c(n)$. Suppose that for all d that agree with c except possibly in the n-th value, $\mathcal{L}^{+d} = \bigcup_m \mathcal{F}^d_m$ where each $\mathcal{F}^d_m$ is a tiny Borel subset of $\mathrm{Hom}(L^d_{n+1}, G)$. For each m and d, let $w^d_m \in V(L^d_{n+1})$ and $k^d_m < \omega$ be such that $\Phi(\mathcal{F}^d_m(w^d_m), k^d_m)$.

Since $w^d_m \in V(L^d_{n+1})$, it is either a *non-path vertex* of the form $u^\frown(i)$ for some $u \in V(L^c_n)$ and $i < 2$, or a *path vertex* p_j for some $0 \le j \le d(n)$ in the joining path. We claim that in either case, there exists a vertex $u^d_m \in V(L^c_n)$ and $\ell^d_m < \omega$ such that $\Phi(\mathcal{F}^d_m({u^d_m}^\frown(i^d_m)), \ell^d_m)$ for some $i^d_m < 2$ where ${u^d_m}^\frown(i^d_m)$ is a non-path vertex of L^d_{n+1}.

Indeed, if $w^d_m = u^\frown(i)$ is already non-path, take $u^d_m = u$, $i^d_m = i$, $\ell^d_m = k^d_m$. If $w^d_m = p_j$ is a path vertex of L^d_{n+1}, then the joining path connects $e^{n\frown}_1(0)$ to $e^{n\frown}_1(1)$ via $p_0, \dots, p_{d(n)}$. Since p_j and $e^{n\frown}_1(0)$ are at distance $j+1$ in L^d_{n+1}, every homomorphism $\varphi \in \mathcal{F}^d_m$ maps them to vertices connected by a walk of lengh $j+1$ in G. Given any odd walk $(a_0, \dots, a_r)$ in G with $a_0, a_r \in \mathcal{F}^d_m(e^{n\frown}_1(0))$, witnessed by $\varphi, \varphi' \in \mathcal{F}^d_m$, the walk $(\varphi(p_j), \dots, \varphi(e^{n\frown}_1(0)) = a_0, \dots, a_r = \varphi'(e^{n\frown}_1(0)), \dots, \varphi'(p_j))$ has length $r + 2(j+1)$, which has the same parity as r (hence is also odd), and has endpoints in $\mathcal{F}^d_m(p_j)$. By $\Phi(\mathcal{F}^d_m(p_j), k^d_m)$, $r + 2(j+1) \le 2k^d_m - 1$, so $r \le 2(k^d_m - j - 1) - 1$. It follows that $\Phi(\mathcal{F}^d_m(e^{n\frown}_1(0)), \max\{0, k^d_m - j - 1\})$. Thus, set $u^d_m = e^n_1$, $i^d_m = 0$, $\ell^d_m = \max\{0, k^d_m - j - 1\}$.

Claim 2.10.1. *For each m and d, there is a Borel set $B^d_m \supseteq \mathcal{F}^d_m({u^d_m}^\frown(i^d_m))$ such that $\Phi(B^d_m, \ell^d_m)$.*

Proof of Claim. Use the First Reflection Theorem (Lemma 2.5). Fix $\ell = \ell^d_m$. It suffices to verify that $\Phi_\ell := \{A \subseteq X : \Phi(A, \ell)\}$ is Π^1_1 on Σ^1_1. Let Y be Polish and $A \subseteq Y \times X$ be Σ^1_1. Fix a Polish space N and a continuous surjection $g\colon N \to A$. For each odd $r \ge 2\ell + 1$, define

$$C_r := \{(y, x_0, \dots, x_r, n_0, n_1, \hat{e}_1, \dots, \hat{e}_r) \in Y \times X^{r+1} \times N^2 \times E^r : \\ g(n_0) = (y, x_0) \wedge g(n_1) = (y, x_r) \wedge \textstyle\bigwedge_{j<r} \pi(\hat{e}_{j+1}) = (x_j, x_{j+1})\}.$$

Then C_r is closed (by the continuity of g and π), and $D_r := \mathrm{proj}_Y(C_r)$ is Σ^1_1. Hence $D := \bigcup\{D_r : r \text{ odd}, r \ge 2\ell + 1\}$ is Σ^1_1, and $A_{\Phi_\ell} = \{y \in Y : A_y \in \Phi_\ell\} = Y \setminus D$ is Π^1_1. Since $\mathcal{F}^d_m({u^d_m}^\frown(i^d_m))$ is analytic and satisfies $\Phi(\cdot, \ell)$, the First Reflection Theorem yields the desired Borel. ⊣

Take

$$\mathcal{H}^d_m = \{\varphi \in \mathcal{L} : \varphi(u^d_m) \in B^d_m\},$$

which is also a Borel subset of $\mathrm{Hom}(L^c_n, G)$. We claim that $\mathcal{H}^d_m$ is tiny. Indeed, $\mathcal{H}^d_m(u^d_m) = \{\varphi(u^d_m) : \varphi \in \mathcal{L} \wedge \varphi(u^d_m) \in B^d_m\} \subseteq B^d_m$, so $\Phi(\mathcal{H}^d_m(u^d_m), \ell^d_m)$.

Now, the set of pairs (u^d_m, i^d_m) ranges over the finite set $V(L^c_n) \times 2$ (which does not depend on d), and (m, d) range over countably many values. Thus $\bigcup_{m,d} \mathcal{H}^d_m$ is a countable union of tiny sets, hence small. The set $\mathcal{L}_- := \mathcal{L} \setminus \bigcup_{m,d} \mathcal{H}^d_m$ is therefore large (and Borel). By part (b), there exist φ and d such that $\varphi \in \mathcal{L}^{+d}_- \subseteq \mathcal{L}^{+d}$. Then $\varphi \in \mathcal{F}^d_m$ for some m. It follows that $\varphi^{i^d_m}(u^d_m) = \varphi({u^d_m}^\frown(i^d_m)) \in \mathcal{F}^d_m({u^d_m}^\frown(i^d_m)) \subseteq B^d_m$, hence $\varphi^{i^d_m} \in \mathcal{H}^d_m$. But $\varphi \in \mathcal{L}^{+d}_-$ implies $\varphi^{i^d_m} \in \mathcal{L}_-$, contradicting $\mathcal{L}_- \cap \mathcal{H}^d_m = \emptyset$. □

2.2.3 Construction of a minimal graph of Borel chromatic number at least 3

The goal of this subsection is to construct a family of graphs $\mathbb{L}_c$ indexed by $c \in \omega^\omega$.

Fix c and consider X_c as the set of all tuples $(m, k, x) \in \mathbb{N} \times \mathbb{N} \times 2^{\mathbb{N}}$ such that either $m = 0$ and $k = 0$, or $m \geq 1$ and $k \leq c(m-1)$; this is a closed subspace of the product space, and hence Polish. Define, for each (m, k, x) with $m \leq n$, $\pi_n(m, k, x)$ as the vertex of L_n^c determined by these parameters. Formally, $\pi_n(m, k, x) := (p_k)^\frown x\restriction_{n-m}$. For example, $\pi_3(1, 2, 0110\cdots) = (p_2, 0)$ can be seen in Figure 2.2 labeled as $(2, 0)$.

Finally, $\mathbb{L}_c$ is the graph on X_c where two (n_i, k_i, x_i), for $i < 2$, are adjacent if and only if the $\pi_n(n_i, k_i, x_i)$ are adjacent in all L_n^c with $n \geq \max\{n_0, n_1\}$. Some basic properties of $\mathbb{L}_c$ follow.

Lemma 2.11.

(a) Two vertices $(n_i, k_i, x_i) \in X_c$, for $i < 2$, are in the same connected component of $\mathbb{L}_c$ if and only if there are $t_i \in 2^{<\omega}$ and $x \in 2^\omega$ such that $|t_0| - |t_1| = n_1 - n_0$ and $x_i = t_i^\frown x$.

(b) $\mathbb{L}_c$ has no cycles and all vertices of $\mathbb{L}_c$ have degree 2 except for the vertex $(0, 0, \{0\}^\omega)$, which has degree 1.

(c) If, additionally, c takes only odd values, then $\chi_B(\mathbb{L}_c) = 3$.

Proof. For (a), let $(n_i, k_i, x_i) \in X_c$ for $i < 2$. If they are in the same connected component, then for all large enough n, $\pi_n(n_0, k_0, x_0)$ and $\pi_n(n_1, k_1, x_1)$ lie in the same copy of L_{n-1}^c inside L_n^c. By the construction of L_n^c, two vertices $(p_{k_0})^\frown x_0\restriction_{n-n_0}$ and $(p_{k_1})^\frown x_1\restriction_{n-n_1}$ are in the same copy of L_{n-1}^c in L_n^c if and only if their last binary digits agree, that is, $x_0(n - n_0 - 1) = x_1(n - n_1 - 1)$ for all large n. Writing $x_0 = t_0^\frown x$ and $x_1 = t_1^\frown x$ with $|t_0| = n_1 - n_0 + |t_1|$ (assuming $n_0 \leq n_1$), one obtains the stated condition. Conversely, if $x_i = t_i^\frown x$ with $|t_0| - |t_1| = n_1 - n_0$, then for all large n, $\pi_n(n_0, k_0, x_0) = (p_{k_0})^\frown t_0^\frown x\restriction_r$ and $\pi_n(n_1, k_1, x_1) = (p_{k_1})^\frown t_1^\frown x\restriction_r$ for some r, so they share a common tail and lie in the same connected component of L_n^c, hence of $\mathbb{L}_c$.

For (b), every vertex of L_n^c has degree at most 2, so the same holds for $\mathbb{L}_c$. It follows from acyclicity of the finite paths L_n^c that $\mathbb{L}_c$ is acyclic. The vertex $(0, 0, 0^\omega)$ maps under π_n to $e_0^n = (p_0)^\frown 0^n$, which is an endpoint of L_n^c, hence has degree 1 in L_n^c for all $n \geq 1$. Thus $(0, 0, 0^\omega)$ has degree 1 in $\mathbb{L}_c$. For any other vertex $(m, k, x) \in X_c$, the image $\pi_n(m, k, x)$ is an interior vertex of L_n^c for all sufficiently large n, so it has degree exactly 2.

(c) That $\chi_B(\mathbb{L}_c) \leq 3$ follows from Proposition 2.2 and part (b). Argue the other inequality by contradiction. Suppose that there is a Borel partition of X_c into two independent sets $X_c = I_0 \cup I_1$. Since X_c is Polish, the Baire category theorem guarantees that some I_ε is non-meager, hence comeager in a basic open set $U = \{(m, k, x) \in X_c : x\restriction_r = t\}$ for some m, k, r, and $t \in 2^r$. In particular, for a comeager set of $x \in 2^{\mathbb{N}}$ extending t, $(m, k, x) \in I_\varepsilon$. Since $(m, k, t^\frown(0)^\frown y)$ and $(m, k, t^\frown(1)^\frown y)$ are in the same connected component (by part (a)) for any $y \in 2^{\mathbb{N}}$, and I_ε is comeager in U, one can find y such that both $(m, k, t^\frown(0)^\frown y)$ and $(m, k, t^\frown(1)^\frown y)$ lie in I_ε. But by Lemma 2.10(a), these two vertices are an odd distance apart in $\mathbb{L}_c$ (as $(p_k)^\frown t^\frown(0)$ and $(p_k)^\frown t^\frown(1)$ are an odd distance apart in every L_n^c for large enough n), contradicting I_ε being independent. $\square$

The main result is then split into two parts.

Theorem 2.12. *For any analytic graph G, either $\chi_B(G) \leq 2$ or there is a c such that $\mathbb{L}_c \to_c G$.*

Moreover, c can be chosen to be unbounded and take only odd values.

Theorem 2.13. *Let $c, d \in \omega^\omega$ be unbounded and only taking odd values, then $\mathbb{L}_c$ and $\mathbb{L}_d$ are continuously homomorphically equivalent.*

The proof of Theorem 2.13 is purely combinatorial: one constructs homomorphisms between $\mathbb{L}_c$ and $\mathbb{L}_d$ by iteratively extending partial maps along finite path components, using the "large gap property" guaranteed by unboundedness. The argument does not depend on the method used to establish Theorem 2.12; see [CMSV21, Proposition 4.2 and Claim 4.1] for details.

Combining Theorems 2.12 and 2.13, and fixing any unbounded odd c_0 (e.g., $c_0(0) = 1$ and $c_0(n) = 2n - 1$ for $n > 0$)[3], the L_0 dichotomy of [CMSV21] is recovered:

Corollary 2.14. *Let $\mathbb{L}_0 := \mathbb{L}_{c_0}$. For any analytic graph G on a Polish space, exactly one of the following holds:*

(a) $\chi_B(G) \leq 2$;

(b) $\mathbb{L}_0 \to_c G$.

2.2.4 Proof of Theorem 2.12

By Lemma 2.11(c) and Fact 2.1, it is clear that both conditions cannot hold simultaneously. Now suppose that G is an analytic graph of Borel chromatic number at least three. The theorem is proved by constructing a valid c and a continuous homomorphism witnessing $\mathbb{L}_c \to_c G$.

By Proposition 2.9, X is large. Fix compatible complete metrics ρ on X and δ on E. Since $L_0^c = \bullet$, $\mathrm{Hom}(L_0^c, G)$ may be identified with X and set $\mathcal{L}_0 := X$, which is large. Recursively construct a function $c \in \omega^\omega$ (taking only odd values) and a sequence of large Borel sets $\mathcal{L}_n \subseteq \mathrm{Hom}(L_n^c, G)$ such that for all n,

(1) $\mathcal{L}_{n+1} \subseteq \mathcal{L}_n^{+c}$;

(2) for all $u \in V(L_n^c)$, $\mathrm{diam}_\rho(\overline{\mathcal{L}_n(u)}) < 2^{-n}$; and

(3) for all $(u, v) \in E(L_n^c)$, $\mathrm{diam}_\delta(\overline{\mathcal{L}_n(u, v)}) < 2^{-n}$.

Given $\mathcal{L}_n$ large, apply Lemma 2.10(c) to obtain $d \in \omega^\omega$ with $d\restriction_n = c\restriction_n$ and $\mathcal{L}_n^{+d}$ large; set $c(n) := d(n)$. To arrange conditions (2) and (3) for $n + 1$: since X and E can each be covered by countably many closed sets of ρ-diameter (respectively δ-diameter) less than $2^{-(n+1)}$, and small sets form a σ-ideal, one may intersect with the preimages of these covers and discard the resulting small pieces, passing to a large Borel subset $\mathcal{L}_{n+1} \subseteq \mathcal{L}_n^{+c}$ satisfying (2) and (3) at stage $n + 1$. Moreover, by choosing $c(n)$ sufficiently large using part (b) at each stage, c can be made unbounded.

By (1), for all $n \geq m$,

$$\mathcal{L}_{n+1}(\pi_{n+1}(m, k, x)) \subseteq \mathcal{L}_n(\pi_n(m, k, x));$$

indeed, given that $\pi_{n+1}(m, k, x) = \pi_n(m, k, x)^\frown(x(n - m))$, each $\varphi \in \mathcal{L}_{n+1}$ satisfies $\varphi^{x(n-m)} \in \mathcal{L}_n$ and

$$\varphi^{x(n-m)}(\pi_n(m, k, x)) = \varphi(\pi_{n+1}(m, k, x)).$$

By the same reasoning, if $\pi_n(m_0, k_0, x_0)$ and $\pi_n(m_1, k_1, x_1)$ are adjacent in L_n^c for all large n, then

$$\begin{aligned}&\mathcal{L}_{n+1}(\pi_{n+1}(m_0, k_0, x_0), \pi_{n+1}(m_1, k_1, x_1))\\ &\quad\subseteq \mathcal{L}_n(\pi_n(m_0, k_0, x_0), \pi_n(m_1, k_1, x_1)).\end{aligned}$$

[3] It is perhaps worth noting that I picked a *different* parameter for Figure 2.2 purely for aesthetic purposes.

Thus, the sets $\overline{\mathcal{L}_n(\pi_n(m,k,x))}$ are closed and nested, with diameters tending to zero (see (2) from before). Since ρ is a complete metric, Cantor's intersection theorem applies. For $(m,k,x) \in X_c$, let $f(m,k,x)$ be the unique point in

$$\bigcap_{n=m}^{\infty} \overline{\mathcal{L}_n(\pi_n(m,k,x))} = \bigcap_{n=0}^{\infty} \overline{\mathcal{L}_{n+m}((p_k)^\frown x\restriction_n)}.$$

The map $f\colon X_c \to X$ is continuous.

It remains to check that it is a homomorphism from $\mathbb{L}_c$ to G. Suppose that $(m_i,k_i,x_i)_{i<2}$ are adjacent in $\mathbb{L}_c$, that is, for all $n \geq n_0 := \max\{m_0,m_1\}$, $\pi_n(m_i,k_i,x_i)$ are adjacent in L_n^c. Similar to the preceding paragraph, the fact that δ is a complete metric implies that there is a unique edge $e \in E$ inside

$$\bigcap_{n\geq n_0} \overline{\mathcal{L}_n(\pi_n(m_0,k_0,x_0),\pi_n(m_1,k_1,x_1))}.$$

By the continuity of π, $(f(m_0,k_0,x_0),f(m_1,k_1,x_1)) = \pi(e) \in E(G)$, in other words, the images form an edge of G, and this concludes the proof.

The hereby presented argument shows that the core of the L_0 dichotomy, namely the construction of a continuous homomorphism from an $\mathbb{L}_c$ graph into any analytic graph of Borel chromatic number at least three, admits a proof that closely parallels Bernshteyn's argument for the G_0 dichotomy, with the challenge of adding a dynamically-changing parameter to the construction.

The main new ingredient is the replacement of the independence-based σ (sets whose vertex projections are independent) with the odd-walk-bounded σ-ideal defined by the Φ property. It remains an interesting question whether analogous adaptations can yield simplified proofs of other dichotomy results in Borel combinatorics, or even entirely new ones.

2.3 A Ramsey-type theorem for E_0 trees

I present a Ramsey-type theorem for E_0 trees, which is a key combinatorial ingredient in ongoing work.

2.4 No countable basis for Borel directed graphs of dichromatic number at least three

This section proves that the Borel directed graphs whose vertex set admits a partition into two Borel acyclic sets form a $\mathbf{\Sigma}^1_2$-complete set; equivalently, that deciding whether a Borel directed graph has Borel dichromatic number at least 3 is a $\mathbf{\Pi}^1_2$-complete problem. It follows that no countable family of Borel directed graphs can serve as a basis for this class under Borel homomorphism and, more generally, that any basis must be at least as complex as $\mathbf{\Pi}^1_2$.

The proof lifts the classical NP-completeness reduction of Bokal, Fijavž, Juvan, Kayll, and Mohar to the Borel setting, using the coding framework of Thornton. Combined with a straightforward reduction from undirected to directed colouring problems, this completes the picture for finite Borel chromatic and dichromatic thresholds: for every finite k, the set of Borel (directed) graphs admitting a Borel k-(di)colouring is $\mathbf{\Sigma}^1_2$-complete, and in particular admits no countable basis. This

contrasts with the uncountable threshold, where a single-element basis exists for Borel chromatic number (Kechris–Solecki–Todorčević) and a continuum-size basis exists for Borel dichromatic number (Raghavan–Xiao).

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We work throughout with *nice codes* in the sense of Thornton [Tho22a, Tho22b]; see Subsection 2.4.1 for a brief review.

Proposition 2.15. *The set of nice codes for Borel directed graphs admitting a Borel* 2*-dicolouring is* $\mathbf{\Sigma}^1_2$*-complete. Equivalently, the set of nice codes for Borel directed graphs with Borel dichromatic number at least* 3 *is* $\mathbf{\Pi}^1_2$*-complete.*

Corollary 2.16. *There is no countable family* $\mathbf{B}$ *of Borel directed graphs that forms a basis for Borel directed graphs of Borel dichromatic number at least* 3 *under Borel homomorphism.*

The proof of Proposition 2.15 proceeds by adapting the classical NP-completeness reduction from 2-colouring of 3-uniform hypergraphs to 2-dicolouring of directed graphs [BFJ$^+$04] to the Borel setting. The key step is verifying that this reduction is Δ^1_1 in the codes, following the template established by Thornton in [Tho22b, Appendix A] for a similar result about edge 3-colouring. I remark that 2-dicolouring of directed graphs is *not* a CSP in the sense of Thornton's algebraic framework [Tho22a], since acyclicity is not a constraint expressible as a homomorphism to a fixed finite template; thus Proposition 2.15 does not follow directly from Thornton's general results.

The picture reveals a notable difference between the undirected and directed settings at the uncountable threshold: a single graph suffices as a basis in the undirected case, while a continuum-size family is needed for directed graphs. With finitely many colours, both settings exhibit the same completeness obstruction to a countable basis, though at different thresholds: the phenomenon begins at $k = 3$ for undirected graphs and already at $k = 2$ for directed graphs, mirroring the classical NP-completeness landscape. The case $k = 2$ for directed graphs, established in this paper, is the last remaining piece of this picture (see Remark 2.6).

Several questions are posed in Section 5.1.

2.4.1 Coding framework

Briefly recall the coding apparatus from [Tho22b, Appendix A]; see also [Tho22a, Section 4]. The reader already familiar with this framework may skip to Subsection 2.4.2.

Fix a good ω-parametrization $U \subseteq \omega \times \mathcal{N}$ of Π^1_1 [Tho22a, Theorem A.5].

Definition 2.17 ([Tho22a, Definition A.6])**.** A *simple code* for a Δ^1_1 set $B \subseteq \mathcal{N}$ is a pair $(e, i) \in \omega^2$ with $U_e = \mathcal{N} \setminus U_i = B$.

A *nice coding* is a triple $(\mathcal{C}, \mathcal{D}^\Pi, \mathcal{D}^\Sigma)$ where:

(i) $\mathcal{C} \subseteq \omega$ is Π^1_1 (the set of *codes*);

(ii) $\mathcal{D}^{\Pi} \subseteq \omega \times \mathcal{N}$ is Π^1_1 and $\mathcal{D}^{\Sigma} \subseteq \omega \times \mathcal{N}$ is Σ^1_1;

(iii) for every $e \in \mathcal{C}$, $\mathcal{D}^{\Pi}_e = \mathcal{D}^{\Sigma}_e$;

(iv) every Δ^1_1 set $B \subseteq \mathcal{N}$ has a code $e \in \mathcal{C}$ with $B = \mathcal{D}^{\Pi}_e$;

(v) there are recursive functions translating between simple and nice codes in both directions.

Fix nice codings for $\Delta^1_1(\mathcal{N}^k)$ for all k, writing $\mathcal{D}_e$ for $\mathcal{D}^{\Pi}_e$.

Lemma 2.18 ([Tho22a, Lemma A.7])**.** *Fix a Δ^1_1 linear order $\preceq$ on $\mathcal{N}$. The following operations are Δ^1_1 in the codes:*

(1) $(A, B) \mapsto A \cup B$*;*

(2) $(A, B) \mapsto A \cap B$*;*

(3) $(A, B) \mapsto A \times B$*;*

(4) $A \mapsto \mathrm{dom}(A)$*, for relations with countable sections;*

(5) $(A, f) \mapsto f(A)$*, for countable-to-one f;*

(6) $A \mapsto$ *the enumeration function of the finite sections of A along $\preceq$.*

This lemma can typically be used as a black box: to verify that a construction is Δ^1_1 in the codes, it suffices to express it as a composition of the preceding operations.

2.4.2 The classical reduction

Now, I recall the reduction from 2-colouring of 3-uniform hypergraphs to 2-dicolouring of directed graphs. The NP-completeness of 2-dicolouring follows by combining this reduction with the well-known NP-completeness of 2-colouring 3-uniform hypergraphs. The reduction is due to Bokal et al. [BFJ$^+$04]; I present it in a form convenient for the Borel lifting.

Fact 2.19. *The problem of deciding whether a 3-uniform hypergraph admits a 2-colouring is NP-complete.*

Definition 2.20 (Template gadget)**.** Let

$$T_{\text{shared}} := \{a, b, c\} \quad T_{\text{aux}} := \{a', b', c'\} \quad T := T_{\text{shared}} \cup T_{\text{aux}}.$$

The *template gadget* F is the directed graph on vertex set T with arc set E_T as depicted in Figure 2.3.

Lemma 2.21 (Gadget property)**.** *A map $\sigma\colon \{a, b, c\} \to 2$ extends to a 2-dicolouring of F if and only if σ is not constant (i.e., not all three shared vertices receive the same colour).*

Proof. This is a finite verification; see [BFJ$^+$04, Theorem 3.1]. □

Lemma 2.22 (Classical reduction)**.** *For every 3-uniform hypergraph H, there is a directed graph $D(H)$ such that H is 2-colourable if and only if $D(H)$ is 2-dicolourable.*

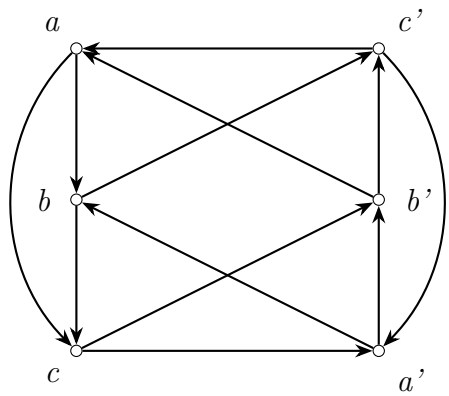


Figure 2.3: The template gadget F. Shared vertices $\{a, b, c\}$ correspond to hypergraph vertices; auxiliary vertices $\{a', b', c'\}$ are local to each gadget copy.

Proof. Let $V = V(H)$ and fix a linear order on V. Define

$$E_{\text{sort}} = \big\{(v_1, v_2, v_3) \in E(H) : v_1 < v_2 < v_3\big\},$$

a canonical listing of hyperedges as ordered triples. For each

$$e = (v_1, v_2, v_3) \in E_{\text{sort}},$$

place a copy F_e of the template gadget, identifying the shared labels a, b, c with the hypergraph vertices v_1, v_2, v_3 and introducing fresh auxiliary vertices $(e, a'), (e, b'), (e, c')$. The directed graph $D(H)$ has vertex set

$$V_D = V \cup \big\{(e, t) : e \in E_{\text{sort}},\ t \in T_{\text{aux}}\big\}$$

and arcs given by the union of the arcs of all copies F_e, with each shared label mapped to the corresponding vertex of V.

First, suppose that $c\colon V \to 2$ is a proper 2-colouring of the hypergraph H. For each e, the induced colouring on the shared vertices of F_e is non-constant, so by Lemma 2.21 it extends to a 2-dicolouring of F_e. Since auxiliary vertices of distinct gadgets are disjoint, combining any choice of extensions with c on shared vertices gives a global 2-dicolouring of $D(H)$.

Conversely, suppose that $\psi\colon V_D \to 2$ be a 2-dicolouring of $D(H)$ and set $\varphi = \psi\restriction_V$. For any $e = (v_1, v_2, v_3) \in E_{\text{sort}}$, the restriction $\psi\restriction_{V(F_e)}$ is a 2-dicolouring of F_e, so by the contrapositive of Lemma 2.21, $\varphi(v_1), \varphi(v_2), \varphi(v_3)$ are not all equal. Hence φ is a 2-colouring of H. □

Remark 2.23. The converse direction requires no selection or choice: φ is simply the restriction of ψ to the original vertex set V, which is a subset of V_D by construction. This is be important in Subsection 2.4.3, where it means that no uniformization argument is needed for the reverse direction of the Borel reduction.

2.4.3 The Borel reduction

I now verify that the construction of Subsection 2.4.2 lifts to the Borel setting. The argument follows the template of [Tho22a, Theorem A.9], where a similar verification for edge 3-colouring is carried

out.

Fact 2.24 ([Tho22a, Corollary 4.6])**.** *The set of nice codes for locally finite Borel 3-uniform hypergraphs admitting a Borel 2-colouring is $\mathbf{\Sigma}^1_2$-complete.*

It therefore suffices to produce a Δ^1_1 reduction from the set of codes for Borel 2-colourable 3-uniform hypergraphs to the set of codes for Borel 2-dicolourable directed graphs.

Now I verify that the construction is Δ^1_1 in the codes. Let H be a locally finite Borel 3-uniform hypergraph with vertex set $V \subseteq \mathcal{N}$ and hyperedge set $E \subseteq V^3$. We fix a Δ^1_1 linear order $\preceq$ on $\mathcal{N}$ and encode the template gadget F, its label set T, and its arc set E_T by canonical finite sequences in $\mathcal{N}$.

Begin by sorting the hyperedges.

$$E_{\text{sort}} = \{(a, b, c) \in E : a \preceq b \preceq c\}.$$

This is obtained from E by intersection with the graph of $\preceq$, hence is Δ^1_1 on codes by Lemma 2.18(2).

Then, build the vertex set. Define $V_D = V \cup \big(E_{\text{sort}} \times T_{\text{aux}}\big)$, which is a union of V with a product, hence Δ^1_1 on codes by Lemma 2.18(1),(3).

Finally, build the arc set. For each $e = (v_1, v_2, v_3) \in E_{\text{sort}}$ and each template arc $(u, v) \in E_T$, define the *substitution map* $\varphi_H \colon E_{\text{sort}} \times E_T \to V_D^2$ that sends a pair $(e, (u, v))$ to the corresponding arc of $D(H)$, by replacing each label $\ell \in T$ with its realization in $D(H)$:

- if $\ell \in T_{\text{shared}}$ is the shared label in position i (i.e., $\ell = a, b, c$ for $i = 1, 2, 3$ respectively), it maps to the vertex $v_i \in V$ occurring in $e = (v_1, v_2, v_3)$;
- if $\ell \in T_{\text{aux}}$, it maps to $(e, \ell) \in E_{\text{sort}} \times T_{\text{aux}}$.

This substitution is a recursive operation on the components of the tuple $(e, (u, v))$: it extracts coordinates of e (a projection) or forms pairs with e (a product), both of which are Δ^1_1 operations. Hence φ_H is Δ^1_1 in the code of H and in the (fixed, computable) code of E_T.

The arc set is $A_D = \varphi_H(E_{\text{sort}} \times E_T)$. To apply Lemma 2.18(5), we need φ_H to be countable-to-one. This holds: for any arc $\alpha \in A_D$, the preimage $\varphi_H^{-1}(\alpha)$ is finite, since E_T is finite and, by local finiteness of H, each vertex of V appears in only finitely many hyperedges in E_{sort}. By Lemma 2.18(1),(3),(5), the arc set A_D is Δ^1_1 on codes.

Applying the simple-to-nice code translation from the coding framework (Definition 2.17(v)), we obtain:

Lemma 2.25. *The function that maps the code of H to the code of $D(H)$ is Δ^1_1.*

Now I show that Borel colourability is preserved.

Claim 2.25.1. *If H is Borel 2-colourable, then $D(H)$ is Borel 2-dicolourable.*

Proof. Let $c \colon V \to 2$ be a Borel 2-colouring of H. For each $e \in E_{\text{sort}}$, the restriction of c to the shared vertices of F_e is non-constant, so by the gadget property (Lemma 2.21) there exists at least one extension to a 2-dicolouring of F_e. Denote by $\mathcal{C}_F$ the finite set of maps $T \to 2$ and define S as the set of all $(e, \sigma) \in E_{\text{sort}} \times \mathcal{C}_F$ such that σ is a 2-dicolouring of F_e extending the shared-vertex colours. The set S is Borel since the defining condition involves only finitely many acyclicity checks and colour-matching conditions, all determined by the Borel colouring c and the fixed finite template.

Furthermore, S has finite nonempty sections S_e. By the Luzin–Novikov uniformization theorem, there is a Borel selector $e \mapsto \sigma(e) \in S_e$.

Since auxiliary vertices of distinct gadgets are disjoint, the map $\psi\colon V_D \to 2$ defined by

$$\psi(v) = \begin{cases} c(v), & v \in V; \\ \sigma(e)(t), & (e,t) \in E_{\text{sort}} \times T_{\text{aux}} \end{cases}$$

is a well-defined Borel 2-dicolouring of $D(H)$. □

Claim 2.25.2. *If $D(H)$ is Borel 2-dicolourable, then H is Borel 2-colourable.*

Proof. Let $\psi\colon V_D \to 2$ be a Borel 2-dicolouring of $D(H)$ and define $\varphi = \psi\restriction_V$. Since $V \subseteq V_D$ is Borel, φ is Borel. For any hyperedge $e = (v_1, v_2, v_3) \in E_{\text{sort}}$, the gadget F_e is a directed subgraph of $D(H)$, so $\psi\restriction_{V(F_e)}$ is a 2-dicolouring of F_e, and the gadget property implies that $\varphi(v_1), \varphi(v_2), \varphi(v_3)$ are not all equal. Hence φ is a Borel 2-colouring of H. □

Remark 2.26. Note that the reverse direction (Claim 2.25.2) uses neither the local finiteness of H nor any uniformization. Local finiteness is needed only for two purposes: it is a hypothesis of Fact 2.24, and it guarantees that the sections S_e in the forward direction are finite (enabling the application of Luzin–Novikov).

I can finally prove the main result of the section.

Proof of Proposition 2.15. By Fact 2.24, the set of nice codes for locally finite Borel 3-uniform hypergraphs with Borel 2-colourings is $\mathbf{\Sigma}^1_2$-complete. By Lemma 2.25, the map $\text{code}(H) \mapsto \text{code}(D(H))$ is Δ^1_1. By Claims 2.25.1 and 2.25.2, H is Borel 2-colourable if and only if $D(H)$ is Borel 2-dicolourable. Thus the set of nice codes for Borel 2-dicolourable directed graphs is $\mathbf{\Sigma}^1_2$-hard.

For the upper bound, note that a directed graph D (given by a nice code) admits a Borel 2-dicolouring if and only if there exists a nice code e_c such that $\mathcal{D}_{e_c}$ defines a valid 2-dicolouring of D. The condition "$\mathcal{D}_{e_c}$ is a 2-dicolouring" is Π^1_1 in the codes e_c and e_D, as it requires totality and that each colour class induces no directed cycle, both of which are universal conditions. The existential quantification over e_c gives a Σ^1_2 condition. Hence the set of codes for Borel 2-dicolourable directed graphs is $\mathbf{\Sigma}^1_2$-complete, and its complement, which are codes for directed graphs with $\vec{\chi}_B(D) \geq 3$, is $\mathbf{\Pi}^1_2$-complete. □

Remark 2.27. The convention used in [TV21] and [Tho22a] is to state the $\mathbf{\Sigma}^1_2$-completeness of the class of graphs *admitting* a colouring. We follow this convention.

Proof of Corollary 2.16. Suppose a countable basis $\mathbf{B} = \{H_1, H_2, \dots\}$ exists, so that

$$\vec{\chi}_B(D) \geq 3 \quad \iff \quad \exists\, H_n \in \mathbf{B},\ \exists \text{ Borel homomorphism } H_n \to D.$$

For each fixed H_n, the condition "there exists a Borel homomorphism $H_n \to D$" is $\mathbf{\Sigma}^1_2$ in the code for D: one existential quantifier over a nice code e_f for a Borel function $f\colon V(H_n) \to V(D)$, and the requirement that f preserves arcs is Π^1_1 in the codes.

The countable union $\bigcup_n \{D : H_n \to_B D\}$ is therefore $\mathbf{\Sigma}^1_2$. And Proposition 2.15 asserts that the $\{D : \vec{\chi}_B(D) \geq 3\}$ is $\mathbf{\Pi}^1_2$-complete. A set that is simultaneously $\mathbf{\Sigma}^1_2$ and $\mathbf{\Pi}^1_2$-hard would give $\mathbf{\Pi}^1_2 \subseteq \mathbf{\Sigma}^1_2$, contradicting the fact that the projective hierarchy is strictly ordered. □

Remark 2.28. The argument proves more than the non-existence of a countable basis. If $\mathbf{B}$ is any basis for the class of Borel directed graphs with $\vec{\chi}_B(D) \geq 3$, then the right-hand side of

$$\vec{\chi}_B(D) \geq 3 \quad \iff \quad \exists\, H \in \mathbf{B},\ \exists \text{ Borel homomorphism } H \to D$$

must define a $\mathbf{\Pi}^1_2$-complete set, since the left-hand side is $\mathbf{\Pi}^1_2$-complete by Proposition 2.15. If $\mathbf{B} \in \mathbf{\Sigma}^1_2$ (viewed as a subset of the code space), then "$H \in \mathbf{B}$" is a $\mathbf{\Sigma}^1_2$ condition on the code of H, and "there exists a Borel homomorphism $H \to D$" is $\mathbf{\Sigma}^1_2$ in the codes of H and D (as in the proof of Corollary 2.16). The conjunction of two $\mathbf{\Sigma}^1_2$ conditions is $\mathbf{\Sigma}^1_2$, and existential quantification preserves $\mathbf{\Sigma}^1_2$, so the right-hand side would be $\mathbf{\Sigma}^1_2$ in the code of D. But this contradicts $\mathbf{\Pi}^1_2$-completeness of the left-hand side, since $\mathbf{\Sigma}^1_2 \neq \mathbf{\Pi}^1_2$. Hence $\mathbf{B} \notin \mathbf{\Sigma}^1_2$: any basis must be at least $\mathbf{\Pi}^1_2$-hard as a subset of the code space, and is therefore at least as complex to describe as the set of all directed graphs without a Borel acyclic 2-colouring. The separation $\mathbf{\Sigma}^1_2 \neq \mathbf{\Pi}^1_2$ used here follows from Π^1_1-determinacy, which is provable from standard large cardinal hypotheses.

2.5 Uncountable sets and an infinite linear order game

This section addresses a question by Will Brian and Steven Clontz regarding an infinite game of perfect information with two players, called *Baker's game*.

Games have many useful applications in fields like topology and set theory. Leaving many details out of this famous example, the Banach–Mazur game (see [Oxt57]) is played by two players who alternate turns picking subsets of ω^ω, forming a descending chain $W_0 \supseteq W_1 \supseteq \cdots$. A previously agreed upon set $S \subseteq \omega^\omega$ constitutes the winning condition: Player I wins if there is an element $x \in S$ such that $x \in W_n$ for all n; otherwise, Player II wins. A well-known result is that Player II has a winning strategy if and only if S is of the first category. In particular, Player II wins whenever S is countable.

The game I study in this section, Baker's game, is similar: two players take turns selecting real numbers; Player I tries to ensure that some element in the fixed set S remains in between all of these real numbers. In [BC23], the authors show that Player II has a winning strategy if and only if the set S is countable. It makes sense to play this game in more general linear orders, and so they ask if it is possible for Player II to have a winning strategy on an uncountable set S instead, or if this property somehow only holds in the reals, thus providing a characterization of the reals in these terms.

In collaboration with the other author of [MWS25], we showed the existence of a dense linear order of arbitrary infinite cardinality on which Player II has a winning strategy on **every** set S, including uncountable ones. This construction demonstrates that the outcome of Baker's game is highly sensitive to the order type of the underlying space and not merely a matter of cardinality. Concretely, the Brian–Clontz characterization does not extend to arbitrary dense linear orders.

2.5.1 The Cantor Game

In [Bak07], Matthew Baker introduced the following game on the real numbers. Fix a subset $S \subseteq \mathbb{R}$. Player I starts by picking a real number a_0. Then, Player II picks a real number b_0 such that $a_0 < b_0$. In the n-th turn, Player I picks a real number a_n such that $a_{n-1} < a_n < b_{n-1}$ and then Player II

picks a real number b_n such that $a_n < b_n < b_{n-1}$. After ω-many turns they form two sequences of real numbers $\{a_n\}$ and $\{b_n\}$ such that $a_0 < a_1 < \cdots < a_n < \cdots < b_n < \cdots < b_1 < b_0$. Player I wins if and only if there exists $x \in S$ such that $a_n < x < b_n$ for all $n < \omega$. Otherwise, Player II wins.

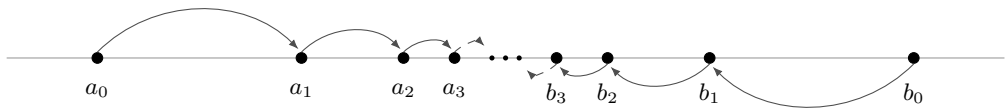


Figure 2.4: The nested construction of intervals in the Cantor Game.

It should be noted that the $[a_n, b_n]$ form a nested set of intervals, and thus there always exists real number x with the property that for all $n < \omega$, $a_n < x < b_n$. Proposition 2.32 generalizes the following observation when playing the game in an arbitrary dense linear order.

Proposition 2.29 ([Bak07])**.** *If S is countable, then Player II has a winning strategy in the Cantor Game.*

The so-called *Cantor Game* was introduced as a game-theoretic way of proving that $\mathbb{R}$ is uncountable, which follows by the previous observation. Baker also proved that if S contains a perfect set, then Player I has a winning strategy. Later in [LaD17], M.D. Ladue (a student of Baker) proved that Player I has a winning strategy if and only if the set S contains a perfect set. The question of whether the converse of Proposition 2.29 is true remained open until 2022. In [BC23], Brian and Clontz used elementary submodels to prove the converse statement of proposition 2.29 above.

Question 2.1 ([BC23])**.** Is there a linear order $(X, <)$ and an uncountable $S \subseteq X$ on which Player II has a winning strategy?

2.5.2 Scattered and well-ordered sets

Given an ordinal α, denote by α^* the converse of α, i.e., the conversely well-ordered set $(\alpha, >)$. Recall that a linearly ordered set is *scattered* if it does not contain a copy of the rationals. A linearly ordered set is said to be *σ-scattered* if it is a countable union of scattered sets. Given an order type γ, say that a linearly ordered set is *γ-free* if it does not contain a copy of γ. The following result can be found in [Tod85]. We include the proof for the reader's convenience.

Theorem 2.30 ([Tod84])**.** *Let X be a linearly ordered set. The following are equivalent.*

(i) X is a countable union of well-ordered sets.

(ii) X is σ-scattered and ω_1^*-free.

Proof. $(i) \Rightarrow (ii)$ is clear, so we focus on $(ii) \Rightarrow (i)$. It suffices to assume that X is scattered (and ω_1^*-free). Define the following relation on X: $x \sim y$ if and only if the closed interval $[x, y]$ is a countable union of well-ordered sets. Routine arguments show that $\sim$ is an equivalence relation. Also, every equivalence class $\mathcal{C}$ is convex, i.e., if $x, y \in \mathcal{C}$ and $x < z < y$, then $z \in \mathcal{C}$. Let $[x]$ be the equivalence class of x. Since every equivalence class is convex, the order $[x] < [y]$ if and only if $x < y$ is well-defined. Hence, the quotient order $X/\sim$ is isomorphic to a subspace of X. It follows that $X/\sim$ is scattered by assumption.

Claim 2.30.1. *$[x]_{\geq} := \{y \in [x] : y \geq x\}$ is a countable union of well-ordered sets for all x.*

Proof of Claim. Let $\{y_\alpha : \alpha < \kappa\}$ be a cofinal subset of $[x]_{\geq}$ with $y_0 = x$. So,

$$[x]_{\geq} = \bigcup_{\alpha<\kappa} [y_\alpha, y_{\alpha+1}].$$

Since $[y_\alpha, y_{\alpha+1}]$ is a countable union of well-ordered sets, then write $[y_\alpha, y_{\alpha+1}] = \bigcup_{n<\omega} A_n^\alpha$, where A_n^α is ω^*-free. Let $A_n := \bigcup_{\alpha<\kappa} A_n^\alpha$ and notice that A_n is ω^*-free since κ is ω^*-free. Then, $[x]_{\geq} = \bigcup_{n<\omega} A_n$, so $[x]_{\geq}$ is a countable union of well-ordered sets. ⊣

Claim 2.30.2. *$[x]_{\leq} := \{y \in [x] : y \leq x\}$ is a countable union of well-ordered sets for all x.*

Proof of Claim. Let $\{y_\alpha : \alpha < \kappa\}$ be a coinitial subset of $[x]_{\leq}$ with $y_0 = x$. Since X is ω_1^*-free, then κ is countable, so

$$[x]_{\leq} = \bigcup_{n<\omega} [y_{n+1}, y_n].$$

Since $[y_{n+1}, y_n]$ is a countable union of well-ordered sets for all $n < \omega$, then $[x]_{\leq}$ is a countable union of well-ordered sets. ⊣

The previous two claims imply that $[x]$ is a countable union of well-ordered sets for all $x \in X$. Now we show that actually $X = [x]$ for some x.

Claim 2.30.3. *$(X/\sim, <)$ is dense, i.e., if $[x] < [y]$ then there is $z \in Z$ such that $[x] < [z] < [y]$.*

Proof of Claim. Let $[x] < [y]$ and suppose there is no z such that $[x] < [z] < [y]$. Then, $[x, y] \subseteq [x]_{\geq} \cup [y]_{\leq}$ which implies that $[x, y]$ is a countable union of well-ordered sets by the previous two claims. Therefore, $[x] = [y]$. ⊣

This last claim shows that X must have a single equivalence class, since a nontrivial scattered dense order contains a copy of $\mathbb{Q}$. □

If S is a countable union of well-ordered subsets of an ω_1-free set X, then S must be countable. Also, it is clear that a subset S of $\mathbb{R}$ is countable if and only if S is a countable union of well-ordered sets. This idea, in combination with the preceding theorem, is generalized by the following.

Corollary 2.31. *If X is an ω_1-free and ω_1^*-free linearly ordered set, then the following are equivalent for any $S \subseteq X$.*

1. *S is countable.*
2. *S is a countable union of well-ordered sets.*
3. *S is σ-scattered.*

2.5.3 Baker's game on linear orders

We define a variation of Baker's game in an arbitrary dense linear order $(X, <)$. Given $S \subseteq X$, denote by $BG(X, S)$ the game where two players take turns selecting elements a_n, b_n of X as in the Cantor Game for the reals. Player I wins if after ω-many turns there is an $x \in S$ such that for all $n < \omega$, $a_n < x < b_n$, and Player II wins otherwise. The density condition ensures the game never reaches a stalemate. Thus, for the rest of the section, X is a dense linearly ordered set with a fixed subset S (the winning conditions). Observe that every countable set S satisfies the subsequent proposition.

Proposition 2.32. *If S is a countable union of well-ordered sets, then Player II has a winning strategy for $BG(X, S)$.*

Proof. Let $S = \bigcup_{n<\omega} S_n$ where each S_n is ω^*-free (that is, well-founded). In the n-th turn, Player II plays $b_n = \min(S_n \cap (a_n, b_{n-1}))$ if $S_n \cap (a_n, b_{n-1}) \neq \emptyset$ and any legal move otherwise (by density, there must always be one). Now, suppose that for all $n < \omega$, $a_n < x < b_n$. This implies that $x \notin S_n$ for all $n < \omega$, and hence, $x \notin S$. In other words, this is a winning strategy for Player II. □

In general, S being a countable union of well-ordered sets does not imply that S is countable (unless it contains no copies of ω_1). The previous proposition implies that if S is an (uncountable) well-ordered set, then Player II has a winning strategy for $BG(X, S)$. We now show that if S is conversely well-ordered, then Player II has a winning strategy for $BG(X, S)$. Notice that an uncountable conversely well-ordered set S has the property that S cannot be written as a countable union of well-ordered sets, so this result shows that, in general, the converse of Proposition 2.32 is false.

Theorem 2.33. *Suppose that γ is an ordinal and $\{S_\alpha : \alpha < \gamma\}$ is a collection of subsets of X on which Player II has a winning strategy. Furthermore, assume that $S_\beta < S_\alpha$ for all $\alpha < \beta < \gamma$ (i.e. $x < y$ for all $x \in S_\beta$ and $y \in S_\alpha$). Then, Player II has a winning strategy for $BG\left(X, \bigcup_{\alpha<\gamma} S_\alpha\right)$.*

Proof. Player I starts by playing $a_0 \in X$. If there is $b_0 > a_0$ such that $b_0 \leq y$ for all $y \in \bigcup_{\alpha<\gamma} S_\alpha$, then Player II wins by picking b_0. Otherwise, Player II picks any legal move in the first turn. In this case, $a_1 > a_0$, so let $\alpha_1 < \gamma$ be the smallest such that $y_1 < a_1$ for some $y_1 \in S_{\alpha_1}$. If there is a legal move $b_1 > a_1$ such that $b_1 \leq y$ for all $y \in \bigcup_{\alpha<\alpha_1} S_\alpha$, then Player II picks such b_1 and then uses the winning strategy for S_{α_1}. Otherwise, Player II plays any legal move b_1 and then, since $a_2 > a_1$, one can pick the smallest $\alpha_2 < \alpha_1$ such that $y_2 < a_2$ for some $y_2 \in S_{\alpha_2}$.

By repeating this process, since γ is well-ordered, one cannot have a decreasing sequence $\cdots < \alpha_2 < \alpha_1 < \gamma$. Hence, at some turn N, there is a legal move $b_N > a_N$ for Player II such that $b_N \leq y$ for all $y \in \bigcup_{\alpha<\alpha_N} S_\alpha$. After that, Player II continues playing with the winning strategy for $BG(X, S_{\alpha_N})$.

This describes a winning strategy for Player II. Indeed, given any $x \in X$ such that $a_n < x < b_n$ for all $n < \omega$, the following holds: Given that $y_N < a_N < x$ and the hypothesis that $S_\beta < S_\alpha$ whenever $\alpha < \beta$, $x \notin S_\alpha$ for all $\alpha > \alpha_N$; also $x \notin S_\alpha$ for all $\alpha < \alpha_N$ by the way b_N was played. Finally, $x \notin S_{\alpha_N}$, since Player II used a winning strategy for $BG(X, S_{\alpha_N})$. □

Corollary 2.34. *If S is conversely well-ordered, then Player II has a winning strategy on $BG(X, S)$.*

Corollary 2.35. *For any infinite cardinal κ, there is a dense linear order $(X, <)$ of size κ on which Player II has a winning strategy on every subset of X.*

Proof. Take $X = \kappa^* \times \mathbb{Q}$ ordered lexicographically. Concretely, given $x = (\alpha, p)$ and $y = (\beta, q)$ elements of X, $x \leq y$ if either $\beta < \alpha$ or $\alpha = \beta$ and $p <_{\mathbb{Q}} q$. Since the rationals are countable and κ is infinite, X has the desired size. The fact that X is dense follows from the fact that $\mathbb{Q}$ is dense and unbounded.

Now let $S \subseteq X$ and define, for each $\alpha < \kappa$, $S_\alpha = \{\alpha\} \times \mathbb{Q}$. As each S_α is countable, Player II has a winning strategy for $BG(X, S_\alpha)$, by virtue of Proposition 2.32. Since $S = \bigcup_{\alpha<\kappa} S_\alpha$ and the S_α are ordered the way that Theorem 2.33 requires, the result follows. $\square$

Chapter 3

Deep Computations

The contents of this chapter are based on my contributions to and perspectives on [DIMW+26], the second paper in a series started by the authors of [AnIW24] that connects model theory, structural Ramsey theory, and the topology of function spaces to the foundational study of "deep computations." While the initial work in [AnIW24] established the formal framework for ultracomputations, characterizing them as accumulation points of standard computations and proving the existence of deep equilibria, this chapter focuses on the complexity and classification of this asymptotic behaviour.

By utilizing the Todorčević trichotomy for Rosenthal compacta as a "Rosetta stone," we translate abstract topological properties into a hierarchy of computational complexity classes. We demonstrate how model-theoretic concepts, such as Shelah's classification theory and the Non-Independence Property (NIP), provide the necessary medium to distinguish between "tame" computations (characterized by polynomial approximability) and "wild" ones. This approach provides a unified theoretical framework for learning architectures, such as Neural ODEs and Deep Equilibrium Models, by linking their asymptotic behavior to classical results in functional analysis and statistical learning theory.

3.1 Introduction

In this chapter we study the asymptotic behavior of computations as parameters of the computation tend towards a limit, e.g., the depth of a neural network tending to infinity, or the time interval between layers of a time-series network tending toward zero. Recently, particular cases of this concept have attracted considerable attention in deep learning research (e.g., Neural Ordinary Differential Equations [CRBD], Physics-Informed Neural Networks [RPK19], and Deep Equilibrium Models [BKK]). The formal framework introduced here provides a unified setting to study these limit phenomena from a foundational viewpoint.

The main result of the paper is the identification of a hierarchy of four distinct complexity classes for limit computations. This is obtained by invoking a major result about topology of spaces of functions, namely, Todorčević's trichotomy for Rosenthal compacta [Tod99] and transferring it into the realm of computation. The medium that we use to translate between topology and computation is model theory, specifically, Shelah's classification theory [She90].

The translation between topology and computation proceeds as follows. To every computation

in a given computation model, we associate a continuous real-valued function, called the *type* of the computation, that describes the logical properties of this computation with respect to the rest of the model. This association allows us to view computations in any given computational model as elements of a space of real-valued functions, which is called the *space of types* of the model. The idea of embedding models of theories into their type spaces is borrowed from model theory, where type spaces play a central role. The embedding of computations into spaces of types then allows us to utilize the vast theory of topology of function spaces, known as $\mathrm{C_p}$-*theory*, to obtain results about complexity of topological limits of computations. As we shall indicate next, recent classification results for spaces of functions provide an elegant and powerful machinery to classify computations according to their levels of "tameness" or "wildness", with the former corresponding roughly to polynomial approximability and the latter to exponential approximability. The model-theoretic viewpoint of spaces of types thus becomes a "Rosetta stone" that allows us to interconnect various classification programs: in topology, the classification of Rosenthal compacta pioneered by Todorčević [Tod99]; in logic, the classification of theories developed by Shelah [She90]; and in statistical learning, the notions of PAC learning and VC dimension pioneered by Vapnik and Chervonenkis [VC74, VC71].

In a previous paper [AnIW24], the authors introduced the concept of limits of computations, which they called *ultracomputations* (given that they arise as ultrafilter limits of standard computations) and *deep computations* (following usage in machine learning [BKK]). There is a technical difference between both designations, but in this paper, to simplify the nomenclature, we will ignore the difference and use primarily the term "deep computation".

In [AnIW24], a new "tame vs wild" (i.e., polynomially approximable vs nonpolynomially approximable) dichotomy for the complexity of deep computations was proved by invoking a classical result of Grothendieck from the 1950s [Gro52]. Under our model-theoretic Rosetta stone, the property of polynomial approximability of computations is identified with continuous extendibility in the sense of topology, and with the notions of *stability* and *type definability* in the sense of model theory.

In this paper, we refine the dichotomy of [AnIW24], and obtain new complexity classes for deep computation. The "wild" part of the dichotomy from [AnIW24] is further subdivided between "tame" and "wild" classes; three new levels of tameness are identified within this new subdivision, and seven distinct minimal prototypes of tameness are detected.

The finer classification results of this paper are obtained through more contemporary topological and model-theoretic tools than in the previous paper [AnIW24]. From the topological side, instead of Grothendieck's classical criterion, we use the modern theory of *Rosenthal compacta*, and from the model theory side, instead of stability, we use *neostability*. The main tool imported from neostability is Shelah's *Independence Property*, or rather, its negation, which is commonly referred to as "NIP". More on this below.

Deep computations arise as topological limits of standard (continuous) computations. In topology, the *first Baire class*, or *Baire class 1* consists of functions (also called simply *Baire-1*) arising as pointwise limits of sequences of continuous functions. Intuitively, the first Baire class consists of functions with "controlled" discontinuities, and lies just one level of topological complexity above the Baire class 0, which (by definition) consists of continuous functions.

We apply a result of Bourgain, Fremlin and Talagrand from the late 1970s on compact subsets of the first Baire class to obtain a new "tame vs wild" dichotomy for deep computations [BFT78];

then, we apply a trichotomy of Todorčević, from the late 90s, for *Rosenthal compacta* to identify new tame classes [Tod99].

A Rosenthal compactum is a compact topological space that is homeomorphic to a subspace of the space of Baire class 1 functions on some Polish (i.e., separable, completely metrizable) space equipped with the topology of pointwise convergence. Rosenthal compacta exhibit "topological tameness," meaning that they behave in relatively controlled ways. Since the late 70's, they have played a crucial role in understanding the complexity of structures of functional analysis, especially Banach spaces. Todorčević's trichotomy for Rosenthal compacta has been utilized to settle longstanding problems in topological dynamics and topological entropy [GM22]. It is noteworthy that Todorčević's proof relies on sophisticated set-theoretic forcing and infinite Ramsey theory. At the time of writing this paper, decades after his original argument, there are no elementary proofs of the trichotomy; in fact, Todorčević's methods have been analyzed and exploited further [Tod23, HT19].

Through our Rosetta stone, Rosenthal compacta in topology correspond to the important concept of *No-Independence Property* (known as "NIP") in model theory, identified by Shelah [She71, She90], and to the concepts *VC-dimension* and *Probably Approximately Correct* learning (known as "PAC learnability") in statistical learning theory [VC71, VC74, Val84].

Beyond Todorčević's trichotomy, we invoke a more recent heptachotomy for separable Rosenthal compacta obtained by Argyros, Dodos and Kanellopoulos [ADK08]. These authors identify seven fundamental "prototypes" of separable Rosenthal compacta, and show that any non-metrizable separable Rosenthal compactum must contain a "canonical" embedding of one of these prototypes.

From the point of view of approximation theory, it is noteworthy that in the context of Grothendieck [Gro52] and Bourgain–Fremlin–Talagrand [BFT78], compactness and sequential compactness coincide; thus, the results of these papers allow us to ensure that deep computations, which in principle are abstract ultrafilter limits (over a possibly uncountable set), can always be approximated by concrete *sequences* of standard computations.

We believe that the results presented in this paper show practitioners of computation, or topology, or descriptive set theory, or model theory, how classification invariants used in their field translate into classification invariants of other fields. In the interest of accessibility, we do not assume that the reader has previous familiarity with advanced topology, model theory, or computing. The only technical prerequisites to read this paper are undergraduate-level topology and measure theory. The necessary topological background beyond undergraduate topology is covered later in this section.

In this section, we also present basic topological and combinatorial preliminaries. In section 3.2, we introduce the structural/model-theoretic viewpoint (no previous exposure to model theory is needed). Section 3.4 is devoted to the classification of deep computations.

Throughout the paper, our results pertain to classical models of computation (particularly computations involving real-valued quantities that are known and manipulated to a finite degree of precision). The final section, Section 3.5, introduces a probabilistic viewpoint, the development of which we intend to extend in future research.

In this section we recall notions and results from general topology and function space theory. For simplicity, we assume that all spaces are Hausdorff unless explicitly stated otherwise.

A subset of a topological space is F_σ if it is a countable union of closed sets, and G_δ if it is a countable intersection of open sets. In a metric space, every open set is F_σ; equivalently, every closed set is G_δ.

A *Polish space* is a separable and completely metrizable topological space, i.e., one admitting a complete metric that induces its topology. The class of Polish spaces is closed under countable topological products. Classical examples of Polish spaces (besides the real line) include the Cantor space $2^{\mathbb{N}}$ (the set of all infinite binary sequences, endowed with the product topology), the Baire space $\mathbb{N}^{\mathbb{N}}$ (the set of all infinite sequences of naturals, also with the product topology), and the space $\mathbb{R}^{\mathbb{N}}$ of sequences of real numbers.

Fact 3.1. *[Eng89, 4.3.24] A subset of a Polish space is itself Polish in the subspace topology if and only if it is a G_δ set. In particular, closed subsets and open subsets of Polish spaces are also Polish spaces.*

If X topological space and A is a subset of X, the A is said to be *nowhere dense* if its closure has an empty interior, and of the *first category* if A is a countable union of closed nowhere dense sets; otherwise, A is said to be of the *second category*. Sets $A \subseteq X$ of the first category are also called *meager*, and their complements $X \setminus A$ are called *comeager*. Thus, the set of meager subsets of X is the σ-ideal generated by the nowhere dense subsets of X. A topological space X is *Baire* if every comeager subset of X is dense. Equivalently, X is Baire if any countable intersection of open dense sets remains dense. The Baire Category Theorem states that compact Hausdorff spaces and all completely metrizable spaces (hence, all Polish spaces) are Baire.[1]

Given two topological spaces X and Y we denote by $\mathrm{C_p}(X, Y)$ the set of all continuous functions $f : X \to Y$ endowed with the topology of pointwise convergence, i.e., with the relative (subspace) product topology $\mathrm{C_p}(X, Y) \subseteq Y^X$—where Y carries its topology, but X serves merely as a (discrete) index set stripped of any topology. The space $\mathrm{C_p}(X, \mathbb{R})$ of continuous real-valued functions on X is denoted simply $\mathrm{C_p}(X)$.

A function $f \in Y^X$ between topological spaces X, Y is said to be of *Baire class 1*, or a *Baire-1* function, if there is a sequence $(f_n)_{n\in\mathbb{N}} \subseteq \mathrm{C_p}(X, Y)$ such that $f = \lim_{n\to\infty} f_n$ (i.e., $f(x) = \lim_{n\to\infty} f_n(x)$ for all $x \in X$). An elementary argument shows that Baire-1 functions are continuous everywhere except on a set of the first category. If X and Y are topological spaces, the space of Baire-1 functions $f : X \to Y$ endowed with the topology of pointwise convergence is denoted $\mathrm{B}_1(X, Y)$; thus, $\mathrm{C_p}(X, Y) \subseteq \mathrm{B}_1(X, Y) \subseteq Y^X$ are topological inclusions. We abbreviate $\mathrm{B}_1(X, \mathbb{R})$ as $\mathrm{B}_1(X)$. Relatively compact subsets of $\mathrm{C_p}(X)$ and of $\mathrm{B}_1(X)$ (particularly for X Polish) are of interest in analysis and topological dynamics.

A topological space X is *perfectly normal* if it is normal and every closed subset of X is a G_δ set (equivalently, every open subset of X is an F_σ set). Every metrizable space (hence, every Polish space) is perfectly normal.

The following fact was proved by Baire in his groundbreaking thesis. For a proof, see [Tod97, Section 10].

Fact 3.2 (Baire)**.** *If X is perfectly normal, then the following conditions are equivalent for a function $f\colon X \to \mathbb{R}$:*

(1) f is a Baire class 1 function, that is, f is a pointwise limit of continuous functions.

(2) $f^{-1}[U]$ is an F_σ subset of X whenever $U \subseteq \mathbb{R}$ is open.

If X is Baire, then (1) and (2) are equivalent to:

[1] More generally, any Čech-complete space is Baire [Eng89, Theorem 3.9.3].

(3) For every closed $F \subseteq X$, the restriction $f|_F$ has a point of continuity.

Moreover, if X is Polish and $f \notin \mathrm{B}_1(X)$, then there exist countable $D_0, D_1 \subseteq X$ and reals $a < b$ such that

$$D_0 \subseteq f^{-1}(-\infty, a], \quad D_1 \subseteq f^{-1}[b, \infty), \quad \overline{D_0} = \overline{D_1}.$$

3.1.1 From Rosenthal's dichotomy to the Bourgain–Fremlin–Talagrand dichotomy to Shelah's NIP

In metrizable spaces, the closure of a subset consists of limits of convergent sequences therein. In general, such a characterization of the closure fails in remarkable ways.

The n-th coordinate map $p_n \colon 2^{\mathbb{N}} \to \{0,1\}$ on the Cantor space $X = 2^{\mathbb{N}}$ $(= \{0,1\}^{\mathbb{N}})$ is continuous for each $n \in \mathbb{N}$; however, $(p_n)_{n\in\mathbb{N}}$ has *no* convergent subsequences in $\mathbb{R}^X$ ([Tod97, Chapter 1.1]). In a sense, this example exhibits the worst failure of sequential convergence possible: The closure of $\{p_n\}$ in $\{0,1\}^X$ (or in $\mathbb{R}^X$ for that matter) is homeomorphic to the Stone-Čech compactification $\beta\mathbb{N}$ of the (discrete) space of natural numbers. The space $\beta\mathbb{N}$ has cardinality 2^{2^ω} (largest among all separable topological spaces) and, effectively, is the set of ultrafilters on $\mathbb{N}$.[2]

The following theorem, proved by Haskell Rosenthal in 1974, is fundamental in functional analysis and captures a sharp division in the behavior of sequences in a Banach space.

Theorem 3.3 (Rosenthal's Dichotomy, [Ros74])**.** *If X is Polish and $(f_n)_{n\in\mathbb{N}} \subseteq \mathrm{C_p}(X)$ is pointwise bounded, then (f_n) has a convergent subsequence, or a subsequence whose closure in $\mathbb{R}^X$ is homeomorphic to $\beta\mathbb{N}$.*

The genesis of the dichotomy was *Rosenthal's ℓ_1-Theorem*, which states that a Banach space includes an isomorphic copy of ℓ_1 (the space of absolutely summable sequences), or else every bounded sequence therein is weakly Cauchy. The ℓ_1-Theorem connects diverse areas: Banach space geometry, Ramsey theory, set theory, and topology of function spaces.

As we move from $\mathrm{C_p}(X)$ to the larger space $\mathrm{B}_1(X)$, a dichotomy paralleling the ℓ_1-Theorem holds. This result is usually not phrased as a dichotomy in the above manner, but rather as an equivalence (see Theorem 3.4 below).

First, we introduce some useful notation. For any $f \in \mathbb{R}^X$ and any real a, define

$$X^f_{\leq a} \coloneqq f^{-1}(-\infty, a], \qquad\qquad X^f_{\geq a} \coloneqq f^{-1}[a, +\infty).$$

For any set $A \subseteq \mathbb{R}^X$ and any real a, define (by a slight abuse of notation)

$$X^A_{\leq a} \coloneqq \bigcap_{f\in A} X^f_{\leq a} = \{x \in X : f(x) \leq a \text{ for all } f \in A\},$$
$$X^A_{\geq a} \coloneqq \bigcap_{f\in A} X^f_{\geq a} = \{x \in X : f(x) \geq a \text{ for all } f \in A\}.$$

(In case $A = \emptyset$, we define $X^\emptyset_{\geq a} = X = X^\emptyset_{\leq a}$.) For any sequence $\{f_n\} \subseteq \mathbb{R}^X$ and $I \subseteq \mathbb{N}$, define $I^{\complement} \coloneqq \mathbb{N} \setminus I$ and $f_I \coloneqq \{f_i : i \in I\}$.

[2] Our earlier proof [AnIW24] of the existence of deep equilibria in compositional computation structures relies on a classical result on the existence of idempotent elements of $\beta\mathbb{N}$.

Theorem 3.4 ("The BFT Dichotomy". Bourgain–Fremlin–Talagrand [BFT78])**.** *Let X be a Polish space and $A \subseteq \mathrm{C_p}(X)$ be pointwise bounded. The following are equivalent:*

(i) A is relatively compact in $\mathrm{B}_1(X)$.

(ii) For every $(f_n)_{n\in\mathbb{N}} \subseteq A$ and every $a < b$ there is $I \subseteq \mathbb{N}$ such that

$$X^{f_I}_{\leq a} \cap X^{f_{I^\complement}}_{\geq b} = \emptyset.$$

(As stated above, the BFT Dichotomy is a particular case of the equivalence (ii)⇔(v) in [BFT78, Corollary 4G].)

The sets $X^{f_I}_{\leq a}$ and $X^{f_{I^\complement}}_{\geq b}$ appearing in condition Theorem 3.4*(ii)* are defined, respectively, in terms of $|I|$-many inequalities of the form $f_i(x) \leq a$, and $|I^\complement|$-many of the form $f_j(x) \geq b$. Thus, at least one of $X^{f_I}_{\leq a}$ and $X^{f_{I^\complement}}_{\geq b}$ is defined by the satisfaction of infinitely (countably) many inequalities. For our purposes, it is more natural to consider only finitely many inequalities at a time, which motivates the definitions below.

Definition 3.5. We say that a collection of functions $A \subseteq \mathbb{R}^X$ *has the finitary No-Independence Property (NIP)* if, for all sequences $(f_n)_{n\in\mathbb{N}} \subseteq A$ and reals $a < b$, there exist finite disjoint sets $E, F \subseteq \mathbb{N}$ such that $X^{f_E}_{\leq a} \cap X^{f_F}_{\geq b} = \emptyset$. We say that such E, F *witness finitary NIP for (f_n) and a, b.*

A set $A \subseteq \mathbb{R}^X$ *has the finitary Independence Property (IP)* if it does not have finitary NIP, i.e., if there exists a sequence $(f_n)_{n\in\mathbb{N}} \subseteq A$ and reals $a < b$ such that for every pair of finite disjoint sets $E, F \subseteq \mathbb{N}$, we have $X^{f_E}_{\leq a} \cap X^{f_F}_{\geq b} \neq \emptyset$.

If the word "finite" is omitted in the above definitions, we obtain the definitions of *countable* NIP (weaker than finitary NIP) and *countable* IP (stronger than finitary IP), respectively.

If we insist on witnesses $E, F \subseteq \mathbb{N}$ such that $F = E^\complement$, we call the respective properties "BFT-NIP" and "BFT-IP". Thus, Theorem 3.4 becomes (for pointwise bounded function collections $A \subseteq \mathrm{C_p}(X)$) that A is relatively compact in $\mathrm{B}_1(X)$ if and only if A has BFT-NIP.

Unless otherwise unspecified, IP/NIP shall mean *finitary* IP/NIP henceforth.

Proposition 3.6. *If X is compact and $A \subseteq \mathrm{C_p}(X)$, then A has BFT-NIP if and only if it has finitary NIP.*

(No pointwise boundedness is assumed of A.)

Proof. Trivially (as per the preceding discussion), finitary NIP implies BFT-NIP. Reciprocally, assume that X is compact and $A \subseteq \mathrm{C_p}(X)$ has finitary IP. Fix a sequence $(f_n) \subseteq A$ and reals $a < b$ such that $X^{f_E}_{\leq a} \cap X^{f_F}_{\geq b} \neq \emptyset$ for every finite disjoint $E, F \subseteq \mathbb{N}$. Fix $I \subseteq \mathbb{N}$. Since $(f_n) \subseteq A \subseteq \mathrm{C_p}(X)$ is a sequence of continuous functions, the collection $\mathcal{F}_I = \{X^{f_n}_{\leq a} : n \in I\} \cup \{X^{f_m}_{\geq a} : m \in I^\complement\}$ consists of closed (non-empty) sets. Since all finite $E \subseteq I$ and $F \subseteq I^\complement$ witness finitary IP for (f_n) and a, b, we see that the collection $\mathcal{F}_I$ has the finite intersection property. By compactness, $X^{f_I}_{\leq a} \cap X^{f_{I^\complement}}_{\geq b} = \bigcap \mathcal{F}_I \neq \emptyset$. □

Theorem 3.7. *Let X be a compact metrizable (hence Polish) space. For every pointwise bounded $A \subseteq \mathrm{C_p}(X)$, the following properties are all equivalent:*

(i) A is relatively compact in $\mathrm{B}_1(X)$;

(ii) A has BFT-NIP;

(iii) A has countable NIP;

(iv) A has finitary NIP.

(The equivalence *(ii)*⇔*(iv)* holds for arbitrary compact X. The equivalence *(ii)*⇔*(iii)*, for arbitrary X.)

Proof. Corollary of Theorem 3.4 and Proposition 3.6. □

Theorem 3.7 may be stated as the following dichotomy (under the assumptions) for any compact metrizable X: either A is relatively compact in $\mathrm{B}_1(X)$, or A has IP (in either sense).

The Independence Property was first isolated by Saharon Shelah in model theory as a dividing line between theories whose models are "tame" (corresponding to NIP) and theories whose models are "wild" (corresponding to IP). See [She71, Definition 4.1], [She90]. We will discuss this dividing line in more detail in the next section.

3.1.2 NIP as a universal dividing line between polynomial and exponential complexity

The particular case of the BFT dichotomy (Theorem 3.4) when A consists of $\{0,1\}$-valued (i.e., $\{\mathrm{Yes}, \mathrm{No}\}$-valued) strings was discovered independently, around 1971-1972 in many foundational contexts related to polynomial ("tame") vs exponential ("wild") complexity: In model theory, by Saharon Shelah [She71], [She90], in combinatorics, by Norbert Sauer [Sau72], and Shelah [She72], [She90], and in statistical learning, by Vladimir Vapnik and Alexey Chervonenkis [VC71], [VC74].

In model theory Shelah's classification theory is a foundational program in mathematical logic devised to categorize first-order theories based on the complexity and structure of their models. A theory T is considered classifiable in Shelah's sense if the number of non-isomorphic models of T of a given cardinality can be described by a bounded number of numerical invariants. In contrast, a theory T is unclassifiable if the number of models of T of a given cardinality is the maximum possible number. A key fact is that the number of models of T is directly impacted by the number of *types* over sets of parameters in models of T; a controlled number of types is a characteristic of a classifiable theory.

In Shelah's classification program [She90], theories *without* the independence property (called NIP theories, or *dependent* theories) have a well-behaved, "tame" structure; the number of types over a set of parameters of size κ of such a theory is of polynomially or similar "slow" growth on κ.

In contrast, theories with the Independence Property (called IP theories) are considered "intractable" or "wild". A theory with the Independence Property produces the maximum possible number of types over a set of parameters; for a set of parameters of cardinality κ, the theory will have 2^{2^κ}-many distinct types.

In combinatorics Sauer [Sau72] and Shelah [She72] proved the following independently: Let $\mathscr{F}$ be a collection of subsets of some set S. Either: for every $n \in \mathbb{N}$ there is a set $A \subseteq S$ with

$|A| = n$ such that $|\{S_i \cap A : i \in \mathbb{N}\}| = 2^n$ ($\mathscr{F}$ is a collection having "exponential complexity"); or: there exists $N \in \mathbb{N}$ such that for every $A \subseteq S$ with $|A| \geq N$, one has

$$|\{S_i \cap A : i \in \mathbb{N}\}| \leq \sum_{i=0}^{N-1} \binom{|A|}{i}.$$

($\mathscr{F}$ is a collection having "polynomial complexity"). Clearly, any collection $\mathscr{F}$ of subsets of a *finite* set S has polynomial complexity. The "polynomial" name is justified: indeed, for fixed $N > 0$, as a function of the size $|A| = m > 0$, we have

$$\sum_{i=0}^{N-1} \binom{m}{i} \leq \sum_{i=0}^{N-1} \frac{m^i}{i!} \leq \left(\sum_{i=0}^{N-1} \frac{1}{i!} \right) \cdot m^{N-1} < e \cdot m^{N-1} = O\left(m^N\right).$$

(More precisely, the order of magnitude is $O(m^{N-1})$: polynomial in m for N fixed.)

In machine learning Readers familiar with statistical learning may recognize the Sauer-Shelah lemma as the dichotomy discovered and proved slightly earlier (1971) by Vapnik and Chervonenkis [VC71, VC74] to address uniform convergence in statistics. The least integer N given by the preceding paragraph, when it exists, is called the *VC-dimension* of $\mathscr{F}$; it is a core concept in machine learning. If such an integer N does not exist, we say that the VC-dimension of $\mathscr{F}$ is infinite. The lemma provides upper bounds on the number of data points (sample size) needed to learn a concept class of known VC dimension d up to a given admissible error in the statistical sense. The Fundamental Theorem of Statistical Learning states that a hypothesis class is PAC-learnable (PAC stands for "Probably Approximately Correct") if and only if its VC dimension is finite.

3.1.3 Rosenthal compacta

The universal classification implied by Theorem 3.4, as attested by the examples outlined in the preceding section, led to the following definition (by Gilles Godefroy [God80]):

Definition 3.8. A Rosenthal compactum is any topological space realized as a compact subset of the space $\mathrm{B}_1(X) = \mathrm{B}_1(X, \mathbb{R})$ (equipped with the topology of pointwise convergence) of all real functions of the first Baire class on some Polish space X.

A Rosenthal compactum K is necessarily Hausdorff since it is a topological subspace of the Hausdorff product space $\mathbb{R}^X$.

Rosenthal compacta possess significant topological and dynamical tameness properties, and play an important role in functional analysis, measure theory, dynamical systems, descriptive set theory, and model theory. In this paper, we use them to study deep computations.

3.1.4 The special case $\mathrm{B}_1(X, \mathbb{R}^{\mathcal{P}})$ with $\mathcal{P}$ countable

Fix an arbitrary (at most) countable set $\mathcal{P}$ whose elements $P \in \mathcal{P}$ will be called *predicate symbols* or *formal predicates*. Our present goal is to characterize relatively compact subsets of $\mathrm{B}_1(X, \mathbb{R}^{\mathcal{P}})$, where X is always assumed to be a perfectly normal space (typically a Polish space).

The set $\mathcal{P}$ shall be considered discrete whenever regarded as a topological space. Since

$$\mathrm{C_p}(X, \mathbb{R}^{\mathcal{P}}) \subseteq \mathrm{B}_1(X, \mathbb{R}^{\mathcal{P}}) \subseteq (\mathbb{R}^{\mathcal{P}})^X,$$

the "ambient" space $(\mathbb{R}^{\mathcal{P}})^X$ is quite relevant. The product $X \times \mathcal{P}$ will be regarded as either a point set, or as a topological product depending on context. We have natural homeomorphic identifications

$$(\mathbb{R}^{\mathcal{P}})^X \cong \mathbb{R}^{X \times \mathcal{P}} \cong (\mathbb{R}^X)^{\mathcal{P}},$$

given by

$$\begin{aligned} \mathbb{R}^{X \times \mathcal{P}} &\to (\mathbb{R}^{\mathcal{P}})^X : \varphi \mapsto \hat{\varphi} \\ \mathbb{R}^{X \times \mathcal{P}} &\to (\mathbb{R}^X)^{\mathcal{P}} : \varphi \mapsto \hat{\varphi}, \end{aligned}$$

where

$$\hat{\varphi}(x) \coloneqq \varphi(x, \cdot) \in \mathbb{R}^{\mathcal{P}}, \qquad \hat{\varphi}(P) \coloneqq \varphi(\cdot, P) \in \mathbb{R}^X.$$

Such identifications view X and $\mathcal{P}$ as mere point sets (the topology of X in particular plays no role).

For $x \in X$, define the "left projection" map

$$\lambda_x \colon \mathbb{R}^{X \times \mathcal{P}} \to \mathbb{R}^{\mathcal{P}} : \varphi \mapsto \lambda_x(\varphi) \coloneqq \hat{\varphi}(x);$$

for $P \in \mathcal{P}$, the "right projection" map

$$\rho_P \colon \mathbb{R}^{X \times \mathcal{P}} \to \mathbb{R}^X : \varphi \mapsto \rho_P(\varphi) \coloneqq \hat{\varphi}(P).$$

For fixed $x \in X$ and $P \in \mathcal{P}$, we also have canonical projection maps

$$\lambda_x \colon \mathbb{R}^X \to \mathbb{R} : f \mapsto f(x), \qquad \rho_P \colon \mathbb{R}^{\mathcal{P}} \to \mathbb{R} : f \mapsto f(P).$$

When clear from context, rather than using the specific symbols ("λ" for left,"ρ" for right) to denote projections, we may use the generic symbol "π"; thus, π_x may mean λ_x, and π_P may mean ρ_P.

The Proposition below reduces the study of $\mathbb{R}^{\mathcal{P}}$-valued continuous or Baire-1 functions on X to the special case of real-valued ones.

Proposition 3.9. *The identification* $(\mathbb{R}^{\mathcal{P}})^X \cong \mathbb{R}^{X \times \mathcal{P}} \cong (\mathbb{R}^X)^{\mathcal{P}}$ *induces identifications*

$$\mathrm{C_p}(X, \mathbb{R}^{\mathcal{P}}) \cong \mathrm{C_p}(X \times \mathcal{P}) \cong \mathrm{C_p}(X)^{\mathcal{P}}, \qquad \mathrm{B}_1(X, \mathbb{R}^{\mathcal{P}}) \cong \mathrm{B}_1(X \times \mathcal{P}) \cong \mathrm{B}_1(X)^{\mathcal{P}}.$$

(The cardinality of $\mathcal{P}$ plays no role.)

Proof. The identification of $\mathrm{C_p}$-spaces follows trivially from the definition of topological product and the fact that $\mathcal{P}$ is discrete: a continuous map $X \to \mathbb{R}^{\mathcal{P}}$ is precisely a $\mathcal{P}$-indexed family of continuous functions $X \to \mathbb{R}$, and these correspond to continuous functions $X \times \mathcal{P} \to \mathbb{R}$. The identification of Baire-1 spaces follows immediately, since it is defined in terms of the purely topological notion of limit (in the ambient space) of sequences of continuous functions. $\square$

Given $A \subseteq Y^X$ and $K \subseteq X$ we write $A|_K := \{f|_K : f \in A\}$, i.e., the set of all restrictions of functions in A to K. The following Theorem is a slightly more general version of Theorem 3.4.

Theorem 3.10. *Assume that $\mathcal{P}$ is countable, X is a Polish space, and $A \subseteq \mathrm{C_p}(X, \mathbb{R}^{\mathcal{P}})$ is pointwise bounded in the sense that $\pi_P \circ A$ ($\subseteq \mathrm{C_p}(X)$) is pointwise bounded for every $P \in \mathcal{P}$. The following are equivalent for every compact $K \subseteq X$:*

(i) $A|_K$ is relatively compact in $\mathrm{B}_1(K, \mathbb{R}^{\mathcal{P}})$;

(ii) $\pi_P \circ A|_K$ has NIP for every $P \in \mathcal{P}$.

Proof. Compact subsets $K \subseteq X$ are closed, hence also Polish. Therefore, the asserted equivalence follows from Theorems 3.4 and 3.6. □

We conclude this section with an elementary but important observation.

Lemma 3.11. *Assume that X is Hausdorff and that $A \subseteq \mathrm{C_p}(X)$. The following are equivalent for every $L \subseteq X$:*

(i) A_L satisfies the NIP;

(ii) $A|_{\overline{L}}$ satisfies the NIP.

Proof. The implication (i)⇒(i) is trivial. To see the opposite implication, suppose that (ii) does not hold, and choose $\{f_n\}_{n\in\mathbb{N}} \subseteq A$ and $a < b$ such that for all finite disjoint $E, F \subseteq \mathbb{N}$,

$$\overline{L} \cap \bigcap_{n\in E} f_n^{-1}(-\infty, a] \cap \bigcap_{n\in F} f_n^{-1}[b, \infty) \neq \emptyset.$$

But then, for any a', b' with $a < a' < b' < b$ and any finite disjoint $E, F \subseteq \mathbb{N}$, we can choose

$$x \in \overline{L} \cap \bigcap_{n\in E} f_n^{-1}(-\infty, a') \cap \bigcap_{n\in F} f_n^{-1}(b', \infty),$$

which, by the definition of closure, gives

$$L \cap \bigcap_{n\in E} f_n^{-1}(-\infty, a'] \cap \bigcap_{n\in F} f_n^{-1}[b', \infty) \neq \emptyset,$$

contradicting (i). □

Historical remarks

The Baire hierarchy of functions was introduced by René-Louis Baire in his 1899 doctoral thesis, *Sur les fonctions de variables réelles*. The field of $\mathrm{C_p}$-theory was pioneered by A. V. Arhangel'skiĭ and his students in the 1970s and 1980s [Ark92], and it has found many applications in model theory and functional analysis. For a recent survey, see [Tka11].

3.2 Compositional Computation Structures

In this section, we connect function spaces with arbitrary-precision computations. We start by summarizing some basic concepts from [AnIW24].

3.2.1 Computation States Structures

A *computation states structure* is a pair $(L, \mathcal{P})$, where L is a set whose elements we call *states* and $\mathcal{P}$ is a collection of real-valued functions on L that we call *predicates*, also called *features*. For each $P \in \mathcal{P}$, we call the value $P(v)$ the P-th *(observable) feature* of v.

A *transition* of a computation states structure $(L, \mathcal{P})$ is a map $f\colon L \to L$.

The *type* of a state $v \in L$ is the indexed family[3]

$$\mathrm{tp}(v) = (P(v))_{P \in \mathcal{P}} \in \mathbb{R}^{\mathcal{P}}$$

of its features.

Intuitively, L is the set of states of a computation, and the predicates $P \in \mathcal{P}$ are primitives that are given and accepted as computable. (I.e., each symbol $P \in \mathcal{P}$ denotes a function $P(\cdot)$ taken as a computational primitive on L.)

We assume that each state $v \in L$ is uniquely characterized by its type $\mathrm{tp}(v)$, so we may identify L with a topological subspace of $\mathbb{R}^{\mathcal{P}}$ and topologized as such (i.e., L is identified with its forward image $\mathrm{tp}[L] \subseteq \mathbb{R}^{\mathcal{P}}$ under the type map). As a topological space, L is thus endowed with the topology of pointwise convergence induced by the product topology of $\mathbb{R}^{\mathcal{P}}$. In particular, for each $P \in \mathcal{P}$, the projection map $v \mapsto P(v)$ is continuous.

Important state spaces include $L = \mathbb{R}^{\mathbb{N}}$ (resp., $L = \mathbb{R}^n$ for some positive integer n), endowed with one predicate $P_i(v) = v_i$ (giving the i-th coordinate of v) for each $i \in \mathbb{N}$ (resp., for each $0 \leq i < n$). State subspaces $L \subseteq \mathbb{R}^{\mathbb{N}}$ and $L \subseteq \mathbb{R}^n$ are of primary interest in this paper. (Any states structure having a finite or countable family of predicates is effectively such a subspace.)

Definition 3.12. Given a computation states structure $(L, \mathcal{P})$, any type $\mathrm{tp}(v) \in \mathbb{R}^{\mathcal{P}}$ of a state $v \in L$ will be called a *realized* (state) type. The topological closure of the set of realized types in $\mathbb{R}^{\mathcal{P}}$ (endowed with the pointwise convergence topology) will be called the *space of (state) types* of $(L, \mathcal{P})$, denoted $\mathcal{L}$; elements $\mathbf{v} \in \mathcal{L}$ are called *state types*. Elements $\mathbf{v} \in \mathcal{L} \setminus L$ are *unrealized* state types.

Intuitively, state types capture a notion of "limit state". For each $P \in \mathcal{P}$, the P-th coordinate map (natural projection) $\mathbb{R}^{\mathcal{P}} \to \mathbb{R} : \mathbf{v} = (\mathbf{v}_Q)_{Q \in \mathcal{P}} \mapsto \mathbf{v}_P$, gives a continuous extension to $\mathcal{L}$ of the predicate $P(\cdot)$ on L; abusing notation, the extension will still be denoted $P(\cdot)$ (or just P).

Remark 3.13. Just as L is identified with a subset of $\mathcal{L}$, but going in the opposite direction, one may regard unrealized state types as "hyper-states" (when such type is obtained as a sequential limit—or ultralimit—of realized state types, unrealized state types may be called "deep states"). A comment for readers versed in model theory: at least for the purpose of realizing state types (with parameters in the original states space L), the type space $(\mathcal{L}, \mathcal{P})$ serves as a "poor man's" ultraproduct of $(L, \mathcal{P})$ wherein, by the very definition of $\mathcal{L}$, each type $\mathbf{v}$ is realized.

[3] As in §3.9 above, we regard each predicate on L as named by a specific formal label P (a *predicate symbol*) which is given the semantic meaning of a *bona fide* real function $P(\cdot)\colon v \mapsto P(v)$ (the predicate proper) in the computation states structure. Such a formal symbol P becomes *de facto* the name of an observable feature having the real value $P(v)$ at any state $v \in L$. Whenever $\mathcal{P}$ is used as an index set (e.g., as in the topological product $\mathbb{R}^{\mathcal{P}}$ below), it is regarded as a set of formal symbols. This syntactic vs. semantic abuse of nomenclature is commonplace in mathematics, but the model-theoretic framework requires contextual awareness of the distinction. To disambiguate when necessary, we shall write $P(\cdot)$, implying the semantic meaning of a real function on L, in contrast to P, meant as a purely syntactic symbol. Nevertheless, by a standard abuse of nomenclature in appropriate contexts, we may also treat P as a function on states (i.e., take P to mean $P(\cdot)$ proper).

Bounds on precision and magnitude. Shards

In order for computation state structures to serve their purpose as models of real-world computing, we must keep in mind that computations involving real-valued quantities can only ever carried out approximately, to a preset degree of precision (i.e., to within acceptable error bounds). More subtly, yet equally important in practice, one must also bound the magnitude of real quantities involved in all computations: one cannot possibly store—let alone compute with—arbitrarily large numbers! Thus, although the collection $\mathcal{P}$ of predicates may be infinite (possibly, uncountable even), "true" computations must imply choices, at the outset, of bounds for errors and for absolute magnitudes of quantities involved. Heuristically, different parts of a computation may require suitably matching (different) error bounds, as well as dealing with quantities of very different sizes. Focusing for now on the second of these issues, our results shall generally apply to sets of computations for which arbitrary but fixed bounds for each predicate are set in advance. This leads to the concept of *shard* first introduced in [AnIW24], as per the following definition.

Definition 3.14. Fix a computation states structure $\underline{L} = (L, \mathcal{P})$, with $L \subseteq \mathbb{R}^{\mathcal{P}}$ implicitly identified with its type space. A *sizer* is a family $r_\bullet = (r_P)_{P\in\mathcal{P}}$ of positive real numbers, indexed by $\mathcal{P}$. Given a sizer $r_\bullet$, let $\mathbb{R}[r_\bullet] \coloneqq \prod_{P\in\mathcal{P}}[-r_P, r_P]$ (a compact space), and let the $r_\bullet$-*shard* of the states space L be

$$L[r_\bullet] = L \cap \mathbb{R}[r_\bullet].$$

For a sizer $r_\bullet$, the $r_\bullet$-*type shard* is defined as $\mathcal{L}[r_\bullet] = \overline{L[r_\bullet]}$ (a closed, hence compact subset of $\mathbb{R}[r_\bullet]$).

Let also $\mathcal{L}_{\mathrm{sh}}$ (the space of *shard-supported types*) be the union of all type-shards as the sizer $r_\bullet$ varies.

Evidently, $\mathcal{L}_{\mathrm{sh}} \subseteq \mathcal{L}$, although the inclusion may be proper. However, the equality $\mathcal{L}_{\mathrm{sh}} = \mathcal{L}$ holds in the important special case when $\mathcal{P}$ is (at most) countable (see [AnIW24]).

3.2.2 Compositional Computation Structures

Definition 3.15. A *Compositional Computation Structure* (CCS) is a triple $(L, \mathcal{P}, \Gamma)$, where

- $(L, \mathcal{P})$ is a computation states structure, and
- $\Gamma \subseteq L^L$ is a semigroup under composition.

Elements of the semigroup Γ are called the *computations* of the structure $(L, \mathcal{P}, \Gamma)$. The state space L is implicitly identified with a subset of the type space $\mathcal{L} \subseteq \mathbb{R}^{\mathcal{P}}$ of $(L, \mathcal{P})$. We also assume that the identity map id on L is an element of Γ (which is thus not merely a semigroup but a monoid of transformations of L).

We topologize Γ as a subset of the topological product L^L, where (as usual) the "exponent" L serves merely as an index set, but the "base" $L \subseteq \mathbb{R}^{\mathcal{P}}$ has the relative product topology (of pointwise convergence). In this manner, the semigroup Γ is identified with a subset of the topological product $\mathcal{L}^L \subseteq (\mathbb{R}^{\mathcal{P}})^L$, which leads to an inclusion $\overline{\Gamma} \subseteq \mathcal{L}^L$. Elements $\xi \in \overline{\Gamma}$ are called (real-valued) *deep computations* or *ultracomputations*.

A collection R of sizers is *exhaustive* if $L = \bigcup_{r_\bullet \in R} L[r_\bullet]$ (shards $L[r_\bullet]$ exhaust L). A transformation $\gamma \in \Gamma$ is *R-confined* if γ restricts to a map $\gamma|_{L[r_\bullet]} \colon L[r_\bullet] \to L[r_\bullet]$ (into $L[r_\bullet]$ itself) for every $r_\bullet \in R$. A subset $\Delta \subseteq \Gamma$ is *R-confined* if each $\gamma \in \Delta$ is.

Unlike its subspace L^L (a topological semigroup), the space $\mathcal{L}^L$ does not have a natural semigroup structure, so ultracomputations are not generally composable. Intuitively, ultracomputations are "final" or "ultimate".

Proposition 3.16. *If $\Delta \subseteq \Gamma$ is confined by an exhaustive sizer collection, then $\overline{\Delta}$ is a compact subset of $\mathcal{L}^L_{\mathrm{sh}}$.*

Proof. Assume that R confines Δ. For each $v \in L$, let $r_\bullet^{(v)} \in R$ be a sizer such that $v \in L[r_\bullet^{(v)}]$. An arbitrary $\gamma \in \Delta$ restricts to a map $\gamma{\restriction}L[r_\bullet^{(v)}] \colon L[r_\bullet^{(v)}] \to L[r_\bullet^{(v)}]$, so $\Gamma \subseteq K \coloneqq \prod_{v \in L} \mathcal{L}[r_\bullet^{(v)}]$. The space K is a product of compact spaces, hence compact, so $\overline{\Gamma}$ is a closed, hence compact subset thereof, and a subset of $\mathcal{L}^L_{\mathrm{sh}} \supseteq K$ *a fortiori*. □

For a CCS $(L, \mathcal{P}, \Gamma)$, we regard the elements of Γ as "standard" finitary computations, and the elements of $\overline{\Gamma}$, i.e., deep computations, as possibly infinitary limits of standard computations. The main goal of this paper is to study the computability, definability and computational complexity of deep computations. Since deep computations are defined through a combination of topological concepts (namely, topological closure) and structural and model-theoretic concepts (namely, models and types), we will import technology from both topology and model theory.

Remark 3.17. (For readers versed in model theory.)[4]

1. compositional computation structures as defined above are not real-valued structures in a literal sense, because there is no intrinsic structural manner to ensure that Γ is realized as a literal subset of the product space L^L. However, in a semantically transparent manner, any CSS as above may alternatively be seen (per its original definition in [AnIW24]) as a real-valued structure $(L, \mathcal{P}, \Gamma, \circ, \mathrm{ev})$, where

   - $(L, \mathcal{P})$ is a computation states structure,
   - $(\Gamma, \circ)$ is a semigroup,
   - $\mathrm{ev} \colon \Gamma \times L \to L : (\gamma, v) \mapsto \mathrm{ev}(\gamma, v)$ is a semigroup action of Γ on L (i.e., $\mathrm{ev}(\gamma \circ \delta, v) = \mathrm{ev}(\gamma, \mathrm{ev}(\delta, v))$ for $\gamma, \delta \in \Gamma$ and $v \in L$).

   Both definitions are reconciled upon identifying $\gamma \in \Gamma$ with the map $(\gamma, v) \mapsto \mathrm{ev}(\gamma, v)$.

2. In the preceding sense of real-valued structures, the class of compositional computation structures (for a fixed predicate symbol collection $\mathcal{P}$) is first-order elementary. The limited notion of state type we have introduced (essentially, quantifier-free state types with parameters of sort L only) may be enriched to state types involving more formulas (e.g., with parameters in Γ, or including quantified formulas), and extended to "transform-types" (i.e., types of sort Γ). In particular, given a structure $\underline{C} = (L, \mathcal{P}, \Gamma, \circ, \mathrm{ev})$, all types are realized in some ultrapower $\underline{C}^{(\mathcal{U})}$ of $\underline{C}$ (constructed in the usual manner detailed in the references above). [5] This approach to spaces of types was introduced in Banach space theory by Krivine [Kri74, KM81, Kri72].

[4] Refer to [Kei23] where a general framework for real-valued structures is introduced; see also [DnI17, section 6] for a basic tutorial.

[5] By contrast (cf., Remark 3.13), the ultrapower construction is not necessary to realize state types in a computation states structures.

3.2.3 Computability of deep computations and the Extendibility Axiom

Let $(L, \mathcal{P})$ be a computation structure with *countable* predicate collection $\mathcal{P}$.

1. A *computable predicate* is any real function $\varphi\colon L \to \mathbb{R}$ whose restriction to an arbitrary shard is uniformly approximable by polynomials (with real coefficients) in the features $P(\cdot)$.

2. A transform $f\colon L \to \mathcal{L}$ is *computable* if, for each $Q \in \mathcal{P}$, the output feature $Q \circ f\colon L \to \mathbb{R}$ is a computable predicate.

It is shown in [AnIW24] that computable transforms $f\colon L \to \mathcal{L}$ are precisely those that extend to continuous maps $\tilde{f}\colon \mathcal{L} \to \mathcal{L}$.

To summarize, for a function $f\colon L \to \mathcal{L}$, the following conditions are equivalent:

- f is computable
- f has uniformly polynomially approximable features on shards
- f extends to a continuous map $\mathcal{L} \to \mathcal{L}$.

These equivalences motivate the following definition:

A computation $\gamma \in \Gamma$ is *extendable* if it has some extension $\tilde{\gamma}\colon \mathcal{L}_{\mathrm{sh}} \to \mathcal{L}_{\mathrm{sh}}$ such that, for every sizer $r_\bullet$, there is a sizer $s_\bullet$ such that $\tilde{\gamma}|_{\mathcal{L}[r_\bullet]}\colon \mathcal{L}[r_\bullet] \to \mathcal{L}[s_\bullet]$ is continuous.

Because $\tilde{\gamma}$ extends γ continuously on each compact type-shard, we call it the *K-extension* of γ; such extension is evidently unique (if it exists).

Abusing notation, for any $\Delta \subseteq \Gamma$ consisting of extendable computations, the set $\{\tilde{\gamma}\colon \gamma \in \Delta\}$ will be denoted $\tilde{\Delta}$.

Extendibility Axiom. *Every computation $\gamma \in \Gamma$ is extendable.*

For a more detailed discussion of the Extendibility Axiom, we refer the reader to [AnIW24].

For the rest of the paper, the Extendibility Axiom is imposed on all CSSs.

3.3 Examples of Compositional Computation Structures

We begin by formalizing the Newton–Raphson root-finding method as a CCS [Gal00], [BFB16, Section 2.3]. Let $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ be the extended complex plane (the Alexandroff —one-point— compactification of $\mathbb{C}$). We regard the unit sphere $S^2 \subseteq \mathbb{R}^3$ as the Riemann sphere and endowed with the stereographic projection map $S^2 \to \hat{\mathbb{C}}$ —a topological homeomorphism whose *inverse* $\hat{\mathbb{C}} \to S^2$ is given by $z \mapsto (P_1(z), P_2(z), P_3(z))$, where $(P_1(\infty), P_2(\infty), P_3(\infty)) = (0, 0, 1)$ and, for $z \neq \infty$,

$$P_1(z) = \frac{2\,\Re z}{|z|^2 + 1}, \qquad P_2(z) = \frac{2\,\Im z}{|z|^2 + 1}, \qquad P_3(z) = \frac{|z|^2 - 1}{|z|^2 + 1}.$$

The extended complex plane $\hat{\mathbb{C}}$ is thus endowed with the real predicates P_1, P_2, P_3 with values in $[-1, 1]$; its topology is metrized by the pullback of the usual Euclidean metric on S^2.

Newton CCSs: steps and fractals

Let $p = p(z)$ be a non-constant rational function with complex coefficients —i.e., an analytic (meromorphic) map $\hat{\mathbb{C}} \to \hat{\mathbb{C}}$. The *Newton method step* (or *Newton map*) for p is the rational map

$$N_p \colon z \mapsto z - \frac{p(z)}{p'(z)}. \tag{3.1}$$

Successive iterations of the Newton-Raphson method (i.e., compositional powers $N_p^{(n)}$) may be regarded as computations of a CCS whose states space is $L = \hat{\mathbb{C}}$. The natural choice of predicates is $\mathcal{P} = (P_1, P_2, P_3)$, the inverse stereographic projection coordinates. (The real- and imaginary part maps $z \mapsto \Re z$, $z \mapsto \Im z$ are not defined at $z = \infty$.)

Newton CCSs Let $\Gamma_p := \{N_p^{(n)} : n \in \mathbb{N}\}$ be the semigroup of iterates of the Newton map (3.1). The CCS $(\hat{\mathbb{C}}, (P_1, P_2, P_3), \Gamma_p)$ is *Newton's method for* $p(z)$ (or the *Newton CCS for* $p(z)$). Note that the states space $\hat{\mathbb{C}}$ is compact and the Newton map N_p is continuous thereon, so the Extendibility Axiom is satified by Newton CCSs.

Given an initial estimate z_0, one hopes for the sequence of Newton steps applied to z_0 (namely, $N_p^{(\bullet)}(z_0) := (z_n)_{n\in\mathbb{N}}$, where $z_n := N_p^{(n)}(z_0)$) to converge to a root r of p.

Fatou and Julia sets Recall that the *Fatou set* $\mathcal{F}_p$ consists of all points that have a compact neighborhood on which some subsequence of the sequence $p^{(\bullet)}$ of iterates of p converges uniformly; the *Julia set* $\mathcal{J}_p := \hat{\mathbb{C}} \setminus \mathcal{F}_p$ is its complement: it is (necessarily closed and) typically a fractal. The Julia set of the Newton map N_p is called the *Newton fractal* of p.

3.3.1 Example: Newton's CCS for the cubic $\rho(z) = z^3 - 2z + 2$

The polynomial $\rho(z) = z^3 - 2z + 2$ has Newton map

$$N_\rho : z \mapsto z - \frac{z^3 - 2z + 2}{3z^2 - 2} = \frac{2z^3 - 2}{3z^2 - 2}.$$

Denote the real root of ρ by r_0; the roots in the upper- and the lower half plane, by r_+ and r_-, respectively.

Successive Newton steps obtained from the initial guess $z_0 = 0$ yield the divergent sequence $N_\rho^{(\bullet)}(0) = (0, 1, 0, 1, \dots)$ having sequential limits $0, 1$ —neither a root of ρ.

Visualizing Newton iterates

Assign to each complex point $z \in \mathbb{C}$ an "RGB color" $(R, G, B) \in [0, 1]^3$ whose entries give the corresponding (Red/Green/Blue) color intensity; thus, $(1, 0, 0)$ is red, $(0, 1, 0)$ is green, and $(0, 0, 1)$ is blue—while, for example, $(0.5, 0, 0.5)$ is a light purple. Each of the three distinct roots of ρ will be given one of these colors (the real root r_0, red; the roots r_{+1} on the upper- and r_{-1} on the lower half plane, respectively green and blue). The RGB color of all other $z \in \mathbb{C}$ is a convex combination whose relative weights are $1/d_i$ where $d_i = |z - r_i|$ for $i = 0, \pm 1$; this will be called the "standard"

color of z: the closer z is to root r_i, the closer the colors of these points are.[6]

For any $n \in \mathbb{N}$, define a coloring of $\mathbb{C}$ in the natural way: color each $z_0 \in \mathbb{C}$ using the standard color of $z_n := N_\rho^{(n)}(z_0)$. At any stage, particularly early ones, the coloring looks "fuzzy" (out of focus) because the coloring function is continuous; however, as n increases, the coloring becomes "sharper" (the coloring function has large regions of near-constancy, in the transition areas between which color varies rapidly —i.e., is "nearly discontinuous"). For one thing, for many z_0 in the Fatou set, the (entire) sequence (z_n) converges to one of the three roots: the RGB colors of (z_n) tend towards (a single one of) red, green or blue correspondingly. The Fatou set also has an open region consisting of points z_0 for which $(z_n)_{n\in\mathbb{N}}$ has a subsequence converging to 0 or 1 (uniformly on some compact neighborhood of z_0). The four open regions are immediately recognizable in Figure 3.1.

Let B_0, B_+, B_- be the respective basins of the roots r_0, r_+, r_-, and let B_{01} be the basin of the attractive cycle 0, 1, 0, 1, ... It may be shown that the Fatou set of N_ρ is $F_\rho = B_{01} \cup B_0 \cup B_+ \cup B_-$, each of these four sets being nonempty open. Since B_{01} is nonempty, $N_\rho^{(\bullet)}$ does not converge pointwise; however, its subsequences of even/odd iterates are pointwise convergent on B_{01} to two distinct deep computations f_0 (constant 0 on B_{01}) and f_1 (constant 1 on B_{01}). [7] The behavior of iterates $N_\rho^{(n)}$ on the Newton fractal of ρ (the Julia set of ρ) is chaotic.

3.3.2 Example: Newton's CCS for 3-cyclotomic $\kappa(z) = z^3 - 1$

Consider next the 3-cyclotomic polynomial $\kappa(z) = z^3 - 1$. The roots of the 3-cyclotomic polynomial $\kappa(z) = z^3 - 1$ are the cube roots of unity $r_0 = 1$ and $r_\pm = \exp(\pm 2\pi i/3)$; the Newton map is

$$N_\kappa \colon z \mapsto z - \frac{z^3 - 1}{3z^2} = \frac{2z^3 + 1}{3z^2}.$$

In Figure 3.2(C), one sees a Fatou set consisting of solid-color (disconnected) open sets, one in each of the colors red (for $r_0 = 1$), green and blue (for $r_{\pm 1}$, respectively), where the sequence of all iterates $z_n := N_\kappa^{(n)}(z_0)$ starting with a arbitrary initial point z_0 converges to the corresponding root. (Thus, each solid-color region is the basin of attraction of the respective root).

3.3.3 Example: Square roots

When $p \colon \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ has real coefficients, it restricts to a map $\mathbb{RP}^1 \to \mathbb{RP}^1$ on the real-projective line $\mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$. In this case, the Newton CSS for p may be regarded as having states space $\mathbb{RP}^1$. We shall explain the details in the simplest case of a polynomial $p(x) = x^2 - a$ (real $a > 0$) to illustrate iterative square-root approximations.

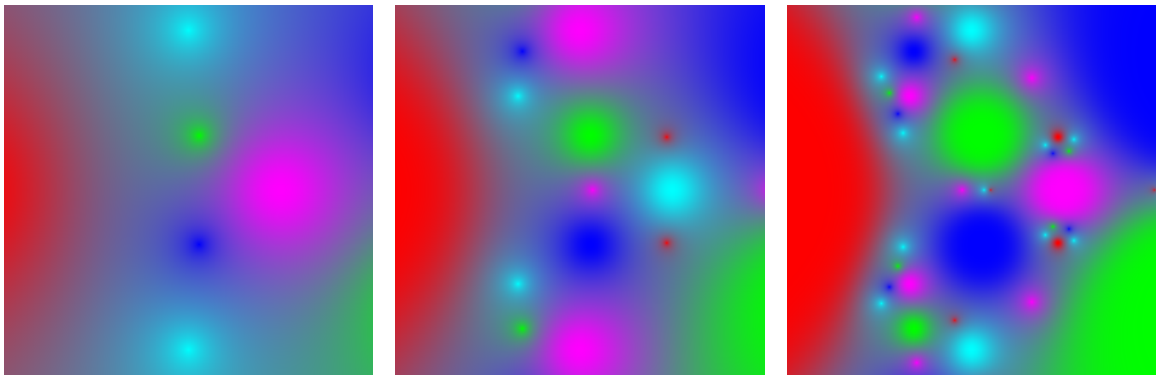
Let $\mathbb{R} \to S^1 : x \mapsto (P_1(x), P_2(x))$ be the (inverse) stereographic projection of the real line onto $S^1 \subseteq \mathbb{R}^2$, i.e.,

$$P_1(x) = \frac{2x}{x^2 + 1}, \qquad P_2(x) = \frac{x^2 - 1}{x^2 + 1},$$

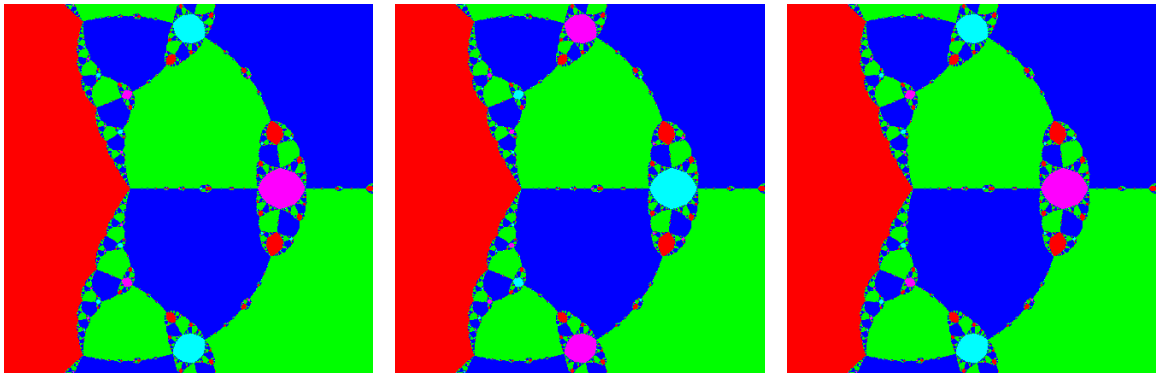
extended to a projective transformation of the real projective line $\mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$ by $\infty \mapsto$

[6] While ∞ is a fixed point of N_ρ, it is a repeller: if $|z|$ is large, then $N_\rho(z) = \frac{2}{3}z + O(1)$ has roughly two-thirds its magnitude. Therefore, ∞ is not the limit of any subsequence of $N_\rho^{(\bullet)}(z_0)$ with z_0 in the Fatou set of N_ρ; in particular, ∞ is a point of the Newton fractal of $\rho(z)$.

[7] As remarked above, ∞ is a repelling fixed point of N_ρ, hence $f_0(\infty) = \infty = f_1(\infty)$.

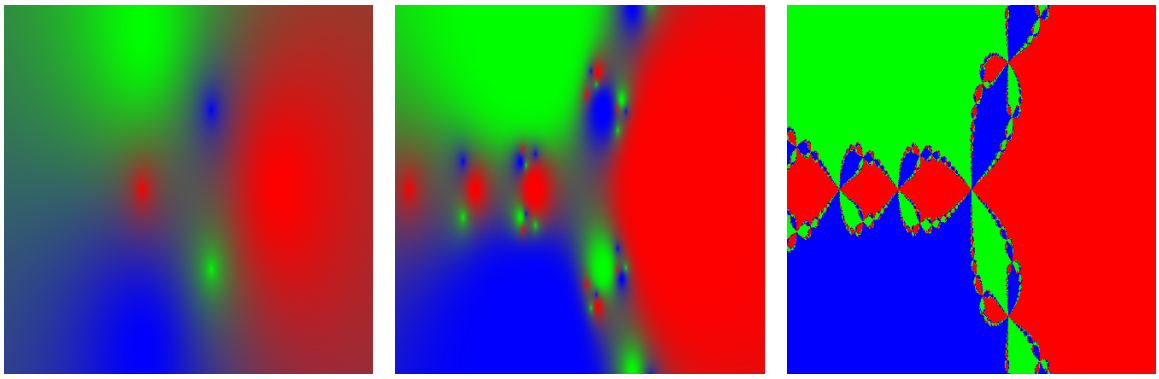


(a) $n = 1$ (b) $n = 2$ (c) $n = 3$



(d) $n = 99$ (odd) (e) $n = 100$ (even) (f) $n = 101$ (odd)

Figure 3.1: Iterates of the Newton map N_ρ of $\rho(z) = z^3 - 2z + 2$. Points in the basin of each root $r_0, r_{\pm 1}$ (respectively, red, green and blue) converge to the root. The basin of attraction of the 2-cycle $\{0, 1\}$ is the union of magenta and cyan regions; these two colors alternate like semaphore lights: points z in the cyan region are those whose subsequence $N_\rho^{(2\cdot\bullet)}(z)$ of even Newton iterates (resp., $N_\rho^{(2\cdot\bullet+1)}(z)$ of odd iterates) has limit 1 (resp., limit 0). For points z in the magenta region, the limits of the even and odd iterate subsequences are exchanged. The boundaries of colored regions constitute the Newton fractal of ρ.



(a) After 1 iteration (b) After 3 iterations (c) After 100 iterations

Figure 3.2: Visualization of the Newton iterates of $\kappa(z) = z^3 - 1$.

$(P_1(\infty), P_2(\infty)) = (0,1)$; this gives a compact CCS $\mathfrak{L} = (\mathbb{RP}^1, (P_1, P_2))$. As above, consider the quadratic polynomial $p(x) = x^2 - a$ for some fixed real number $a > 0$. The Newton map N_p is

$$x \mapsto x - \frac{x^2 - a}{2x} = \begin{cases} \frac{x^2+a}{2x}, & x \neq 0, \infty; \\ \infty, & x = 0, \infty. \end{cases}$$

These Newton steps converge to one of the square roots $\pm\sqrt{a}$ starting from any initial point $x_0 \neq 0, \infty$. More explicitly, the unique deep equilibrium $\mathfrak{f}\colon \mathbb{RP}^1 \to \mathbb{RP}^1$ of (the sequence of iterates of) N_p is given by

$$\mathfrak{f}(x) = \lim_{n\to\infty} N_p^n(x) = \begin{cases} \sqrt{a}, & x \in (0, \infty); \\ -\sqrt{a}, & x \in (\infty, 0); \\ \infty. & x = 0, \infty. \end{cases}$$

Thus, $\mathfrak{f}$ is discontinuous at $x = 0, \infty$.

3.3.4 Example: Threshold classifiers on Cantor's set

Consider the states space $L = 2^{\mathbb{N}}$ $(= \{0,1\}^{\mathbb{N}})$ consisting of all bit sequences $x = (x_n)_{n\in\mathbb{N}} \subseteq \{0,1\}$. The product space $2^{\mathbb{N}}$ is compact[8] and homeomorphic to the Cantor (ternary) subset of the interval $[0,1]$. Consider the predicate collection $\mathcal{P} = (\bullet_n)_{n\in\mathbb{N}}$ of projections (coordinates)

$$\bullet_n \colon 2^{\mathbb{N}} \to \{0,1\} : x \mapsto x_n.$$

Since $2^{\mathbb{N}} \subseteq \mathbb{R}^{\mathbb{N}}$ $(\cong \mathbb{R}^{\mathcal{P}})$ is closed, the type space of $(2^{\mathbb{N}}, (\bullet_n)_{n\in\mathbb{N}})$ is effectively still $2^{\mathbb{N}}$. The constant 0 and 1 elements of $2^{\mathbb{N}}$ will be denoted $0_\bullet$ and $1_\bullet$, respectively. In this way, we obtain the *Cantor computation states structure (CSS)*

$$\mathcal{C} = (2^{\mathbb{N}}, (\bullet_i)_{i\in\mathbb{N}}).$$

The concatenation action of words

For fixed $n \in \mathbb{N}$, we call elements of $2^n = \{0,1\}^n$ *finite bit sequences* (or *binary words*) *of length* n; the finite set 2^n is topologized discretely.

The set of all finite-length words is the disjoint union $2^{<\mathbb{N}} := \bigsqcup_{n\in\mathbb{N}} 2^n$ (a discrete space); the (unambiguously defined) length n of any word $w \in 2^{<\mathbb{N}}$ is denoted $|w|$. For $m \geq n$, we also have natural continuous restriction maps (projections) $2^{\mathbb{N}} \to 2^m \to 2^n$. The restriction of x — either a sequence or a word of length $m \geq n$— to a length-n word is denoted $x{\restriction}_n$ (i.e., $\cdot{\restriction}_n$ projects onto 2^n). Since the index set $\mathbb{N}$ and the value space $2 = \{0,1\}$ are well-ordered, the spaces $2^{\mathbb{N}}$ as well as $2^{<\mathbb{N}}$ and 2^n for $n \in \mathbb{N}$ carry a natural lexicographical order henceforth denoted $\preceq$; the order topology coincides with the (product) topology of $2^{\mathbb{N}}$. All restriction maps $\cdot{\restriction}_n$ are continuous and order-preserving.

Given lengths $n, m \in \mathbb{N}$, we have a concatenation operation $2^m \times 2^n \to 2^{m+n} : (v, w) \mapsto \widehat{vw}$; such concatenations altogether yield a semigroup structure on $2^{<\mathbb{N}}$ —more precisely, a monoid with the empty word $() \in 2^0$ as identity. The more general concatenation operation $2^{<\mathbb{N}} \times 2^{\mathbb{N}} \to 2^{\mathbb{N}} : (w, x) \mapsto \widehat{wx}$ is a continuous and order-preserving action of $2^{<\mathbb{N}}$ on $2^{\mathbb{N}}$; we shall call it the *prefixing*

[8] The topology on the space $2 = \{0,1\}$ is discrete.

action. Thus, *prefixing by a given word* $w \in 2^{<\mathbb{N}}$ means applying the operation $\widehat{w\cdot} : x \mapsto \widehat{wx}$. In this manner, we obtain the *Cantor prefixes CCS*

$$\widehat{\mathcal{C}} = \langle 2^{\mathbb{N}}, (\bullet_i)_{i\in\mathbb{N}}, 2^{<\mathbb{N}}, \widehat{\cdot\cdot} \rangle$$

where, by an abuse of nomenclature, the operation $\widehat{\cdot\cdot}$ denotes both the concatenation $(v, w) \mapsto \widehat{vw}$ on $2^{<\mathbb{N}}$ and the prefixing $(w, x) \mapsto \widehat{wx}$ of $2^{<\mathbb{N}}$ on $2^{\mathbb{N}}$.

Shift action

An altogether different action on $2^{\mathbb{N}}$ is given by shift operations ("Bernoulli shifts"); namely, an arbitrary $m \in \mathbb{N}$ acts on $2^{\mathbb{N}}$ by the map

$$\overleftarrow{m} : x = (x_i)_{i\in\mathbb{N}} \mapsto \overleftarrow{m}x := (x_{m+i})_{i\in\mathbb{N}}.$$

Here, $\overleftarrow{0} : 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ is the identity $x \mapsto x$, and the equality $(\overleftarrow{m+n})x = \overleftarrow{m}(\overleftarrow{n}\,x)$ for $m, n \in \mathbb{N}$ holds identically (i.e., the additive semigroup (monoid) $(\mathbb{N}, 0, +)$ is represented as the semigroup of shifts of $2^{\mathbb{N}}$). Let $\overleftarrow{\mathbb{N}} = \{\overleftarrow{m} : m \in \mathbb{N}\}$ be the semigroup of all shifts. This gives the *Cantor shifts* (or *Cantor-Bernoulli*) CCS

$$\mathcal{B} = \langle 2^{\mathbb{N}}, (\bullet_n)_{n\in\mathbb{N}}, \mathbb{N}, (\overleftarrow{n})_{n\in\mathbb{N}} \rangle.$$

Finite-precision classifiers and definability

Notation 3.18. When useful, we will regard $0, 1$ as logical constants (0 for falsehood, 1 for truth). In view of such identification, we shall call a $\{0, 1\}$-valued function on a set X (resp., in a finite power X^n) an *(n-ary) relation on* X. The propositional connectives of negation "$\neg$", disjunction "$\vee$" and conjunction "$\wedge$" are realized using (linear lattice) operations on $\mathbb{R}$; explicitly, as $r \mapsto 1-r$ (negation), binary maximum (max) and binary minimum (min), respectively.

Let $[\![\cdot]\!]$ be Iverson's bracket; [9] thus, for any statement $\mathtt{S}$, the value of $[\![\mathtt{S}]\!]$ is 1 if $\mathtt{S}$ is true, 0 if false.

For any fixed word $w \in 2^n$, let $\preccurlyeq_n w$ be the relation $[\![\cdot\restriction_n \preceq w]\!] : x \mapsto [\![x\restriction_n \preceq w]\!]$ on $2^{\mathbb{N}}$. We shall write $x \preccurlyeq_n w$ with the meaning of $x\restriction_n w$ or $[\![x\restriction_n w]\!]$, depending on context (thus, "$\preccurlyeq_n w$" names a unary relation taking its argument x on the left). The subscript n is used only for emphasis and optional, since its value is determined uniquely by the condition $w \in 2^n$. Since both $[\![\cdot \preceq w]\!]$ and $\cdot\restriction_n$ are continuous, so is $\preccurlyeq_n w$, which will be called a *precision-n threshold classifier*; any such relation $\preccurlyeq_n w$ is a *finite-precision threshold classifier*. For $n = 0$, there is only one "zeroth-precision" classifier $\preccurlyeq_0()$: the identically true relation $\mathbf{1}$ on $2^{\mathbb{N}}$. Although insubstantial for our purposes, the subsequent discussion is simpler if we regard (as we shall) the identically false relation $\mathbf{0}$ as finite-precision threshold classifier, denoted $\prec 0_\bullet$, intuitively, a threshold classifier with negative precision $n = -1$.

We remark that the statement "$x \preccurlyeq_n w$" has identical semantics to "$x \preceq \widehat{w1_\bullet}$" (*not* to "$x \preceq \widehat{w0_\bullet}$") in the usual ordering semantics of $2^{\mathbb{N}}$ which depend on all entries of x, not just those of $x\restriction_n$.

A moment's reflection shows that any finite-precision classifier can be expressed as a Boolean combination [10] of the first n coordinates $P_i(x) = x_i$ $(i < n)$ of its argument $x \in 2^{\mathbb{N}}$ (i.e., $x \preccurlyeq_n w$

[9] We choose square brackets *à la* Knuth, but typeset in double-strike "blackboard bold" style to preclude ambiguity.
[10] In other words, finite-precision classifiers are *definable* in the sense of real-valued logic (sans quantifiers) whose

may be obtained from entries x_i of $x\restriction_n$ using the operations of maximum, minimum and negation $\neg : r \mapsto 1 - r$ in place of classical Boolean connectives). [11]

Infinite-precision threshold classifiers

Threshold ultra-classifiers are relations $2^{\mathbb{N}} \to \{0,1\}$ in the closure of the set of finite-precision classifiers. Since each finite-precision classifier is a non-increasing function on $2^{\mathbb{N}}$, the same holds for any ultra-classifier. It is easily shown that such ultra-classifiers have, for some sequence $z \in 2^{\mathbb{N}}$, one of the forms

- $[\![\cdot \preceq z]\!]$, denoted simply "$\preceq z$", or
- $[\![\cdot \prec z]\!]$, denoted "$\prec z$".

Any finite-precision classifier is an ultra-classifier. Indeed, $\prec 0_\bullet$ is the false relation $\mathbf{0}$ (in agreement with our earlier convention); all other finite-precision classifiers are of the form $\preccurlyeq_n w$ and hence identical to the ultra-classifier $\preceq \widehat{w1_\bullet}$ as remarked above. Since (countably many) sequences of the form $\widehat{w1_\bullet}$ for words $w \in 2^{<\mathbb{N}}$ are dense in $2^{\mathbb{N}}$, one sees that ultra-classifiers are *sequential* limits of finite classifiers, hence are Baire-1 functions on $2^{\mathbb{N}}$.

An ultra-classifier not of finite precision is called *of infinite-precision*; such infinite-precision classifiers are *discontinuous* Baire-1 functions on $2^{\mathbb{N}}$: the sole point of discontinuity of an infinite-precision classifier $\preccurlyeq x$ or $\prec x$ is x. Unlike finite-precision classifiers, those of infinite precision are not definable, i.e., admit no (finite) Boolean expression in the coordinates x_i.

Split Cantor When threshold classifiers are regarded as elements of $\mathrm{B}_1(2^{\mathbb{N}})$, we will denote $\preceq x$ by $\bar{x}$, and $\prec x$ by $\hat{x}$. Per the preceding discussion, such classifiers constitute a set

$$\mathscr{C} = \widehat{C} \cup \overline{C} \subseteq \mathrm{B}_1(2^{\mathbb{N}})$$

consisting of two parts

$$\widehat{C} = \{\hat{x} : x \in 2^{\mathbb{N}}\}, \qquad \overline{C} = \{\bar{x} : x \in 2^{\mathbb{N}}\}.$$

In the topology of pointwise convergence, $\mathscr{C}$ is homeomorphically identified with the (topologized by the lexicographic order) [12] *split Cantor space* $2^{\mathbb{N}} \times \{0,1\}$ by

$$\widehat{C} \cong 2^{\mathbb{N}} \times \{0\} : \hat{x} \mapsto (x, 0), \qquad \overline{C} \cong 2^{\mathbb{N}} \times \{1\} : \bar{x} \mapsto (x, 1).$$

Each of these two parts is homeomorphic to the Cantor set $2^{\mathbb{N}}$, but they are *not* open; indeed, each is essentially dense:[13] $\mathrm{Cl}(\widehat{C}) = \mathscr{C} \setminus \{\mathbf{1}\}$ and $\mathrm{Cl}(\overline{C}) = \mathscr{C} \setminus \{\mathbf{0}\}$.

connectives are vector-lattice operations (maximum, minimum, and linear combinations), plus logical constants (i.e., real constants) as nullary connectives.

[11] The constant relations $\mathbf{0}$, $\mathbf{1}$ are obtained per the usual convention: the conjunction (resp., disjunction) of an empty set of propositions is $\mathbf{1}$ (resp., $\mathbf{0}$).

[12] The primary key of a pair $(x, b) \in 2^{\mathbb{N}} \times \{0,1\}$ is the sequence $x \in 2^{\mathbb{N}}$; the secondary key is the bit $b \in \{0,1\}$. Thus, $\mathscr{C}$ is *not* homeomorphic to the product $\{0,1\} \times 2^{\mathbb{N}}$ (i.e., to the "disjoint union" $2^{\mathbb{N}} \sqcup 2^{\mathbb{N}}$ of two open subspaces).

[13] The constant relations $\mathbf{0}$, $\mathbf{1}$ are isolated points of $\mathscr{C}$.

The upshot is that, starting from a countable set of quite simple finite-precision classifiers (non-increasing Boolean functions) on Cantor space, ultraclassifiers form a much more complicated split Cantor $\mathscr{C} \subseteq \mathrm{B}_1(2^{\mathbb{N}})$.

3.3.5 Example: Prefix tests

Let $\mathcal{C} = (2^{\mathbb{N}}, (\bullet_i)_{i\in\mathbb{N}})$ be the Cantor CSS of Section §3.3.4. For each $w \in 2^{<\mathbb{N}}$, let $\equiv_n w$ be the unary $\{0,1\}$-valued relation $[\![\cdot\restriction_n = w]\!] : x \mapsto [\![x\restriction_n = w]\!]$ which holds if and only if w is the n-prefix of x (i.e., the initial n-word of x); any relation $\equiv_n w$ will be called an *n-prefix test*, or simply a *finite-prefix test*, when n is otherwise unspecified. As for finite-precision classifiers in §3.3.4 above, the subscript n in $\equiv_n w$ is determined as $n = |w|$ and thus optional. One sees that all finite-prefix tests are continuous maps $2^{\mathbb{N}} \to \{0,1\}$; moreover, given w (of length n), the formula $x \equiv_n w$ is explicitly definable as suitable Boolean combination of the entries of $x\restriction_n$. The test $\equiv w$ will also be denoted $\widetilde{w}$ when thought of as a $\{0,1\}$ valued function on $2^{\mathbb{N}}$.

Ultra-prefix tests and equality

Ultra-prefix tests are relations $2^{\mathbb{N}} \to \{0,1\}$ that belong to the closure of prefix tests in the topology of pointwise convergence.

Proposition 3.19. *The identically false relation* $\mathbf{0}$ *is an ultra-prefix test.*

Proof. It suffices to show that the sets

$$W_x^{\not\equiv} \coloneqq \{w \in 2^{<\mathbb{N}} : x \not\equiv w\} \qquad\qquad (x \in 2^{\mathbb{N}})$$

are a filter sub-base on $2^{<\mathbb{N}}$, i.e., that for any finite collection $(x_i)_{i<m} \subseteq 2^{\mathbb{N}}$, the corresponding family $(W_i^{\not\equiv})_{i<m} \coloneqq (W_{x_i}^{\not\equiv})_{i<m}$ has nonempty intersection.[14] Let $k \in \mathbb{N}$ satisfy[15] $2^k > m$; thus, at most m-many distinct k-words $w \in 2^k$ are of the form $w = x_i$ for some $i < m$; some k-word $w^* \in 2^k$ is not of that form; the intersection $\bigcap_{i<m} W_i^{\not\equiv}$ contains w^*, hence is nonempty. □

Proposition 3.20. *Any ultra-prefix test is either the constant* $0_\bullet$*, a finite-prefix test, or an* equality test

$$\overline{\overline{x}} \colon 2^{\mathbb{N}} \to \{0,1\} : y \mapsto [\![y = x]\!]$$

for some fixed $x \in 2^{\mathbb{N}}$*. Such equality tests are Baire-*1 *on* $2^{\mathbb{N}}$*.*

Proof. From the proof of Proposition 3.19 and the remark in footnote[14], one sees that any non-zero ultra-prefix φ is necessarily an accumulation point of a finite union of sets of the form $\widetilde{W}_x \coloneqq \{\widetilde{w} : x \equiv w \in 2^{\mathbb{N}}\}$, and therefore must be an accumulation point of some (single) such set $\widetilde{W}_{x^*}$ for some $x^* \in 2^{\mathbb{N}}$.

One readily sees that $\widetilde{W}_{x^*} = \{\widetilde{w_n} : n \in \mathbb{N}\}$ where $w_n \coloneqq x^*\restriction_n$. It remains to show that $\overline{\overline{x^*}}$ is the sole limit point of $(\widetilde{w_n})_{n\in\mathbb{N}}$, and sequentially so in the pointwise topology. Indeed, for all $y \in 2^{\mathbb{N}}$, we have

$$\lim_{n\to\infty} \widetilde{w_n}(y) = \lim_{n\to\infty} [\![y \equiv_n x^*\restriction_n]\!] = [\![y = x^*]\!] = \overline{\overline{x^*}}(y). \qquad \square$$

[14] From the definition of the sets $W_x^{\not\equiv}$, it should be evident that $\mathbf{0}$ is an ultra-prefix if *and only if* they altogether form a filter sub-base.

[15] One may trivially take $k = m$, but the flow of the proof is obscured.

Unlike finite-prefix tests, equality tests are discontinuous.

Extended Alexandroff compactification

The set $\mathcal{D}_{\mathfrak{c}} := \{\overline{\overline{x}} : x \in 2^{\mathbb{N}}\} \subseteq \mathrm{B}_1(2^{\mathbb{N}})$ is *discrete* of size $\mathfrak{c}$ (continuum); its Alexandroff (one-point) compactification is $\mathcal{A} := \mathcal{D}_{\mathfrak{c}} \cup \{\mathbf{0}\}$. Let $\mathcal{A}^{\complement} := \{\widetilde{w} : w \in 2^{<\mathbb{N}}\}$. The compact space $\mathcal{A}_{\mathrm{E}} = \mathcal{A} \cup \mathcal{A}^{\complement} \subseteq \mathrm{B}_1(2^{\mathbb{N}})$ extends the Alexandroff compactification $\mathcal{A}$ by the countable set $\mathcal{A}^{\complement}$. The point $\mathbf{0} \in \mathcal{A}_{\mathrm{E}}$ is not a G_δ point, i.e., $\{\mathbf{0}\}$ is not a countable intersection of open sets; $\mathbf{0}$ is the only such non-G_δ point. This key topological property implies that the space of ultra-prefix tests is not first-countable (in a first-countable Hausdorff space all points are G_δ).

3.4 Classifying deep computations

3.4.1 NIP, Rosenthal compacta, and deep computations

This is the main section of the chapter. We use topology of function spaces to obtain a new hierarchy of complexity classes for deep computations. The main topological tool is Todorčević's trichotomy for Rosenthal compacta [Tod99]. Our inspiration is the work of Glasner and Megrelishvili on tame semigropus in topological dynamics [GM22].

Theorem 3.21 below is the cornerstone of the section; it establishes a powerful link between deep computations and Rosenthal compacta. It extends the results of [AnIW24] from the setting of continuous deep computations to the more general (and realistic) setting of deep computations of the first Baire class. Informally, Theorem 3.21 states that if $\mathcal{P}$ and $\Delta \subseteq \Gamma$ are countable, then the space of deep computations generated by Δ is a Rosenthal compactum (when restricted to shards) if and only if Δ satisfies the NIP feature by feature. Furthermore, Theorem 3.21 shows that every deep computation generated by Δ is a sequential limit (rather than an abstract ultrafilter limit) of computations in Δ.

Theorem 3.21. *Let $(L, \mathcal{P}, \Gamma)$ be a compositional computation structure (Definition 3.15) satisfying the Extendibility Axiom (see Definition 3.2.3) with $\mathcal{P}$ countable. Let R be an exhaustive collection of sizers, and suppose that $\Delta \subseteq \Gamma$ is R-confined. Then the following conditions are equivalent:*

(i) $\overline{\tilde{\Delta}\restriction_{\mathcal{L}[r_\bullet]}} \subseteq \mathrm{B}_1(\mathcal{L}[r_\bullet], \mathcal{L}[r_\bullet])$ *for all $r_\bullet \in R$;*

(ii) $\pi_P \circ \Delta\restriction_{L[r_\bullet]}$ *satisfies the NIP for all $P \in \mathcal{P}$ and $r_\bullet \in R$; that is, for all $P \in \mathcal{P}$, $r_\bullet \in R$, $a < b$, $\{\gamma_n\}_{n\in\mathbb{N}} \subseteq \Delta$ there are finite disjoint $E, F \subseteq \mathbb{N}$ such that*

$$L[r_\bullet] \cap \bigcap_{n\in E} (\pi_P \circ \gamma_n)^{-1}(-\infty, a] \cap \bigcap_{n\in F} (\pi_P \circ \gamma_n)^{-1}[b, \infty) = \emptyset.$$

Furthermore, if any (hence all) of the preceding conditions hold, then every deep computation $f \in \overline{\Delta}$ can be extended to a Baire-1 function on shards: there is $\tilde{f} \colon \mathcal{L}_{sh} \to \mathcal{L}_{sh}$ such that $\tilde{f}\restriction_{\mathcal{L}[r_\bullet]} \in \mathrm{B}_1(\mathcal{L}[r_\bullet], \mathcal{L}[r_\bullet])$ for all $r_\bullet \in R$. In particular, on each shard, every deep computation is the pointwise limit of a countable sequence of computations.

Proof. Since $\mathcal{P}$ is countable, $\mathcal{L}[r_\bullet] \subseteq \mathbb{R}^{\mathcal{P}}$ is Polish. The Extendibility Axiom implies that $\pi_P \circ \tilde{\Delta}\restriction_{\mathcal{L}[r_\bullet]}$ is a pointwise bounded set of continuous functions for all $P \in \mathcal{P}$. Hence, Theorem 3.10 and Lemma

3.11 prove the equivalence of (i) and (ii). If (i) holds and $f \in \overline{\Delta}$, then we can write as an ultralimit of computations in Δ, say $f = \mathcal{U}\lim_i \gamma_i$. Define $\tilde{f} := \mathcal{U}\lim_i \tilde{\gamma}_i$. Then, for all $r_\bullet \in R$ we have $\tilde{f}\restriction_{\mathcal{L}[r_\bullet]} \in \overline{\tilde{\Delta}\restriction_{\mathcal{L}[r_\bullet]}} \subseteq \mathrm{B}_1(\mathcal{L}[r_\bullet], \mathcal{L}[r_\bullet])$. The assertion that every deep computation is a pointwise limit of a countable sequence of computations follows from the fact that for a Polish space X, every compact subset of $\mathrm{B}_1(X)$ is Fréchet-Urysohn (that is, a space where any point in the closure of a set A is the limit of a sequence of points in A; see Theorem 3F in [BFT78] or Theorem 4.1 in [Deb13]). □

3.4.2 The Todorčević trichotomy, and levels of NIP and PAC learnability

Recall that a topological space X is *hereditarily separable* if every subspace is separable, and that X is *first countable* if every point in X has a countable local basis. Every separable metrizable space is hereditarily separable, and R. Pol proved that every hereditarily separable Rosenthal compactum is first countable (see section 10 of [Deb13]). This suggests the following definition:

Definition 3.22. Let $(L, \mathcal{P}, \Gamma)$ be a CCS satisfying the Extendibility Axiom and R be an exhaustive collection of sizers. Let $\Delta \subseteq \Gamma$ be an R-confined countable set of computations satisfying the NIP on shards and features (condition (ii) in Theorem 3.21). We say that Δ is:

(i) NIP_1 if $\overline{\tilde{\Delta}\restriction_{\mathcal{L}[r_\bullet]}}$ is first countable for every $r_\bullet \in R$;

(ii) NIP_2 if $\overline{\tilde{\Delta}\restriction_{\mathcal{L}[r_\bullet]}}$ is hereditarily separable for every $r_\bullet \in R$;

(iii) NIP_3 if $\overline{\tilde{\Delta}\restriction_{\mathcal{L}[r_\bullet]}}$ is metrizable for every $r_\bullet \in R$.

Observe that $\mathrm{NIP}_3 \Rightarrow \mathrm{NIP}_2 \Rightarrow \mathrm{NIP}_1 \Rightarrow \mathrm{NIP}$. We now present examples of separable and non-separable examples Rosenthal compacta that show that implications are strict. These examples are due to Todorčević [Tod99].

Example 3.23.

1. *Alexandroff compactification of a discrete space of size continuum.* The subset $\mathcal{A} = \mathcal{D}_{\mathfrak{c}} \cup \{\mathbf{0}\} \subseteq \mathrm{B}_1(2^{\mathbb{N}})$ in §3.3.5 is the Alexandroff (one-point) compactification of the continuum-size discrete set $\mathcal{D}_{\mathfrak{c}} = \{\overline{\overline{x}} : x \in 2^{\mathbb{N}}\}$. Hence, $\mathcal{A}$ is a Rosenthal compactum that is not hereditarily separable (and therefore not first countable). In particular, $\mathcal{A}$ is not separable.

2. *Extended Alexandroff compactification.* The Alexandroff compactification $\mathcal{A}$ of $\mathcal{D}_{\mathfrak{c}}$ above can be embedded into a separable compactum of $\mathrm{B}_1(2^{\mathbb{N}})$ by adding a countable set $\mathcal{A}^{\complement}$ of new points that "soften" neighborhoods in $\mathcal{A}$, yielding the extended compactification $\mathcal{A}_{\mathrm{E}} = \mathcal{A} \cup \mathcal{A}^{\complement}$ of §3.3.5. However, the separable Rosenthal compactum $\mathcal{A}_{\mathrm{E}}$ is not first countable. It is an example of a CCS that is NIP but not NIP_1.

3. *Split Cantor.* The split Cantor set $\not{C}$ is a separable Rosenthal compactum that is NIP_2 but not NIP_3; it is hereditarily separable, but it is not metrizable. The set of all finite-precision classifiers is countable dense in $\not{C}$.

4. *Alexandroff Duplicate.* Let K be any compact metric space. Construct the space $X = K \sqcup \mathrm{C}(K)$, i.e., the disjoint union of K and the space $\mathrm{C}(K)$ of real-valued continuous functions on

K with the Banach topology of uniform convergence (supremum-norm topology). Clearly, X is Polish. For each $a \in K$, define $g_a^0, g_a^1 \colon X \to \mathbb{R}$ as follows:

$$g_a^0(x) = \begin{cases} 0, & \text{if } x \in K \\ x(a), & \text{if } x \in \mathrm{C}(K); \end{cases}$$

$$g_a^1(x) = \begin{cases} [\![a = x]\!], & \text{if } x \in K; \\ x(a), & \text{if } x \in \mathrm{C}(K). \end{cases}$$

Let $\mathrm{D}(K) = \{g_a^0 : a \in K\} \cup \{g_a^1 : a \in K\}$. Observe that all points g_a^1 are isolated and that open neighborhoods of $g_{a_0}^0$ are of the form $\{g_a^i : a \in U, i \in \{0,1\}\} \setminus \{g_a^1 : a \in F\}$ where $U \subseteq K$ is an open neighborhood of a_0 and $F \subseteq K$ is a finite set.

Another way to present this space is as the space $K \times \{0,1\}$ with the identifications $(a,0) \mapsto g_a^0$ and $(a,1) \mapsto g_a^1$ and the basic open neighborhoods are given asabove. We can also embed $\mathrm{D}(K)$ into the product $A(K) \times K$ by identifying $(a,0) \mapsto (\mathbf{0}, a)$ and $(a,1) \mapsto (\delta_a, a)$.

Notice that $\mathrm{D}(K)$ is a first countable Rosenthal compactum. It is not separable if K is uncountable; thus, we typically study the interesting case when $K = 2^{\mathbb{N}}$.

As with the Alexandroff compactification $A(2^{\mathbb{N}})$, we can make the space $\mathrm{D}(2^{\mathbb{N}})$ separable by "softening" the neighborhoods of each point through the addition of a countable set. For example, the closure of the set $\{(v_s, s^\frown 0^\infty) : s \in 2^{<\mathbb{N}}\} \subseteq \hat{A}(2^{\mathbb{N}}) \times 2^{\mathbb{N}}$ is $\{(\mathbf{0}, a) : a \in 2^{\mathbb{N}}\} \cup \{(\delta_a, a) : a \in 2^{\mathbb{N}}\} \cup \{(v_s, s^\frown 0^\infty) : s \in 2^{<\mathbb{N}}\}$, where $\{(\mathbf{0}, a) : a \in 2^{\mathbb{N}}\} \cup \{(\delta_a, a) : a \in 2^{\mathbb{N}}\}$ is homeomorphic to $\mathrm{D}(2^{\mathbb{N}})$.

5. *Extended Alexandroff Duplicate of the Split Cantor.* For each finite binary sequence $t \in 2^{<\mathbb{N}}$, let $a_t \in 2^{\mathbb{N}}$ be the sequence starting with t and ending with 0's and let $b_t \in 2^{\mathbb{N}}$ be the sequence starting with t and ending with 1's. Define $h_t \colon 2^{\mathbb{N}} \to \mathbb{R}$ by

$$h_t(x) = \begin{cases} 0, & \text{if } x < a_t; \\ 1/2, & \text{if } a_t \leq x \leq b_t; \\ 1, & \text{if } b_t < x. \end{cases}$$

Let $\hat{D}(S(2^{\mathbb{N}}))$ be the pointwise closure of the set $\{h_t : t \in 2^{<\mathbb{N}}\}$. The identification $h_t \mapsto (v_t, f^+_{t^\frown 0^\infty})$ lifts to a homeomorphism between $\hat{D}(S(2^{\mathbb{N}}))$ and the subspace of $\hat{A}(2^{\mathbb{N}}) \times S(2^{\mathbb{N}})$ consisting of $(\mathbf{0}, f_a^+)$, $(\mathbf{0}, f_a^-)$, (δ_a, f_a^+) and $(v_t, f^+_{t^\frown 0^\infty})$ for $a \in 2^{\mathbb{N}}$ and $t \in 2^{<\mathbb{N}}$ (see 4.3.7 in [ADK08]). Hence, $\hat{D}(S(2^{\mathbb{N}}))$ is a separable first countable Rosenthal compactum that is not hereditarily separable. In fact, it contains an uncountable discrete subspace.

Theorem 3.24 (Todorčević's Trichotomy, [Tod99], [ADK08, Theorem 3])**.** *Let K be a separable Rosenthal Compactum.*

(i) If K is hereditarily separable but non-metrizable, then $S(2^{\mathbb{N}})$ embeds into K.

(ii) If K is first countable but not hereditarily separable, then either $\mathrm{D}(2^{\mathbb{N}})$ or $\hat{D}(S(2^{\mathbb{N}}))$ embeds into K.

(iii) If K is not first countable, then $A(2^{\mathbb{N}})$ embeds into K.

We thus have the following classification for separable Rosenthal compacta:

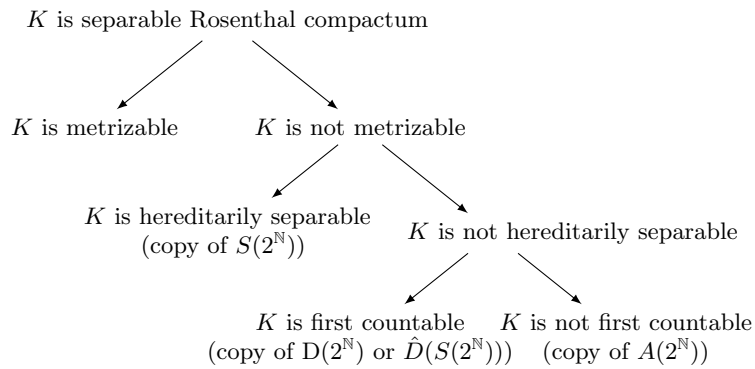


Figure 3.3: Classification of separable Rosenthal compacta.

Todorčević's Trichotomy suggests that in order to distinguish the classes NIP_i, the examples in 3.23 are essential. The following examples show that the levels NIP_i $(i = 1, 2, 3)$ may be distinguished by the topological complexity of deep computations.

Example 3.25.

1. Let $(L, \mathcal{P}, \Gamma)$ be the computation of square root (example 3.3.3 with $\Delta = \Gamma$). We saw that $\overline{\tilde{\Delta}} = \tilde{\Delta} \cup \{\tilde{f}\} \subseteq \mathrm{B}_1(\mathcal{L}, \mathcal{L})$. This corresponds to the Alexandroff compactification of a countable discrete set, which is metrizable. Hence, Δ is NIP_3 but it is not stable, in the sense that $\overline{\tilde{\Delta}} \not\subseteq \mathrm{C}(\mathcal{L}, \mathcal{L})$.

2. Let $(L, \mathcal{P}, \Gamma)$ be the finite precision threshold classifiers (Section 3.3.4) with $\Delta = \{\phi_w : w \in 2^{<\mathbb{N}}\} \cup \{\mathbf{0}^\infty, \mathbf{1}^\infty\}$. We saw that $\overline{\Delta}$ is homeomorphic to the Split Cantor space $S(2^{\mathbb{N}})$ (Example 3.23-(3)), which is hereditarily separable but not metrizable. Hence, Δ is NIP_2 but not NIP_3.

3. Let $(L, \mathcal{P}, \Gamma)$ be the CCS given by $L = 2^{\mathbb{N}}$, $\mathcal{P} = \{P_n : n \in \mathbb{N}\}$ and Γ is the semigroup generated by $\Delta = \{\gamma_t : t \in 2^{<\mathbb{N}}\}$, where $P_n \colon 2^{\mathbb{N}} \to \{0, 1\}$ is the projection map $P_n(x) = x(n)$ and $\gamma_t \colon L \to L$ is given by

$$\gamma_t(x) = \begin{cases} 0^\infty, & \text{if } x <_{\text{lex}} t^\frown 0^\infty; \\ (01)^\infty, & \text{if } t^\frown 0^\infty \leq_{\text{lex}} x \leq_{\text{lex}} t^\frown 1^\infty; \\ 1^\infty, & \text{if } t^\frown 1^\infty <_{\text{lex}} x. \end{cases}$$

   where $(01)^\infty$ denotes the sequence of alternating bits: $010101\cdots$. As in the other examples, it is not difficult to see that $(L, \mathcal{P}, \Gamma)$ satisfies the Extendibility Axiom. For example, the condition $t^\frown 0^\infty \leq_{\text{lex}} x \leq_{\text{lex}} t^\frown 1^\infty$ is equivalent to x extending t. Observe that the set of deep computations is homeomorphic to $\hat{D}(S(2^{\mathbb{N}}))$ (see Example 3.23-(5)). Hence, Δ is NIP_1 but not NIP_2.

4. Let $(L, \mathcal{P}, \Gamma)$ be the finite precision prefix test (Section 3.3.5) with $\Delta = \{\psi_w : w \in 2^{<\mathbb{N}}\}$. We saw that $\overline{\tilde{\Delta}}$ is homeomorphic to the Extended Alexandroff compactification $\hat{A}(2^{\mathbb{N}})$ (Example 3.23-(3)), which is separable but not first countable. Hence, Δ is NIP but not NIP_1.

The definitions provided here for NIP_i $(i = 1, 2, 3)$ are topological. This raises the following question:

Question 3.1. Is there a non-topological characterization for NIP_i, $i = 1, 2, 3$?

3.4.3 The Argyros-Dodos-Kanellopoulos heptachotomy, and approximability of deep computation by minimal classes

In the three separable cases given in 3.23, namely, $\hat{A}(2^{\mathbb{N}})$, $S(2^{\mathbb{N}})$ and $\hat{D}(S(2^{\mathbb{N}}))$, the countable dense subsets are indexed by the binary tree $2^{<\mathbb{N}}$. This choice of index is useful because our emphasis is computational. Real numbers can be represented as infinite binary sequences, i.e., infinite branches of the binary tree $2^{\mathbb{N}}$, and we approximate real numbers with elements in $2^{<\mathbb{N}}$, i.e., finite bitstrings.

Definition 3.26. Let X be a Polish space.

1. If I is countable and $\{f_i : i \in I\} \subseteq \mathbb{R}^X$, $\{g_i : i \in I\} \subseteq \mathbb{R}^X$ are two pointwise families by I, we say that $\{f_i : i \in I\}$ and $\{g_i : i \in I\}$ are *equivalent* if and only if the map $f_i \mapsto g_i$ is extended to a homeomorphism from $\overline{\{f_i : i \in I\}}$ to $\overline{\{g_i : i \in I\}}$.

2. If $\{f_t : t \in 2^{<\mathbb{N}}\}$ is a pointwise bounded family, we say that $\{f_t : t \in 2^{<\mathbb{N}}\}$ is *minimal* if and only if for every dyadic subtree $\{s_t : t \in 2^{<\mathbb{N}}\}$ of $2^{<\mathbb{N}}$, $\{f_{s_t} : t \in 2^{<\mathbb{N}}\}$ is equivalent to $\{f_t : t \in 2^{<\mathbb{N}}\}$.

One of the main results in [ADK08] is that, up to equivalence, there are seven minimal families of Rosenthal compacta and that for every relatively compact $\{f_t : t \in 2^{<\mathbb{N}}\} \subseteq \mathrm{B}_1(X)$ there is a dyadic subtree $\{s_t : t \in 2^{<\mathbb{N}}\}$ such that $\{f_{s_t} : t \in 2^{<\mathbb{N}}\}$ is equivalent to one of the minimal families. We shall describe the seven minimal families next. We follow the same notation as in [ADK08]. For any node $t \in 2^{<\mathbb{N}}$, we denote by $t^\frown 0^\infty$ $(t^\frown 1^\infty)$ the infinite binary sequence starting with t and continuing with all 0's (respectively, all 1's). Fix a regular dyadic subtree $R = \{s_t : t \in 2^{<\mathbb{N}}\}$ of $2^{<\mathbb{N}}$ (i.e., a dyadic subtree such that every level of R is contained in a level of $2^{<\mathbb{N}}$) with the property that for all $s, s' \in R$, we have $s^\frown 0^\infty \neq s'^\frown 0^\infty$ and $s^\frown 1^\infty \neq s'^\frown 1^\infty$. Given $t \in 2^{<\mathbb{N}}$, let v_t be the characteristic function of the set $\{x \in 2^{\mathbb{N}} : x \text{ extends } t\}$. Let $<$ be the lexicographic order in $2^{\mathbb{N}}$. Given $a \in 2^{\mathbb{N}}$, let $f_a^+ \colon 2^{\mathbb{N}} \to \{0, 1\}$ be the characteristic function of $\{x \in 2^{\mathbb{N}} : a \leq x\}$ and let $f_a^- \colon 2^{\mathbb{N}} \to \{0, 1\}$ be the characteristic function of $\{x \in 2^{\mathbb{N}} : a > x\}$. Given two maps $f, g \colon 2^{\mathbb{N}} \to \mathbb{R}$ we denote by $(f, g) \colon 2^{\mathbb{N}} \sqcup 2^{\mathbb{N}} \to \mathbb{R}$ the function which is f on the first copy of $2^{\mathbb{N}}$ and g on the second copy of $2^{\mathbb{N}}$.

1. $D_1 = \{\frac{1}{|t|+1} v_t : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_1} = A(2^{\mathbb{N}})$.

2. $D_2 = \{s_t^\frown 0^\infty : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_2} = 2^{\leq N}$.

3. $D_3 = \{f^+_{s_t^\frown 0^\infty} : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_3} = S(2^{\mathbb{N}})$.

4. $D_4 = \{f^-_{s_t^\frown 1^\infty} : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_4} = S(2^{\mathbb{N}})$.

5. $D_5 = \{v_t : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_5} = \hat{A}(2^{\mathbb{N}})$.

6. $D_6 = \{(v_{s_t}, s_t^\frown 0^\infty) : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_6} = \hat{D}(2^{\mathbb{N}})$.

7. $D_7 = \{(v_{s_t}, x^+_{s_t^\frown 0^\infty}) : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_7} = \hat{D}(S(2^{\mathbb{N}}))$.

Theorem 3.27 (Heptachotomy of minimal families, Theorem 2 in [ADK08])**.** *Let X be Polish. For every relatively compact $\{f_t : t \in 2^{<\mathbb{N}}\} \subseteq \mathrm{B}_1(X)$, there exists $i = 1, 2, \ldots, 7$ and a regular dyadic subtree $\{s_t : t \in 2^{<\mathbb{N}}\}$ of $2^{<\mathbb{N}}$ such that $\{f_{s_t} : t \in 2^{<\mathbb{N}}\}$ is equivalent to D_i. Moreover, the families D_i are minimal and mutually non-equivalent.*

The implication of this result for deep computations is the following: for any countable set of computations Δ satisfying the NIP (for some CCS $(L, \mathcal{P}, \Gamma)$), we can always find a countable discrete set of deep computations that approximates all the other deep computations. For example: in the finite precision prefix test example (subsection 3.3.5), the prefix test computations (family D_5) approximate all other deep computations. However, note that this discrete set D_i may not consist of continuous functions (i.e., they will not be computable in general). For example, functions in D_3 are not continuous.

3.5 Monte Carlo computability

In this section, we replace deterministic computability by probabilistic ('Monte Carlo') computability. We do not assume that $\mathcal{P}$ is countable. (The countability assumption on $\mathcal{P}$ played a crucial role in the proof of Theorem 3.10, as it makes $\mathbb{R}^{\mathcal{P}}$ a Polish space.) The main results of the section are Theorem 3.34, which connects NIP and Monte Carlo computability, and 3.41, which connects Talagrand stability and Monte Carlo computability.

Fundamental in this section is a measure-theoretic version of Theorem 3.10, namely, Theorem 3.32.

3.5.1 NIP and Monte Carlo computability of deep computations

By Fact 3.2, for perfectly normal X, a function f is in $\mathrm{B}_1(X, Y)$ if and only if $f^{-1}[U]$ is an F_σ subset of X for every open $U \subseteq Y$. This motivates the following definition.

Definition 3.28. Given a Hausdorff space X and a measurable space (Y, Σ), we say that $f\colon X \to Y$ is *universally measurable* (with respect to Σ) if $f^{-1}(E)$ is μ-measurable for every Radon measure μ on X and every $E \in \Sigma$. When $Y = \mathbb{R}$, we will always take $\Sigma = \mathcal{B}(\mathbb{R})$, the Borel σ-algebra of $\mathbb{R}$.

If X is a compact (Hausdorff) space, then Radon measures on X are finite, and hence, through renormalization, we can regard them as probability measures. We thus have the following remark:

Remark 3.29. If X is compact, then a function $f\colon X \to \mathbb{R}$ is universally measurable if and only if $f^{-1}(U)$ is μ-measurable for every Radon probability measure μ on X and every open set $U \subseteq \mathbb{R}$.

Intuitively, a function is universally measurable if it is "measurable no matter which reasonable way you try to measure things on its domain". The concept of universal measurability emerged from work of Kallianpur and Sazonov, in the late 1950's and 1960s — with later developments by Blackwell, Darst and others — building on earlier ideas of Gnedenko and Kolmogorov from the 1950s. See [Pap02, Chapters 1 and 2].

Notation 3.30. Following [BFT78], the collection of all universally measurable real-valued functions on X will be denoted by $M_r(X)$. Given a fixed Radon measure μ on X, the collection of all μ-measurable real-valued functions on X will be denoted by $\mathscr{M}^0(X, \mu)$.

In the context of deep computations, we are interested in transition maps of a state space $L \subseteq \mathbb{R}^{\mathcal{P}}$ into itself. In the product space $\mathbb{R}^{\mathcal{P}}$, we have two natural σ-algebras: the Borel σ-algebra, i.e., the σ-algebra generated by open sets in $\mathbb{R}^{\mathcal{P}}$, and the cylinder σ-algebra (i.e., the σ-algebra generated by the sub-basic open sets in $\mathbb{R}^{\mathcal{P}}$, that is, the set $\pi_i^{-1}(U)$ with $U \subseteq \mathbb{R}$ open). In general, the cylinder σ-algebra is strictly smaller; however, when $\mathcal{P}$ is countable, both σ-algebras coincide. We will use the cylinder σ-algebra to define universally measurable maps $f\colon \mathbb{R}^{\mathcal{P}} \to \mathbb{R}^{\mathcal{P}}$. The reason for this choice is the following characterization:

Proposition 3.31. *Let X be a Hausdorff space and $Y = \prod_{i\in I} Y_i$ be any product of measurable spaces (Y_i, Σ_i) for $i \in I$. Let Σ_Y be the cylinder σ-algebra generated by the measurable spaces (Y_i, Σ_i). The following are equivalent for $f\colon X \to Y$:*

(i) $f\colon X \to Y$ is universally measurable (with respect to Σ_Y).

(ii) $\pi_i \circ f\colon X \to Y_i$ is universally measurable (with respect to Σ_i) for all $i \in I$.

Proof. (i)⇒(ii) is clear since the projection maps π_i are measurable and the composition of measurable functions is measurable. To prove (ii)⇒(i), note that if $C = \prod_{i\in I} C_i$ is a measurable cylinder and J is the finite subset of I such that $C_i \neq Y_i$ for $i \in J$, then, $f^{-1}(C) = \bigcap_{i\in J}(\pi_i \circ f)^{-1}(C_i)$ is universally measurable by assumption. □

The preceding proposition says that we can check measurability of a transition just by checking measurability feature by feature. This is analogous to the Baire class 1 setting (see Proposition 3.9).

We saw in section 2.5.2 that, when the set $\mathcal{P}$ of predicates is countable, PAC-learning (or NIP) corresponds to relative compactness in the space of Baire class 1 functions (see Theorems 3.4, 3.7, 3.10). The following result (which does not assume countability of $\mathcal{P}$) gives an analogous characterization of the NIP in terms of universal measurability:

Theorem 3.32 (Bourgain-Fremlin-Talagrand, Theorem 2F in [BFT78])**.** *Let X be a Hausdorff space and $A \subseteq \mathrm{C}(X)$ be pointwise bounded. The following are equivalent:*

(i) $\overline{A} \subseteq M_r(X)$.

(ii) For every compact $K \subseteq X$, $A\restriction_K$ satisfies the NIP.

(iii) For every Radon measure μ on X, A is relatively countably compact in $\mathscr{M}^0(X, \mu)$, i.e., every countable subset of A has a limit point in $\mathscr{M}^0(X, \mu)$.

This result allows us to formalize the concept of a deep computation being *Monte Carlo computable*, which we define below (Definition 3.33). To motivate the definition, let us first recall two facts:

1. Littlewoood's second principle states that every Lebesgue measurable function is "nearly continuous." The formal statement of this, which is Luzin's theorem, is that if (X, Σ, μ) a Radon measure space and Y is a second-countable topological space (e.g., $Y = \mathbb{R}^{\mathcal{P}}$ with $\mathcal{P}$ countable) equipped with the Borel σ-algebra, then any given $f : X \to Y$ is measurable if and only if for every $E \in \Sigma$ and every $\varepsilon > 0$ there exists a closed $F \subseteq E$ such that the restriction $f\restriction_F$ is continuous and $\mu(E\backslash F) < \varepsilon$.

2. Computability of deep computations is characterized in terms of continuous extendibility of computations. This is at the core of [AnIW24].

These two facts motivate the following definition:

Definition 3.33. Let $(L, \mathcal{P}, \Gamma)$ be a CCS. We say that a transition $f\colon L \to L$ is *universally Monte Carlo computable* if and only if there exists an extension $\tilde{f} :: \mathcal{L}_{\mathrm{sh}} \to \mathcal{L}_{\mathrm{sh}}$ of f satisfying the following property: for every sizer $r_\bullet$ there is a sizer $s_\bullet$ such that the restriction $\tilde{f}\restriction_{\mathcal{L}[r_\bullet]}\colon \mathcal{L}[r_\bullet] \to \mathcal{L}[s_\bullet]$ is universally measurable, i.e., $\pi_P \circ \tilde{f}\restriction_{\mathcal{L}[r_\bullet]}\colon \mathcal{L}[r_\bullet] \to [-s_P, s_P]$ is μ-measurable for every Radon probability measure μ on $\mathcal{L}[r_\bullet]$ and $P \in \mathcal{P}$.

Note that, since $\mathcal{L}_{\mathrm{sh}}$ is compact, by remark 3.29, in the context of Monte Carlo computability, we restrict our attention to Radon probability measures instead of general Radon measures.

Theorem 3.34. *Let $(L, \mathcal{P}, \Gamma)$ be a CCS satisfying the Extendibility Axiom. Let R be an exhaustive collection of sizers. Let $\Delta \subseteq \Gamma$ be R-confined. If $\pi_P \circ \Delta\restriction_{L[r_\bullet]}$ satisfies the NIP for all $P \in \mathcal{P}$ and all $r_\bullet \in R$, then every deep computation in Δ is universally Monte Carlo computable.*

Proof. Fix $P \in \mathcal{P}$ and $r_\bullet \in R$. By the Extendibility Axiom, $\pi_P \circ \tilde{\Delta}\restriction_{\mathcal{L}[r_\bullet]}$ is a set of pointwise bounded continuous functions on the compact set $\mathcal{L}[r_\bullet]$. Since $\pi_P \circ \Delta\restriction_{L[r_\bullet]}$ has the NIP, so does $\pi_P \circ \tilde{\Delta}\restriction_{L[r_\bullet]} = \pi_P \circ \Delta\restriction_{\mathcal{L}[r_\bullet]}$, by Lemma 3.11. Let $f \in \overline{\Delta}$ be a deep computation; write f as an ultralimit of computations in Δ, say $f = \mathcal{U}\lim_i \gamma_i$, and define an extension $\tilde{f}\colon \mathcal{L}_{\mathrm{sh}} \to \mathcal{L}_{\mathrm{sh}}$ of f naturally by $\tilde{f} := \mathcal{U}\lim_i \tilde{\gamma}_i$. Since, by hypothesis, Δ is R-confined, we have $f\colon L[r_\bullet] \to L[r_\bullet]$ and hence, $\tilde{f}\colon \mathcal{L}[r_\bullet] \to \mathcal{L}[r_\bullet]$. Now, by Theorem 3.32, we have $\overline{\pi_P \circ \tilde{\Delta}\restriction_{\mathcal{L}[r_\bullet]}} \subseteq M_r(\mathcal{L}[r_\bullet])$; therefore, $\pi_P \circ \tilde{f}\restriction_{\mathcal{L}[r_\bullet]} \in \overline{\pi_P \circ \tilde{\Delta}\restriction_{\mathcal{L}[r_\bullet]}} \subseteq M_r(\mathcal{L}[r_\bullet])$. □

Question 3.2. Under the same assumptions of the preceding theorem, suppose that every deep computation of Δ is universally Monte Carlo computable. Must $\pi_P \circ \Delta\restriction_{L[r_\bullet]}$ have the NIP for all $P \in \mathcal{P}$ and all $r_\bullet \in R$?

3.5.2 Talagrand stability and Monte Carlo computability of deep computations

There is another notion closely related to NIP, introduced by Talagrand in [Tal84] while studying Pettis integration. Suppose that X is a compact Hausdorff space and $A \subseteq \mathbb{R}^X$. Let μ be a Radon probability measure on X. Given a μ-measurable set $E \subseteq X$, a positive integer k and real numbers $a < b$, let

$$D_k(A, E, a, b) = \bigcup_{f \in A} \{x \in E^{2k} : f(x_{2i}) \leq a,\ f(x_{2i+1}) \geq b \ \text{ for all } i < k\}.$$

Definition 3.35. We say that A is *Talagrand μ-stable* if and only if for every μ-measurable set $E \subseteq X$ of positive measure and for every $a < b$ there is $k \geq 1$ such that

$$(\mu^{2k})^*(D_k(A, E, a, b)) < (\mu(E))^{2k},$$

where μ^* denotes the outer measure.

Note that the reference to outer measure in Definition 3.35 is necessary since $D_k(A, E, a, b)$ need not be measurable. The inequality certainly holds when A is a countable set of continuous (or μ-measurable) functions.

Remark 3.36. Let X, μ, $A \subseteq \mathbb{R}^X$, and $\subseteq X$ be as above. Then,

1. If $A \subseteq B \subseteq \mathbb{R}^X$, then
$$D_k(A, E, a, b) \subseteq D_k(B, E, a, b).$$

2. If $a < a' < b' < b$, then
$$D_k(\overline{A}, E, a, b) \subseteq D_k(A, E, a', b').$$

   [*Proof*: If $x \in D_k(\overline{A}, E, a, b)$, then there is $f \in \overline{A}$ such that $f(x_{2i}) \leq a < a'$ and $f(x_{2i+1}) \geq b > b'$ for all $i < k$. By definition of pointwise convergence topology, there exists $g \in A$ such that $g(x_{2i}) < a'$ and $g(x_{2i+1}) > b'$ for all $i < k$. Hence, $x \in D_k(A, E, a', b')$.]

3. Any subset of a μ-stable set is μ-stable. (This is immediate from (1).)

4. If A is stable, then $\overline{A}$ is μ-stable. (This is immediate from (2).)

The main result of this section is that deep computations that are limit points of a Talagrand stable set of computations are Monte Carlo computable; this is Corollary 3.41 below. To prove it, we first show that the closure of a Talagrand μ-stable set is μ-measurable (lemma 3.38); but first, we need the following useful characterization of measurable functions (compare with Fact 3.2):

Fact 3.37 (Lemma 413G in [Fre03])**.** *Suppose that (X, Σ, μ) is a measure space and let $\mathcal{K} \subseteq \Sigma$ be a collection of measurable sets satisfying the following conditions:*

- *(X, Σ, μ) is complete, i.e., for all $E \in \Sigma$ with $\mu(E) = 0$ and $F \subseteq E$ we have $F \in \Sigma$.*
- *(X, Σ, μ) is semi-finite, i.e., for all $E \in \Sigma$ with $\mu(E) = \infty$ there exists $F \subseteq E$ such that $F \in \Sigma$ and $0 < \mu(F) < \infty$.*
- *(X, Σ, μ) is saturated, i.e., $E \in \Sigma$ if and only if $E \cap F \in \Sigma$ for all $F \in \Sigma$ with $\mu(F) < \infty$.*
- *(X, Σ, μ) is inner regular with respect to $\mathcal{K}$, i.e., for all $E \in \Sigma$*
$$\mu(E) = \sup\{\mu(K) : K \in \mathcal{K} \text{ and } K \subseteq E\}.$$

(In particular, if X is compact Hausdorff, $\mathcal{K}$ is the collection of compact subsets of X, μ is a Radon probability measure on X, Σ is the completion of the Borel σ-algebra by μ, and all these conditions hold). Then, $f \colon X \to \mathbb{R}$ is measurable if and only if for every $K \in \mathcal{K}$ with $0 < \mu(K) < \infty$ and $a < b$, either $\mu^*(P) < \mu(K)$ or $\mu^*(Q) < \mu(K)$, where $P = \{x \in K : f(x) \leq a\}$ and $Q = \{x \in K : f(x) \geq b\}$.

The following technical lemma will be instrumental for proving Proposition 3.40, which, in turn, will yield the main result of the subsection, namely Corollary 3.41

Lemma 3.38. *If A is Talagrand μ-stable, then $\overline{A}$ is also Talagrand μ-stable and $\overline{A} \subseteq \mathscr{M}^0(X, \mu)$.*

Proof. If A is Talagrand μ-stable, so is $\overline{A}$, by Remarks 3.36. Suppose that A is Talagrand μ-stable to show that $\overline{A} \subseteq \mathscr{M}^0(X, \mu)$. Assume, for the sake of contradiction that there exists $f \in \overline{A}$ such that $f \notin \mathscr{M}^0(X, \mu)$. By Fact 3.37, there exists a μ-measurable set E of positive measure and $a < b$ such that $\mu^*(P) = \mu^*(Q) = \mu(E)$ where $P = \{x \in E : f(x) \leq a\}$ and $Q = \{x \in E : f(x) \geq b\}$. Then, for any $k \geq 1$: $(P \times Q)^k \subseteq D_k(\{f\}, E, a, b)$, so $(\mu^{2k})^*(D_k(\{f\}, E, a, b)) = (\mu^*(P)\mu^*(Q))^k = (\mu(E))^{2k}$. Thus, the set $\{f\}$ is not μ-stable. However, $\{f\} \subseteq \overline{A}$ and we know (Remarks 3.36) that a subset of a μ-stable set must be μ-stable, so we have a contradiction. □

Definition 3.39. We say that A is *universally Talagrand stable* if A is Talagrand μ-stable for every Radon probability measure μ on X.

We first observe that universal Talagrand stability implies NIP:

Proposition 3.40. *Let X be a compact Hausdorff space and $A \subseteq \mathrm{C}(X)$ be pointwise bounded. If A is universally Talagrand stable, then A satisfies the NIP.*

Proof. Suppose that A is universally Talagrand stable. By Theorem 3.32, it suffices to show that A is relatively countably compact in $\mathscr{M}^0(X, \mu)$ for every Radon probability measure μ on X. Since, for any such μ, A is Talagrand μ-stable, Lemma 3.38 gives us $\overline{A} \subseteq \mathscr{M}^0(X, \mu)$. In particular, A is relatively countably compact in $\mathscr{M}^0(X, \mu)$. □

Corollary 3.41. *Let $(L, \mathcal{P}, \Gamma)$ be a CCS satisfying the Extendibility Axiom. If $\pi_P \circ \Delta\restriction_{L[r_\bullet]}$ is universally Talagrand stable for all $P \in \mathcal{P}$ and all sizers $r_\bullet$, then every deep computation is universally Monte Carlo computable.*

Proof. This follows directly from Proposition 3.40 and Theorem 3.34. □

In the context of deep computations, we have identified two ways to obtain Monte Carlo computability, namely, NIP/PAC and Talagrand stability. It is natural to ask whether these two notions are equivalent. The following results show that, even in the simple case of countably many computations, this question is sensitive to the set-theoretic axioms. On the one hand, it is consistent (with respect to the standard ZFC axioms of set theory) that these two classes are the same:

Theorem 3.42 (Talagrand, Theorem 9-3-1(a) in [Tal84])**.** *Let X be a compact Hausdorff space and $A \subseteq M_r(X)$ be countable and pointwise bounded. Assume that $[0, 1]$ is not the union of $< \mathfrak{c}$ closed measure zero sets. If A satisfies the NIP, then A is universally Talagrand stable.*

(The assumption that $[0, 1]$ is not the union of $< \mathfrak{c}$ closed measure zero sets is a consequence of, for example, the Continuum Hypothesis.)

On the other hand, the opposite is also consistent for the particular case of the Lebesgue measure:

Theorem 3.43 (Fremlin, Shelah, [FS93])**.** *It is consistent with the usual axioms of set theory that there exists a countable pointwise bounded set of Lebesgue measurable functions with the NIP which is not Talagrand stable with respect to Lebesgue measure.*

Notice that the preceding two results apply to sets of measurable functions, a class of functions larger than the class of continuous functions. However, by the Extendibility Axiom, finitary computations are continuous, i.e., if A is a set of computations, then $A \subseteq \mathrm{C_p}(X)$.

Chapter 4

Group Actions on Graphs

This is the group actions on graphs chapter of the thesis.

4.1 Introduction

4.2 Border tracing in oriented adjacency graphs of polygonal tilings

The work presented in this section is a departure from the algebraic perspective of the rest of this chapter; it arose as a side project, and the full details, including proofs of all stated results, appear in [WMW26a]. The connection to the present chapter is that the underlying combinatorial objects are graphs with additional structure, specifically, a cyclic ordering of the neighbours of each vertex, and the algorithms we describe amount to a systematic traversal of these structures.

Border tracing is a classical segmentation technique in digital image processing: given a binary image, one wishes to determine a closed path of pixels that traces the boundary or contour of an object. The standard setting is the integer lattice $\mathbb{Z}^2$ with the well-known 4- and 8-adjacencies, and algorithms for this case have been described in numerous textbooks (see [GW18, KR04] for example).

A natural generalization replaces square pixels by the tiles of an arbitrary edge-to-edge polygonal tiling of $\mathbb{R}^2$, for example, a triangular or hexagonal tiling, and equips the set of tiles with an adjacency relation. This generalization is motivated both by theoretical considerations in digital topology and by practical applications: triangular and hexagonal pixel grids are used extensively in sensors, computer graphics, and data visualization. In [WMW26a], we develop a border tracing algorithm that works for objects in any such tiling, adapting and correcting the *Border Mesh Determination* method of Voss [Vos93].

4.2.1 Oriented adjacency graphs, border meshes and Freeman codes

Let $\mathcal{T}$ be an edge-to-edge polygonal tiling of $\mathbb{R}^2$ consisting of convex polygons, and let A be an irreflexive, symmetric relation on $\mathcal{T}$ such that the graph $G = (\mathcal{T}, A)$ is connected and any two

A-adjacent tiles intersect. We call A an *adjacency* on $\mathcal{T}$; the two most natural choices are *edge-adjacency* (where tiles share a side) and *vertex-adjacency* (where tiles share a vertex, and possibly more).

An *orientation* of G assigns to each tile t a cyclic ordering $z(t) = (q_1, q_2, \ldots, q_{v(t)})$ of its $v(t)$ neighbours. Starting from a directed edge (t, s), one can repeatedly move to the next edge by following the neighbour of s that comes immediately after t in the cyclic order at s. Repeating this rule produces a directed cycle or path, called a *mesh*. Because the rule is deterministic, each directed edge belongs to a unique maximal mesh. The set of all maximal meshes, we denote by M.

A finite subset $\mathcal{C} \subset \mathcal{T}$ (an *object*) induces a subgraph whose restricted orientation may create meshes not present in M. Any such new mesh is called a *border mesh* of $\mathcal{C}$. Intuitively, border meshes trace the boundary of the object within the tiling.

When the adjacency graph is drawn in $\mathbb{R}^2$ with each tile represented by its centroid and each edge by a straight segment, one obtains a natural *counterclockwise* (respectively, *clockwise*) orientation by ordering the neighbours of each tile according to the angular order of the corresponding segments. For a connected, hole-free object under such an orientation, the border mesh found by our algorithm is called the *contour*; it circumscribes the entire object in a precise sense made rigorous in [WMW26a, Theorem 1].

Freeman code. When A is a k-adjacency (every tile has exactly k neighbours), the cyclic orderings can be encoded uniformly. This is important since it turns the cyclic order into a simple integer lookup so the algorithm can move from one boundary tile to the next. For adjacent tiles t and t' with $z(t) = (q_1, q_2, \ldots, q_k)$ and $t' = q_i$, the *Freeman code* of the ordered pair (t, t') is the integer $f(t, t') = i - 1$. Thus the cyclic order $z(t)$ is represented by the sequence $(0, 1, 2, \ldots, k-1)$, and any closed path $(t_1, t_2, \ldots, t_n)$ has a *Freeman chain code* $(f(t_1, t_2),\ f(t_2, t_3),\ \ldots,\ f(t_{n-1}, t_n),\ f(t_n, t_1))$. This encoding is the natural extension of the classical Freeman chain code for 4- and 8-neighbours on the square grid, introduced in [GW18].

Figure 4.1 shows the Freeman codes for a triangular tiling.

4.2.2 The algorithm `BorderTracer`

The general border tracing procedure [WMW26a, Algorithm 1] works for any oriented adjacency graph and any object whose components each have at least two tiles. When the adjacency is a k-adjacency, the algorithm simplifies to the version presented below as Algorithm 1, which uses the Freeman code to drive the neighbour search.

A key correction over the original method of Voss [Vos93] is the **End Condition Test**: a border mesh may pass through the starting tile t_0 multiple times before closing, and hence it is not sufficient to stop upon first return to t_0. Instead, the algorithm verifies that both the first *directed edge* (t_0, t_1) has been repeated, that is, that $t' = t_0$, and that the subsequent tile equals t_1.

The algorithm has linear time complexity in the length of the output list Cont: at each step, locating the previous tile in the cyclic order and scanning for the next object tile each require at most k comparisons.

Figure 4.2 illustrates the border meshes obtained by Algorithm 1 for a connected object with a hole, using the counterclockwise orientation.

Fig. 12

Figure 4.1: Freeman codes for the edge- and vertex-neighbours in the standard triangular tiling, providing the counterclockwise orientation. Reproduced from [WMW26a, Figure 12].

Under the counterclockwise (respectively, clockwise) orientation, and for an A-connected object $\mathcal{C}$ with edge-connected background, the contour produced by Algorithm 1 (starting from edgeneighbours $t_0 \in \mathcal{C}$, $q \notin \mathcal{C}$) satisfies the following properties proved in [WMW26a]:

1. **Circumscription.** The contour visits every tile of $\mathcal{C}$ that meets the topological frontier $\mathrm{fr}(|\mathcal{C}|)$, and the sequence of frontier segments contributed by the contour tiles traces $\mathrm{fr}(|\mathcal{C}|)$ exactly once, counterclockwise (respectively, clockwise).

2. **Separation** (for planar edge-adjacency graphs). Deleting all edges between the contour and the background disconnects $\mathcal{C}$ from its complement.

3. **Boundary coincidence** (under edge-adjacency). The set of tiles appearing in the contour equals the boundary $\mathcal{B}(\mathcal{C})$, defined as all tiles of $\mathcal{C}$ intersecting $\mathrm{fr}(|\mathcal{C}|)$.

Implementation. A complete implementation of Algorithm 1 is available in a public repository [WMW26b]. It supports the edge- and vertex-adjacencies on rectangular, triangular, and hexagonal tilings, and includes routines for generating the Freeman codes, detecting starting pairs (t_0, q), and extracting all border meshes via repeated application of the algorithm with appropriate marking of already-traced borders.

4.3 Recursive constructions

4.3.1 Balanceable graphs

Ramsey's classical theorem for finite graphs states that, for a fixed graph G, there is a large enough complete graph K_n such that no matter how the edges of K_n are coloured (in say two colours), one can find a monochromatic copy of G inside of K_n. Suppose one wishes to find a copy of G inside the coloured graph K_n, but instead of monochromatic, with half of the edges (possibly rounding if

Algorithm 1: Border Tracing for a k-adjacency graph $G = (\mathcal{T}, A, Z, M)$

Input : Object $\mathcal{C} \subset \mathcal{T}$;
a finite set $\mathcal{N}$ with $\mathcal{C} \subset \mathcal{N} \subset \mathcal{T}$ surrounding the object;
the Freeman code sequence $(0, 1, \ldots, k-1)$.
Output: Ordered list Cont of tiles forming a border mesh of $\mathcal{C}$ in G.

```
Step 1.
Scan N to find a tile t = t_0 ∈ C having a neighbour q ∈ N \ C.
Set Cont ← (t_0).
Step 2.
Set b ← f(t, q).
repeat
    b ← (b + 1) mod k
    Let t' be the tile with f(t, t') = b
until a tile t' ∈ C is found (at most k iterations)
if no object tile was found then return error: "C is an isolated tile"
Append t' to Cont, so Cont = (t_0, t_1).
Step 3.
Let Cont = (t_0, t_1, ..., t_n).
Set b ← f(t_n, t_{n-1}).
repeat
    b ← (b + 1) mod k
    Let t' be the tile with f(t_n, t') = b
until a tile t' ∈ C is found
End Condition Test:
if t' ≠ t_0 then
    Append t' to Cont; go to Step 3.
else
    Append t' to Cont. Run Step 3 once more to obtain the next tile t' ∈ C.
    if t' = t_1 then
        Remove the last element of Cont.
        return Cont.
    else
        Append t' to Cont; go to Step 3.
```

G has an odd number of edges) in one colour, and the other half in the remaining colour. Clearly, one needs to impose additional conditions on the colourings of K_n, namely, that each colour class is "well represented" (since otherwise one could colour K_n in a single colour, making it impossible to find a non-trivial graph with the desired property) and one needs additional conditions on the structure of G. I do not go into the details of what those properties are, beyond what is needed to state and expand the results from my own research, as well as tie these concepts to the other ideas in this section. The interested reader will find a thorough discussion and complete characterization of these graphs in [CHM21, Section 2].

Definition 4.1. Suppose that a finite graph G has an edge-colouring $c\colon E(G) \to 2$ in two colours. This coloured graph is called *balanced* if the number of edges of each colour differ by at most one; that is, if G has an even number of edges, there are exactly as many edges of each colour.

A finite graph G is called *balanceable* if in any 2-colouring of the complete graph K_n, with sufficiently many edges of each colour and n sufficiently large, one can find a balanced copy of G.

The *balancing number* $\mathrm{bal}(n, G)$ of a balanceable graph G is the maximum number k of edges

Fig. 11/13

Figure 4.2: Border meshes of a connected object in the edge-adjacency graph of a hexagonal tiling under the counterclockwise orientation. Reproduced from [WMW26a, Figure 11/13].

for which there is a 2-coloring of K_n with exactly k edges in one colour without balanced copies of G.

In other words, any 2-coloring of K_n with more than $\mathrm{bal}(n, G)$ edges in each color contains a balanced copy of G.

Theorem 4.2 ([CHM21, Corollary 2.5])**.** *A finite graph G is balanceable if and only if:*

1. *there are disjoint $X, Y \subseteq V(G)$ such that $V(G) = X \cup Y$ and there are either $\lfloor \frac{1}{2}k \rfloor$ or $\lceil \frac{1}{2}k \rceil$ edges between the vertices of X and Y; and*
2. *there is $W \subseteq V(G)$ such that there are either $\lfloor \frac{1}{2}k \rfloor$ or $\lceil \frac{1}{2}k \rceil$ edges between vertices of W.*

We can study more specific aspects of graphs than whether they are balanceable or not. Interestingly, the balancing number may or may not depend on n, as the following examples demonstrate.

Proposition 4.3. *The 4-cycle, C_4, has balancing number 1. More specifically, $\mathrm{bal}(n, C_4) = 1$ whenever $n \geq 4$.*

Proof. Indeed, it is obvious that the balancing number cannot be less than 1, as C_4 has four edges, and hence a balanced copy needs two edges of each colour.

As for the upper bound, let $n \geq 4$ and fix an arbitrary edge colouring $c\colon E(K_n) \to \{\mathrm{red}, \mathrm{blue}\}$ with at least two edges of each colour. First, suppose there is a vertex $x \in V(K_n)$ all of whose neighbours are joined with edges of the same colour (say red). Then, if there are two blue edges in K_n that intersect, they form a blue path of vertices p_0, p_1, p_2, neither of which can be x by assumption. The edges $(x, p_0), (p_0, p_1), (p_1, p_2), (p_2, x)$ are red, blue, blue, red, respectively, and thus form a balanced copy of C_4. If no two blue edges intersect, then the edges joining the endpoints of two particular blue edges (of which there must be at least two) must be all red, which we use to form a balanced copy of C_4.

Now, consider the case when every vertex has red and blue incident edges. Continue by fixing a red edge xy and partitioning the vertices $w \in V(K_n) \setminus \{x, y\}$ into four classses depending on the

value of the pair $(c(wx), c(wy))$. There are two colours, hence four possible values for this pair; let us abbreviate the classes as RR for those vertices w with pair (red, red) and similarly, RB, BR and BB. By assumption, x and y have blue incident edges and thus $V(K_n) \setminus \{x, y\} \notin \{RR, RB, BR\}$. If $V(K_n) \setminus \{x, y\} = BB$, then any red edge different from xy must be disjoint from xy and forms a balanced copy of C_4 by connecting said edge to xy (via blue edges). We can therefore assume that at least two of the four classes are nonempty, giving six sub-cases to analyze:

If there are $w \in RR$ and $z \in BB$, or if there are $w \in RB$ and $z \in BR$, then (x, w, y, z) is a balanced 4-cycle; if $w \in RB$ and $z \in BB$, then either (x, w, z, y) (whenever wz is blue) or (x, y, w, z) (whenever wz is red) is a balanced 4-cycle; the case $w \in BR$ and $z \in BB$ is symmetric by flipping the possible colours of wz; if $w \in RR$, $z \in RB \cup BR$ and wz is blue, then either (x, y, z, w) (whenever $z \in RB$) or (x, y, w, z) (whenever $z \in BR$) is a balanced 4-cycle.

The only case that is not yet covered so far is when $w \in RR$, $z \in RB$ (or, respectively, when $z \in BR$) and wz is a red edge. Invoking our assumption one last time, there must be a blue edge incident in w, say ww'. Crucially, we may assume that $w' \in RR$ since any other possibility is covered by the preceding discussions. Finally, the cycle (w, w', y, z) (or, respectively (w, w', x, z), when $z \in BR$) is balanced. □

Example 4.4.

1. The path on three edges, P_4, has balancing number 0.

   Suppose that the edges of K_n are coloured in red and blue with at least some edge of each colour. In particular, a red edge and a blue edge must intersect, forming a path of two edges of different colour. Taking any third edge that completes the copy of P_4, successfully builds a balanced copy regardless of the colour of the third edge.

2. The graph consisting of four disjoint edges, $4K_2$, has balancing number $n - 1$. Concretely, when $n \geq 10$, $\text{bal}(n, 4K_2) = n - 1$. This is [CLZ20, Theorem 2.3].

3. When $n \geq 5$, $\text{bal}(n, K_{1,4}) = n-1$. This is an immediate consequence of [CHM21, Theorem 3.2].

4. The complete graph K_5 is not balanceable.

   Consider a colouring of K_{2n} where we split the vertex set into two disjoint sets A and B of size n. We colour every edge connecting vertices in A red, and edges connecting vertices in B blue; edges connecting A with B are also red. Suppose a balanced copy C of K_5 sits in K_{2n} with vertex set $V(C) \subseteq V(K_{2n})$; then five of its ten edges must be blue. The vertex set $V(C) \cap B$ induces a complete graph (a clique in K_{2n}) such that an edge $xy \in E(C)$ is blue if and only if $\{x, y\} \subseteq B$. Since no complete graph has five edges, this is a contradiction.

   This example illustrates the characterization provided by Theorem 4.2. The preceding argument shows that that K_5 fails the second condition of the theorem; it is straightforward (albeit unnecessary) to show that it also fails the first.

Let me elucidate the case when the value of $\text{bal}(n, G)$ is independent of the parameter n. So, suppose that G is a graph on k edges (for simplicity, let's say k is always even), such that $\text{bal}(n, G) = C(G)$. In such a case, I say that G has *constant balancing number*, such as in Example 4.4(1). The graph $\frac{k}{2}K_2$ is a set of $k/2$ disjoint edges (a *matching*), and $K_{1,k/2}$ is a *star graph*, that is, $k/2$ edges incident on a single vertex. Now, fix the complete graph K_n. One can always pick an arbitrary

vertex v_0 and colour all edges incident in v_0 red and every other edge of K_n blue, as done in the proof of Proposition 4.3. Provided one makes n large enough, one can guarantee that K_n has more than $C(G)$ red edges. By definition, there is a balanced copy of G inside K_n. In particular, all of its red edges (which account for exactly $k/2$ edges of G) intersect at some vertex. This vertex in G can be then chosen as the image of the center of the star graph $K_{1,k/2}$ under an injective graph homomorphism $K_{1,k/2} \to G$.

Consider a different colouring of K_n. This time, pick a maximal matching in K_n and colour those edges red, and everything else blue. The size of the matching increases with n, and so once again, for large enough n, we can guarantee that G satisfies Definition 4.1. Consequently, (at least) half of the edges of G must be pairwise disjoint. In other words, one can construct an injective homomorphism $\frac{k}{2}K_2 \to G$.

As it turns out, these two aforementioned properties fully characterize graphs of constant balancing number. In [CGH$^+$23], we proved this result in parallel to the same characterization published in [CHM26, Section 3.2], where the value of $C(G)$ found involves Ramsey numbers, which grow exponentially. In our paper, the characterization offers a deeper understanding of the structure of the graphs with constant balancing number, and furthermore shows that $\mathrm{bal}(n, G)$ is quadratic in k (and not just bounded by a function exponential in k). Moreover, we extend the characterization to graphs with an odd number of edges.

Theorem 4.5. *Let $k > 0$ be an even integer and G a graph with k edges and no isolated vertices. Then, the following are equivalent.*

1. *G has constant balancing number.*
2. *There are injective graph homomorphisms $\frac{k}{2}K_2 \to G$ and $K_{1,k/2} \to G$.*
3. $$\mathrm{bal}(n, G) \leq \frac{(|V(G)| - 3)(|V(G)| + k)}{2}.$$

Moreover, if G has either k or $k+1$ edges, then

$$\frac{1}{18}k^2 - \frac{11}{18}k + \frac{14}{9} \leq \mathrm{bal}(n, G) \leq \frac{15}{8}k^2 - \frac{7}{4}k - 1.$$

Implication (1) $\implies$ (2) is [CHM26, Theorem 3.6] proved using Ramsey numbers and also [CGH$^+$23, Theorem 1.1] proved by an involved structural induction. The implication (2) $\implies$ (3) is [CGH$^+$23, Theorem 3.7] and is the only part that requires that G has no isolated vertices. The moreover part is [CGH$^+$23, Theorem 5.1]. The implication (3) $\implies$ (1) is immediate, as the bound in (3) does not depend on n.

If one wishes to study an analogue notion for infinite graphs, one has to define what it means for a large complete graph to have "sufficiently many edges of each colour." In [BLM20, Section 4], the authors study a somewhat similar notion of balanceability and suggest an approach to generalizing these ideas to the infinite case. I draw inspiration from this approach and adapt some of the ideas to our theory. Namely, that "sufficiently many edges of each colour" corresponds to non-meagre colour classes in a Polish space. The basic notions regarding graphs defined on Polish spaces is covered in Section 2.1.

I prove an analog of Theorem 4.5 that says that if the colours are well represented in a complete graph on a perfect Polish space, then one can find a star graph (in this context: a fixed centre connected to a Cantor set through edges) in one colour, joined with perfectly-many pairwise disjoint edges in the other colour; I believe this to be the right interpretation of the uncountable version of a graph with constant balancing number.

Lemma 4.6. *Let X be a perfect Polish space. Given non-meagre disjoint sets $A, B \subseteq [X]^2$ with the Baire property, there are disjoint open sets $U_A, U_B \subseteq X$ such that A and B are comeagre in $[U_A]^2$ and $[U_B]^2$, respectively.*

Proof. Since A and B possess the Baire property, there are open sets of edges $O_A, O_B \subseteq [X]^2$ such that $A \Delta O_A$ and $B \Delta O_B$ are meagre. The fact that A and B are disjoint implies that $O_A \cap O_B \subseteq (A \Delta O_A) \cup (B \Delta O_B)$, and hence $O_A \cap O_B$ is meagre as well. Thus, by the Baire category theorem, O_A and O_B are disjoint open sets.

Now let $\pi \colon X^2 \to [X]^2$ be the quotient map and appreciate that

$$\mathcal{O}_A := \pi^{-1}[A] \quad \text{and} \quad \mathcal{O}_B := \pi^{-1}[B]$$

are disjoint, symmetric open subsets of the Polish space $X^2 \setminus \{(x,x) : x \in X\}$. Since X is perfect, there are open pairwise disjoint sets $U_A^0, V_A^0, U_B^0, V_B^0 \subseteq X$ such that

$$V_A^0 \times U_A^0,\ U_A^0 \times V_A^0 \subseteq \mathcal{O}_A \quad \text{and} \quad V_B^0 \times U_B^0,\ U_B^0 \times V_B^0 \subseteq \mathcal{O}_B.$$

So set $U_A := U_A^0 \cap V_A^0$ and $U_B := U_B^0 \cap V_B^0$, and notice that since $\mathcal{O}_A$ and $\mathcal{O}_B$ are open and disjoint, so are U_A and U_B open, disjoint and nonempty.

Also, notice that $[U_A]^2 \subseteq O_A$ and $[U_B]^2 \subseteq O_B$. Finally, $[U_A]^2 \setminus A \subseteq O_A \setminus A \subseteq A \Delta O_A$, and similarly $[U_B]^2 \setminus B$, is meagre. □

Theorem 4.7. *Let X be a perfect Polish space and suppose that $\chi \colon [X]^2 \to 2$ is a Baire measurable function with non-meager fibres (the colour classes). Then, there is a colour $i < 2$, a vertex $c \in X$, and a Cantor set $C \subseteq X$ such that for all $x \in C$, $\chi(\{c, x\}) = i$, and such that $[C]^2$ contains an uncountable matching of colour $1 - i$.*

Proof. Suppose that $\chi \colon [X]^2 \to \{\text{red}, \text{blue}\}$ is a Baire measurable function with non-meager fibres. Apply Lemma 4.6 to $A := \chi^{-1}\{\text{red}\}$ and $B := \chi^{-1}\{\text{blue}\}$ to obtain disjoint open sets U_R and U_B with A comeagre in $[U_R]^2$ and B comeagre in $[U_B]^2$.

Since χ is Baire measurable, the sets of edges of each colour in $U_R \times U_B$ have the Baire property, so on some nonempty open set $U_R' \times U_B' \subseteq U_R \times U_B$, either the set of blue edges or the set of red edges is comeager. Without loss of generality, suppose it is blue.

By the Kuratowski-Ulam theorem, the set

$$B := \{y \in U_B' : \{x \in U_R' : \chi(x,y) = \text{blue}\} \text{ is comeagre in } U_R'\}$$

is comeagre in U_B', in particular nonempty, so choose $c \in B$ such that $S_c := \{x \in U_R' : \chi(x,y) = \text{blue}\}$ is comeagre in U_R'.

Since the set of red edges is comeagre in U'_R and χ has the Baire property, Galvin's theorem gives a Cantor set $C_0 \subseteq U'_R$ such that $[C_0]^2$ is red. Now find a Cantor set C contained in $C_0 \cap S_c$ and observe that $[C]^2$ is red and every edge (c, x) for $x \in C$ is blue.

Fix a homeomorphism $\varphi\colon 2^\omega \to C$. It follows that $\{(\varphi(0^\frown x), \varphi(1^\frown x) : x \in 2^\omega)\}$ is a red matching. $\square$

4.3.2 The group of feasible edge replacements

I start this subsection by setting up a combinatorial puzzle. A graph is given, and one may remove an edge and add it wherever there is no edge (trivially, one can add it back to its original position), but the new graph must be isomorphic to the starting graph. Such an action is called a *feasible edge replacement*, which is formalized after these examples. Can one attain every possible configuration of the graph by repeateadly changing edges like this? It is clear that the answer depends on the structure of the particular starting graph. For instance, a cycle of four edges has no valid non-trivial movements since, having removed an edge, the only possibility that preserves the cycle is putting the edge back from where it was taken. Graphically, Figure 4.3 depicts two isomorphic graphs that cannot be transformed into each other by feasible edge replacements.

Figure 4.3: Two 4-cycles on the same vertex set.

Now for the puzzle: the two paths in Figure 4.4 are isomorphic. Is it possible to transform one into the other by a sequence of feasible edge replacements?

Figure 4.4: Two paths of length 3 on the same vertex set.

Definition 4.8 (Labeled graph)**.** For a set X, whose elements are called *labels*, denote by S_X the set of all bijections, called *label permutations*, $X \to X$. Under the operation of composition, S_X forms a group called the *symmetric group on* X. As usual, we use the notation $\rho \cdot \sigma := \rho \circ \sigma$ for the group operation. Endow this set with the topology inherited from the Tychonoff product X^X, with X taken as discrete. Let G be a graph defined on a vertex set V and let $\lambda\colon V \to X$ be a bijection, called a *labeling* of G. For the sake of simplicity, I will always write vertices as $v_\ell \in V$ where $\lambda(v_\ell) = \ell \in X$ is the label.

The pair (G, λ) is called a *labeled graph*, and the paremeters V, E and X are considered to be implicit.

Every label permutation $\sigma \in S_X$ induces the graph isomorphism $\varphi_\sigma := \lambda^{-1} \circ \sigma \circ \lambda\colon V \to V$ of G, and every graph isomorphism of G of the form $\varphi\colon V \to V$ (that is, on the same vertex set) induces

a label permutation $\sigma_\varphi := \lambda \circ \varphi \circ \lambda^{-1} \in S_X$. For this reason, label permutations and isomorphisms are often used interchangeably, but I do not follow this convention for the sake of clarity.

$$\begin{tikzcd} X \arrow[r, "\sigma"] & X \arrow[d, "\lambda^{-1}"] \\ V \arrow[u, "\lambda"] \arrow[r, "\varphi_\sigma"'] & V \end{tikzcd}$$

Figure 4.5: Relationship between label permutations and isomorphisms.

Lemma 4.9. *Let G and G' be graphs on the same vertex set V. Given a bijection $\varphi\colon V \to V$, φ is an isomorphism between G and G' if and only if*

$$\varphi[E(G)] = E(G').$$

Proof. This is immediate from the definition of graph isomorphism: φ is an isomorphism if and only if $v_iv_j \in E(G) \Leftrightarrow \varphi(v_i)\varphi(v_j) \in E(G')$, which is precisely the statement $\varphi[E(G)] = E(G')$. □

We adapt the notation from [CHM23] and all subsequent papers. Given a labeled graph (G, λ), the *edge-label set of G* is defined as

$$L_G := \{ij : v_iv_j \in E(G)\},$$

where ij and ji are not distinguished. Thus, L_G is the set of labels induced on the edges of G by λ. For each $\sigma \in S_X$, define G_σ to be the graph on the same vertex set V with edge set

$$E(G_\sigma) := \{v_iv_j : \sigma(i)\sigma(j) \in L_G\}.$$

In words, G_σ is the unique graph on V satisfying $L_{G_\sigma} = \sigma^{-1}[L_G]$, where $\sigma^{-1}[L_G] := \{\sigma^{-1}(i)\sigma^{-1}(j) : ij \in L_G\}$. In particular, $L_{G_\sigma} = L_G$ for every $\sigma \in S_X$ (the set of edge labels is invariant across all copies), and $G_{\mathrm{id}} = G$.

The relationship between G_σ and φ_σ is as follows. By Lemma 4.9, φ_σ is an isomorphism $G \to G'$ if and only if $\varphi_\sigma[E(G)] = E(G')$. Since $\varphi_\sigma(v_i) = v_{\sigma^{-1}(i)}$, this image equals $\{v_{\sigma^{-1}(i)}v_{\sigma^{-1}(j)} : v_iv_j \in E(G)\}$, whose edge-label set is exactly $\sigma^{-1}[L_G] = L_{G_\sigma}$. In summary:

$$\varphi_\sigma \text{ is an isomorphism } G \to G' \quad \Longleftrightarrow \quad G_\sigma = G' \quad \Longleftrightarrow \quad \sigma^{-1}[L_G] = L_{G'}. \tag{4.1}$$

$$\begin{tikzcd} L_G \arrow[r, mapsto, "\sigma^{-1}"] \arrow[d, leftrightarrow, "\lambda"'] & L_{G_\sigma} = \sigma^{-1}[L_G] \arrow[d, leftrightarrow, "\lambda"'] \\ G \arrow[r, "\varphi_\sigma"'] & G_\sigma \end{tikzcd}$$

Figure 4.6: Relationship between label permutations isomorphisms, and the set L_G.

Suppose that G^c denotes the graph complement of G (that is, the graph obtained from switching every adjacency and every non-adjacency in G) and let $(e, f) \in E(G) \times E(G^c)$, and $\sigma \in S_X$. I use the phrase "$\sigma :: (e \to f)$ is a *feasible edge replacement*" to abbreviate the fact that φ_σ is a graph isomorphism between G and $G - e + f$, or equivalently, that $\sigma^{-1}[L_G] = L_{G-e+f}$.[1] If $\varphi_\sigma \in \mathrm{Aut}(G)$, I employ the symbol $\sigma :: (\emptyset \to \emptyset)$ to signal that no edge is being moved. Denote by $\mathrm{Fer}(G)$ the group generated by all σ for which there are e and f such that $\sigma :: (e \to f)$ is a feasible edge replacement. These label permutations σ are called the *canonical generators* of $\mathrm{Fer}(G)$, and they include permutations that induce automorphisms on G (which can be visualized as an action that leaves the graph looking identical) and permutations that correspond to moving exactly one edge. Thus, intuitively, each element of the group $\mathrm{Fer}(G)$ corresponds to a "sequence" of feasible edge replacements.

Let

$$A(G) := \{\sigma \in S_X : \varphi_\sigma \in \mathrm{Aut}(G)\} \simeq \mathrm{Aut}(G)$$

be the subgroup of label permutations that correspond to the automorphisms of G. Hence, the following subgroup structure holds.

$$1 \leq A(G) \leq \mathrm{Fer}(G) \leq S_X,$$

where, as usual, the 1 on the left represents the trivial subgroup of S_X containing just the identity on X.

It is often convenient to represent a sequence of feasible edge replacements (that is, an element of $\mathrm{Fer}(G)$) as a product of canonical generators (which, as standard in Algebra, is written right-to-left) paired with a "list of instructions" explaining what edges to replace and in what order (which, as standard in English, is written left-to-right). This requires defining some more symbols. Before doing this, there is a technical caveat: the labels of the graph's vertices change after each canonical generator gets applied (the labels get permuted); but having the list of edge replacements in the graph's original labeling is desirable if, for example, one wishes to execute a solution to a puzzle by moving edges without having to keep track of what the corresponding permutations are (not to mention the extra effort needed to figure out what those permutations are in the first place). To circumvent this, I give the following abstract definition (by "abstract," I mean that the operation does not actually require knowledge of the graph G on which it is defined).

Definition 4.10 (Edge replacements as objects)**.** Suppose that $\sigma\colon X \to X$ is label permutation and that $x, y, z, w \in X$, and possibly adding subscripts, are labels. An *edge replacement* is a tuple of the form $(\sigma, (x, y, z, w))$ and we always use the notation $\sigma :: (xy \to zw)$ to represent it.

If $\{x, y\} = \{z, w\}$, I use the notation $\sigma :: (\emptyset \to \emptyset)$ to emphasize that no edge has been moved.

A *sequence of edge replacements* is a tuple $(\sigma, (x_i, y_i, z_i, w_i)_{i \leq n})$ where σ is again a label permutation, and $\{x_i, y_i, z_i, w_i : i \leq n\} \subseteq X$. I employ the symbol

$$\sigma :: (x_0y_0 \to z_0w_0)(x_1y_1 \to z_1w_1) \cdots (x_ny_n \to z_nw_n)$$

to refer to this object.

[1] I delay defining the object $\sigma :: (e \to f)$ (Definition 4.10), for a few paragraphs, purely for expository purposes, as this requires an additional level of abstraction that would otherwise lack motivation.

Intuitively, $\sigma :: (xy \to zw)$ refers to applying the permutation σ to the labels of the vertices and moving the edge $v_x v_y$ to the position of the non-edge $v_z v_w$.[2]

Definition 4.11 (Concatenating edge replacements)**.** Now suppose that

$$f_0 = \sigma :: (x_0y_0 \to w_0z_0)(x_1y_1 \to z_1w_1)\cdots(x_ny_n \to z_nw_n) \qquad \text{and}$$
$$f_1 = \rho :: (a_0b_0 \to c_0d_0)(a_1b_1 \to c_1d_1)\cdots(a_mb_m \to c_md_m)$$

are both sequences of edge replacements. Then the *product* $f_0 * f_1$, is the sequence of edge replacements

$$\begin{aligned} f_0 * f_1 := \rho\cdot\sigma :: (x_0y_0 \to w_0z_0)(x_1y_1 \to z_1w_1)\cdots \\ \cdots(x_ny_n \to z_nw_n)(\sigma^{-1}(a_0)\sigma^{-1}(b_0) \to \sigma^{-1}(c_0)\sigma^{-1}(d_0))\cdots \\ \cdots(\sigma^{-1}(a_m)\sigma^{-1}(b_m) \to \sigma^{-1}(c_m)\sigma^{-1}(d_m)). \end{aligned}$$

In other words, the inverse permutation must be applied to the second sequence of labels to ensure that the final labeling is consistent with the original labeling of the graph. This relabeling also has a very natural categorical interpretation, in the category of labeled graphs (G, λ). Each edge replacement corresponds to a morphism between labeled graphs, and a sequence of feasible edge replacements corresponds to compositions of such morphisms.

Now, I connect the two notions thus far defined in this subsection and prove that they are indeed consistent with one another. The following proposition can be inductively generalized to arbitrary finite sequences of edge replacements.

Proposition 4.12. *Let (G, λ) be a labeled graph and suppose that $f_0 = \sigma :: (xy \to wz)$ and $f_1 = \rho :: (ab \to cd)$ are feasible edge replacements of G. Then, their product*

$$f_0 * f_1 = \rho\cdot\sigma :: (xy \to zw)(\sigma^{-1}(a)\sigma^{-1}(b) \to \sigma^{-1}(c)\sigma^{-1}(d))$$

is such that the induced map $\varphi_{\rho\cdot\sigma}$ is an isomorphism between G and

$$(G - v_xv_y + v_zv_w) - v_{\sigma^{-1}(a)}v_{\sigma^{-1}(b)} + v_{\sigma^{-1}(c)}v_{\sigma^{-1}(d)}.$$

Products of feasible edge replacements are called `fer` *objects*, and their label permutations are precisely the elements of $\mathrm{Fer}(G)$. The preceding proposition shows that $f_0 * f_1$ is a `fer` object, which we prove as follows.

Proof. Denote $G' := G - v_xv_y + v_zv_w$ and $G'' := G - v_av_b + v_cv_d$. At the level of edge labels, this means

$$L_{G'} = (L_G \setminus \{xy\}) \cup \{zw\} \qquad \text{and} \qquad L_{G''} = (L_G \setminus \{ab\}) \cup \{cd\}.$$

Since f_0 is a feasible edge replacement of G, by (4.1), $G_\sigma = G'$, that is,

$$\sigma^{-1}[L_G] = L_{G'} = (L_G \setminus \{xy\}) \cup \{zw\}. \tag{4.2}$$

[2]For readers previously familiar with this notion: the notation $\sigma :: (xy \to zw)$ denotes that $\sigma \in \mathrm{Fer}_G(xy \to zw)$ in the sense of [CHM23].

Since f_1 is a feasible edge replacement of G, again by (4.1), $G_\rho = G''$:

$$\rho^{-1}[L_G] = L_{G''} = (L_G \setminus \{ab\}) \cup \{cd\}. \tag{4.3}$$

The goal is to compute $(\rho \cdot \sigma)^{-1}[L_G]$. Since $(\rho \cdot \sigma)^{-1} = \sigma^{-1} \circ \rho^{-1}$, applying the bijection σ^{-1} to both sides of (4.3) gives

$$\begin{aligned}(\rho \cdot \sigma)^{-1}[L_G] &= \sigma^{-1}\big[\rho^{-1}[L_G]\big] \\ &= \sigma^{-1}\big[(L_G \setminus \{ab\}) \cup \{cd\}\big] \\ &= \big(\sigma^{-1}[L_G] \setminus \{\sigma^{-1}(a)\sigma^{-1}(b)\}\big) \cup \{\sigma^{-1}(c)\sigma^{-1}(d)\}. \end{aligned} \tag{4.4}$$

Substituting (4.2) into (4.4) yields

$$(\rho \cdot \sigma)^{-1}[L_G] = \big(L_{G'} \setminus \{\sigma^{-1}(a)\sigma^{-1}(b)\}\big) \cup \{\sigma^{-1}(c)\sigma^{-1}(d)\} = L_{G'''},$$

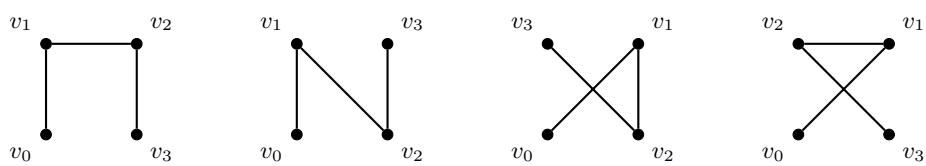
where $G''' := G' - v_{\sigma^{-1}(a)}v_{\sigma^{-1}(b)} + v_{\sigma^{-1}(c)}v_{\sigma^{-1}(d)}$. By (4.1), this means $G_{\rho\cdot\sigma} = G'''$ or, in other words, that $\varphi_{\rho\cdot\sigma}$ is an isomorphism $G \to G'''$. □

With the multiplication I defined in Definition 4.11, `fer` objects form their own group structure, which I call the `Fer` group; clearly, it is isomorphic to $\mathrm{Fer}(G)$. As far as I know, this group is not explicitly defined in the existing body of literature, and doing so has some practical advantages, especially so from an engineering perspective.

For instance, a solution to the puzzle is shown in Figure 4.7, where each caption shows the canonical generator needed to obtain that step of the solution; and the `fer` object that solves the puzzle (with the same labels as in the figure) is the product of the three generators, resulting in

$$(2\ 3) :: (1\ 2 \to 1\ 3)(0\ 1 \to 0\ 2)(2\ 3 \to 1\ 2).$$

It is worth noting that the permutation (2 3) is not a canonical generator, as (at least) two edges must necessarily change their position.



(a) Starting graph (b) $(2\ 3) :: (1\ 2 \to 1\ 3)$ (c) $(1\ 3) :: (0\ 1 \to 0\ 3)$ (d) $(2\ 3) :: (1\ 2 \to 1\ 3)$

Figure 4.7: A solution to the `fer` puzzle.

For the remainder of the chapter, topological closures are always taken in the containing space X^X. In symbols, if $A \subseteq X^X$, $\overline{A} := \mathrm{cl}(A, X^X)$.

Definition 4.13 (Local Amoeba)**.** A graph G is called a *local amoeba* if $\overline{\mathrm{Fer}(G)} = S_X$.

Hence, a local amoeba is precisely a graph that can be transformed into any copy of itself on the same vertex set by a (possibly infinite) sequence of feasible edge replacements. This graph-theoretic

notion and the name *amoeba* first appeared in [CHM21] as an interesting combinatorial object with strong balanceability properties, described in Subsection 4.3.1. When X is finite, all subsets of X^X are closed and thus the preceding definition simplifies to $\mathrm{Fer}(G) = S_X$, which is the algebraic notion introduced in [CHM23] as a tool to study these graphs and their properties. The currently published body of research studies the case when X is finite.

If C_4 and P_4 are the examples depicted in Figure 4.3 and Figure 4.4, respectively, then $\mathrm{Fer}(C_4) = \mathrm{Aut}(C_4)$ is the dihedral group D_4 of order $2 \cdot 4$, and $\mathrm{Fer}(P_4) = S_4$ is the symmetric group. Therefore, C_4 is not a local amoeba, while P_4 is.

Complete graphs are all local amoebas for a rather uninteresting reason: their automorphism group is S_X (any permutation preserves the adjecencies trivially, since any two distinct vertices are adjacent).

Definition 4.14. A graph G is called *rigid* if $\overline{\mathrm{Fer}(G)} = \mathrm{Aut}(G)$. G is called *asymmetric* if $\mathrm{Aut}(G)$ is the trivial group consisting of the identity on X.

One of the main concerns of this active field of research is finding and classifying interesting examples of graphs that are local amoebas. For instance, constructing trees with arbitrary large maximum degree that are local amoebas is a curious problem, whose solution was the main precursor of [EHMWV25], which I present in Subsection 4.3.3.

In terms of Group Theory, $\mathrm{Fer}(G)$ carries all of the expressive power of finite groups. Precisely, one can show that there is no theorem of the form "for every finite graph G, the group $\mathrm{Fer}(G)$ has property φ," where φ is a non-trivial property about groups.

Proposition 4.15. *For any finite group Γ, there is a graph G such that Γ is isomorphic to* $\mathrm{Fer}(G)$.

The heavy lifting of this result is an old theorem of Frucht. After Frucht's 1939 breakthrough showing that every finite group is the automorphism group of some finite graph, he extended this result to the more restrictive class of cubic graphs (graphs in which every vertex has degree 3).

Lemma 4.16 (Frucht's theorem [Fru49])**.** *For any finite group Γ, there exists a cubic graph G such that* $\mathrm{Aut}(G) \simeq \Gamma$.

Proof of Proposition 4.15. Let Γ be an arbitrary finite group and G a cubic graph given by Frucht's theorem. Then observe that G is rigid, that is $\mathrm{Fer}(G) = \mathrm{Aut}(G) \simeq \Gamma$. This is a direct consequence of the fact that applying a non-trivial feasible edge replacement changes the degree of at least one vertex (it increases or decreases by exactly one, depending on whether an edge was added or removed to a particular vertex). □

4.3.3 Local amoeba trees with large maximum degree

Intuitively, if one is trying to transform one graph into another by means of feasible edge replacements, it may come in handy to have isolated vertices that allow new positions for the graph. I previously mentioned that complete graphs are rigid graphs, meaning they have no non-trivial feasible edge replacements. Adding isolated vertices to a complete graph destroys its rigidity, since now new movements are possible.

Definition 4.17. A finite graph G is called a *global amoeba* if there is a $t \geq 1$ such that $G \cup tK_1$ (that is, the graph obtained as the disjoint union of G and t many isolated vertices) is a local amoeba.

Perhaps surprisingly, adding isolated vertices to an arbitrary graph does not necessarily increase the amount of feasible edge replacements; in fact, there is no general logical implication between a graph G being a local amoeba and being a global amoeba.[3] One case where such an implication does exist is for trees (see Corollary 4.19).

The main result of [CGH$^+$23] is the following classification of global amoebas. Notably, if a value of t exists such that Definition 4.17 holds, then $t = 1$ is always the least such value, as the following theorem states.

Theorem 4.18 ([CGH$^+$23, Theorem 4.1])**.** *For any non-empy graph G, the following are equivalent.*

1. *G is a global amoeba.*
2. *$G \cup K_1$ is a local amoeba.*
3. *Every orbit of* $\mathrm{Fer}(G)$ *has a label that corresponds to some vertex of degree 1.*

A well-known fact about any finite tree T with at least an edge is that it must have at least two vertices of degree 1. Recalling that if T is a local amoeba, then $\mathrm{Fer}(T)$ consists of a single orbit (it is in fact, the whole symmetric group), it is clear that condition (3) of Theorem 4.18 is satisfied for any tree T that happens to be a local amoeba.

Corollary 4.19. *If T is a finite tree for which* $\mathrm{Fer}(T)$ *acts transitively on T (for example if T is a local amoeba), then T is a global amoeba.*

The connectedness of T is indispensable. For example, a matching (a set of pairwise disjoint edges) forms a global amoeba, since every vertex has degree 1, but not a local amoeba (it is, in fact, rigid). The acyclic hypothesis on T is also crucial: any cycle C is rigid but $\mathrm{Fer}(C)$ acts transitively on it.

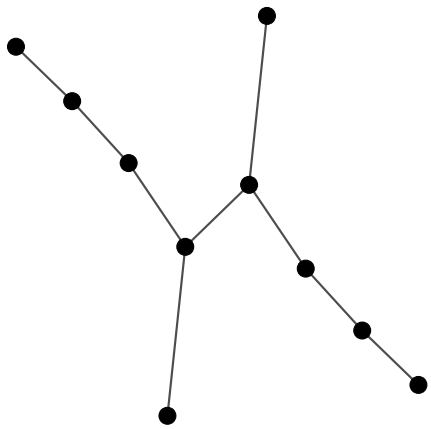
Figure 4.8a depicts the smallest tree T that is a global amoeba but not a local amoeba. Interestingly, its `Fer` group is transitive (that is, it consists of a single orbit). There are also global amoeba trees whose `Fer` group is not transitive, and the smallest such example has two orbits, depicted in Figure 4.8b with vertices coloured by orbit under the action of the `Fer` group.[4] The main conjecture of Section 4.4 is that the `Fer` groups of all global amoeba trees have at most two orbits.

The construction I describe now (see Figure 4.9), loosely inspired by the Fibonacci sequence, illustrates the main results of [EHMWV25]. Start by defining T_1 and T_2 to each be the tree consisting of a single edge, in other words, $T_1 \simeq T_2 \simeq K_2$. Now, suppose that T_{k-1} and T_{k-2} have already been constructed. In each of these trees, select a vertex of maximum degree. The tree T_k is the result of joining T_{k-1} and T_{k-2} by the edge connecting these two vertices. The choice of the vertex of maximum degree is irrelevant, as one can easily verify: either such vertex is unique, or there are exactly two such vertices that are similar (that is, they can be interchanged via an automorphism), and thus the resulting sequence of graphs is unique up to isomorphisms.

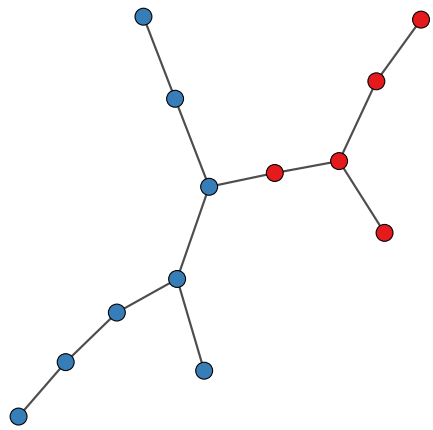
The sequence of graphs $\{T_k : k \in \mathbb{N}\}$ is a particular case of a more general scheme of recursive constructions studied in [EHMWV25, Section 3].

[3] With this in mind, one might try to suggest more descriptive names for these graphs that do not hint at this inexisting causal link, but the author chose to respect the terminology from the well-established literature.

[4] Recall from elementary Group Theory that the set of orbits of a group action forms a partition of the set. Thus, in this example, two vertices have the same colour if and only if there is a `fer` object that sends one to the other.



(a) The smallest non-local global amoeba tree.



(b) The smallest global amoeba tree on which `Fer` does not act transitively.

Figure 4.8: The smallest global amoeba trees with these properties.

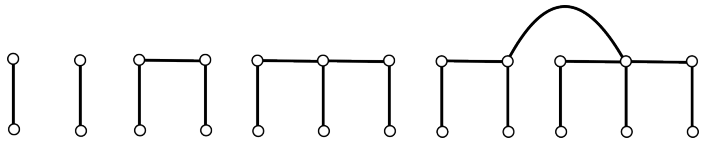


Figure 4.9: Fibonacci-type trees T_k, for $1 \leq k \leq 5$.

Theorem 4.20 ([EHMWV25, Theorem 9])**.** *For every $k \in \mathbb{N}$, T_k is a local amoeba.*

An independent contribution of mine answers the following question which I developed as Section 5 of [EHMWV25]. The proof that a finite graph G is a local amoeba often involves a counting argument: one shows that $\mathrm{Fer}(G)$ has $|X|!$ elements and then, given that $\mathrm{Fer}(G)$ is contained in the symmetric group on X, $\mathrm{Fer}(G)$ must be equal to S_X. There are two disadvantages with this method.

- Given an arbitrary permutation $\sigma \in S_X$ (that is, an arbitrary copy of G on the same vertex set), the proof generally offers no method of finding the sequence of feasible edge replacements. In other words, this proving method shows the existence of a solution to the puzzle, without giving an algorithm to find it. The algorithm presented in Subsection 4.3.4 resolves this for an infinite family of trees.
- Counting methods don't generalize to infinite graphs. I address this in Section 4.5.

A few more notions come in handy not just for the upcoming section, but also for Section 4.5.

Definition 4.21. Let (G, λ) be any labeled graph, $i \in X$ a label, and $\Gamma := \mathrm{Fer}(G)$. The subgroup of Γ generated by the canonical generators of Γ that fix i is denoted by $\mathrm{Fer}^i(G)$. That is, in symbols, $\mathrm{Fer}^i(G) := \Gamma \cap \mathrm{Stab}_\Gamma(i)$.

We say a graph G is *stem-symmetric with respect to the label i* (equivalently, *stem-symmetric with respect to the vertex $\lambda(i)$*) if $\mathrm{Fer}^i(G) = \{\varphi \in S_X : \varphi(i) = i\}$.

In other words, a finite graph $G = (V, E)$ is stem-symmetric with respect to a vertex v if, for any isomorphism $\varphi\colon V \to V$ of G that fixes v, one can attain the graph G_φ by a sequence of `fer` objects that all fix the label of v.

To convince the reader that this defines a non-trivial condition on G, consider the example of a path P_n on an odd number of vertices $n \geq 3$. Let $k = \frac{n-1}{2}$ be the midpoint. If ρ is the automorphism that reflects P_n, that is $\rho\colon i \mapsto n - i - 1$, then ρ fixes the midpoint k and $\mathrm{Aut}(P_n) = \{\mathrm{id}_n, \rho\} \simeq S_2$. Given a canonical generator $\sigma :: (e \to f) \in \mathrm{Fer}^k(P_n)$, the only choice of edges that lets $\sigma(k) = k$ is when $e = f$, and hence $\sigma \in A(P_n)$. This proves that $\mathrm{Fer}^k(P_n) = A(P_n) \simeq S_2$, but as per a previous discussion, $\mathrm{Fer}(P_n) = S_n$ (all paths are local amoebas). For a concrete example, the transposition that interchanges the endpoints, $(0\ (n-1))$ fixes the midpoint, but is not attainable as a sequence of `fer` objects that all fix the midpoint. Therefore, P_n is not stem-symmetric with respect to k. The situation is different when picking a different vertex k in a path.

Lemma 4.22. *For every $n \in \mathbb{N}$, the path P_{n+1} is stem-symmetric with respect to the endpoint n.*

Proof. Let $1 \leq \ell < n$ and notice that the graph $P_{n+1} - (\ell, \ell+1) + (0, \ell+1)$ is a path on $n+1$ vertices and thus isomorphic to P_{n+1}. It is possible to find an explicit isomorphism that fixes n and decompose it into transpositions. In symbols,

$$\prod_{k=0}^{\ell} (k \quad \ell - k) :: ((\ell, \ell+1) \to (0, \ell+1))$$

is a `fer` object that realizes this isomorphism.

The second step is observing that, letting ℓ vary, the preceding permutations generate the symmetric group on (the set) n. This is straightforward and can be done in multiple ways. For example,

every transposition of the form $(0\ \ell)$, for $1 \le \ell < n$, is in $\mathrm{Fer}^n(P_{n+1})$ by an inductive argument on ℓ, taking conjugates, and noticing that the transposition in question is the left-most factor in the ℓ-th permutation above. $\square$

4.3.4 A recursive algorithm for Fibonacci-type trees

I now describe an algorithm that, given any permutation $\sigma \in S_X$ on the vertices of T_k, produces an explicit sequence of feasible edge replacements whose composition realises σ. Throughout this section, fix a labeling of T_k and write $n = n(k) = 2F_k$ for its order, where F_k is the k-th Fibonacci number.

Recursive decomposition of T_k. The proof of Theorem 4.20 proceeds by induction and uses the following structural decomposition, which I now make explicit so that the algorithm can be stated precisely.

For $k \ge 5$, the tree T_k is built from four subtrees H_1, J_1, H_2, J_2 with

$$H_1 \simeq T_{k-3}, \quad J_1 \simeq T_{k-2}, \quad H_2 \simeq T_{k-3}, \quad J_2 \simeq T_{k-4},$$

whose vertex sets are pairwise disjoint, by adding three edges:

$$T_k \;=\; \big((H_1 \cup J_1) + v_a v_b\big) \cup \big((H_2 \cup J_2) + v_c v_d\big) + v_b v_c,$$

where v_a, v_b, v_c, v_d are the vertices of maximum degree in H_1, J_1, H_2, J_2, respectively. Call $a = \lambda(v_a)$, $b = \lambda(v_b)$, $c = \lambda(v_c)$, $d = \lambda(v_d)$ the *roots* of T_k and denote by

$$A = \lambda[V(H_1)], \quad B = \lambda[V(J_1)], \quad C = \lambda[V(H_2)], \quad D = \lambda[V(J_2)]$$

the corresponding label sets. These four sets partition X, and b is the label of the unique vertex of maximum degree in T_k. Now fix once and for all a label $y_0 \in B \setminus \{b\}$.

Two feasible edge replacements in T_k play a distinguished role.

- The replacement $cd \to ad$ induces a bijection $\rho\colon A \to C$ that extends to the isomorphism $H_1 \to H_2$ sending $v_a \mapsto v_c$, while fixing $B \cup D$ pointwise.
- The replacement $ab \to ac$ induces a bijection $\varphi\colon B \to C \cup D$ that extends to the isomorphism $(H_2 \cup J_2) + v_c v_d \to J_1$ sending $v_b \mapsto v_c$, while fixing A pointwise.

Both ρ and φ are therefore canonical generators of $\mathrm{Fer}(T_k)$.

The algorithm. It suffices to produce a `fer` object for every transposition of the form (y_0, x) with $x \in X \setminus \{y_0\}$, since any permutation $\sigma \in S_X$ can be written as a product of such transpositions via its cycle decomposition:

$$(a_0, a_1, \ldots, a_m) \;=\; (y_0, a_0)(y_0, a_1) \cdots (y_0, a_m)(y_0, a_0).$$

The function `RecFer`(x, k), given in Algorithm 2, returns a `fer` object for (y_0, x) by induction on k, using the decomposition above. The four cases correspond to which of the label sets A, B, C, D the

input x belongs to. The key identities driving each case are:

$$\begin{aligned}
x \in A \cup B &\implies (y_0, x) \in \mathrm{Fer}^b(T_{k-1}), \quad \text{by the stem-symmetry of } T_{k-1}, \\
x = b &\implies (y_0, x) = \varphi^{-1}(y_0,\, \varphi^{-1}(b))\, \varphi, \\
x \in C &\implies (y_0, x) = \rho^{-1}(y_0,\, \rho^{-1}(x))\, \rho, \quad \rho^{-1}(x) \in A, \\
x \in D &\implies (y_0, x) = (y_0, x_0)(x, x_0)(y_0, x_0), \quad x_0 \in C \text{ a fixed neighbour of } c.
\end{aligned}$$

Algorithm 2: Factoring a permutation into a `Fer` sequence for T_k.

```
@cache                // Memoize over all (x,k) pairs across the entire execution.
Function RecursiveFer(x, k):
    if k ≤ 4 then return hashMap[(y_0, x)]
    Let a, b, c, d and A, B, C, D be the roots and label sets of T_k
    if x ∈ A ∪ B then
        return RecursiveFer(x, k − 1)
    else if x = b then
        return φ∗ RecursiveFer(φ^{−1}(b), k) ∗φ^{−1}
    else if x ∈ C then
        return ρ∗ RecursiveFer(ρ^{−1}(x), k − 1) ∗ρ^{−1}
    else
        // x ∈ D
        Let x_0 ∈ C be a fixed neighbour of c
        F_0 ← RecursiveFer(x_0, k)
        F_1 ← RecursiveFer(x − c, k − 2) + c          // Shift all labels by c.
        return F_0 ∗ F_1 ∗ F_0
Input: k and σ ∈ S_X.
/* Step 1.  Decompose σ into transpositions (y_0, x).                          */
trans ← []
for (a_0, ..., a_m) cycle of σ do
    Append (y_0, a_i) for i = 0, ..., m, then append (y_0, a_0)
/* Step 2.  Compute a Fer object per transposition (memoized).                 */
for (y_0, x) ∈ trans do
    hashMap[(y_0, x)] ← RecursiveFer(x, k)
/* Step 3.  Form the product.                                                  */
Output:
```

$$F = \prod_{(y_0,x)\,\in\,\mathrm{trans}} \texttt{hashMap}[(y_0, x)]$$

The shallow transposition improvement. The D-case triple product

$$(y_0, x) = (y_0, x_0)(x, x_0)(y_0, x_0),$$

if applied naively, doubles the contribution of (y_0, x_0) at every recursion level and leads to a sequence length satisfying $\ell(k) = \Theta(k^2)$. This can be improved by observing that every transposition (y_0, x) can instead be built by *tracing the unique path* from x to y_0 in T_k. Each step of this path crosses exactly one subtree boundary, and each boundary crossing can be implemented using a constant

number of feasible edge replacements: either a single application of ρ or φ, which are each induced by a single edge replacement. Since the depth of T_k is $O(k)$ and the path from x to y_0 crosses at most $O(k)$ subtree boundaries, the resulting sequence for (y_0, x) has length $O(k)$. We refer to this as the *shallow transposition routine*.

Correctness and complexity.

Theorem 4.23 (Correctness)**.** *For every $k \geq 1$ and every permutation $\sigma \in S_X$ on the vertices of T_k, Algorithm 2 terminates and outputs a sequence of feasible edge replacements of T_k whose permutation is σ.*

Proof. I prove by strong induction on k that $\texttt{RecFer}(x, k)$ returns a `fer` object whose permutation is (y_0, x) and whose edge replacement sequence is feasible in T_k.

For $k \leq 4$ this holds by assumption on the precomputed base cases. Let $k \geq 5$ and suppose the claim holds for all smaller values.

Case $x \in A \cup B$. The labels $A \cup B = \lambda[V(T_{k-1})]$ form the vertex set of the subtree $H = (H_1 \cup J_1) + v_a v_b \cong T_{k-1}$, and $y_0 \in B \subseteq A \cup B$. By the induction hypothesis, $\texttt{RecFer}(x, k-1)$ produces a feasible sequence in T_{k-1} realising (y_0, x). Any feasible edge replacement of a subgraph $H \subseteq T_k$ that fixes the label b of the connecting vertex v_b is also a feasible edge replacement of T_k (this is the content of [EHMWV25, Lemma 5]). Since every element of $\mathrm{Fer}^b(T_{k-1})$ fixes b, the sequence remains feasible in T_k.

Case $x = b$. We have $\varphi^{-1}(b) \in A \cup B$ (since $\varphi\colon B \to C \cup D$ maps $v_b \mapsto v_c$ and hence $b \mapsto c \notin B$, so $\varphi^{-1}(b) \in A$). The recursive call $\texttt{RecFer}(\varphi^{-1}(b), k)$ falls into the $A \cup B$ case and is correct by the above. The conjugation identity $(y_0, b) = \varphi^{-1} \cdot (y_0, \varphi^{-1}(b)) \cdot \varphi$ holds in S_X, and φ corresponds to a single feasible edge replacement, so the concatenated sequence is feasible.

Case $x \in C$. Since $\rho\colon A \to C$ is a bijection and $\rho^{-1}(x) \in A \subseteq A \cup B$, the recursive call falls into the $A \cup B$ case. The conjugation identity $(y_0, x) = \rho^{-1} \cdot (y_0, \rho^{-1}(x)) \cdot \rho$ holds in S_X, and ρ is a single feasible edge replacement.

Case $x \in D$. Since $x_0 \in C$, the object $F_0 = \texttt{RecFer}(x_0, k)$ is correct by the C-case above. The shift $\texttt{RecFer}(x - c, k - 2) + c$ embeds the subtree $J_2 \cong T_{k-2}$ (with labels shifted by c) into D; by the induction hypothesis this produces a feasible sequence for (x_0, x) within D, which remains feasible in T_k by the same extension lemma. The identity $(y_0, x) = (y_0, x_0)(x, x_0)(y_0, x_0)$ holds in S_X since y_0, x_0, x are distinct, and the corresponding concatenation of three feasible sequences is feasible.

Having verified all cases of `RecFer`, the correctness of the main program follows: steps 1–3 apply the standard factorisation of σ into transpositions (y_0, x_i) and compose the corresponding `Fer` objects, yielding a feasible sequence for σ. □

Corollary 4.24 ([EHMWV25, Theorem 9])**.** *For every $k \geq 1$, T_k is a local amoeba.*

Proof. By Theorem 4.23, for every $\sigma \in S_X$ there exists a sequence of feasible edge replacements of T_k whose permutation is σ, and hence $\sigma \in \mathrm{Fer}(T_k)$. Therefore $\mathrm{Fer}(T_k) = S_X$, which is the definition of a local amoeba. □

Theorem 4.25 (Complexity)**.** *Using the shallow transposition routine, Algorithm 2 runs in time and space $O(n \log n)$, where $n = 2F_k = |V(T_k)|$.*

Proof. Since $n = 2F_k$ and $F_k = \Theta(\phi^k)$ for the golden ratio ϕ, we have $k = \Theta(\log n)$.

Template count per transposition. With the shallow transposition routine, $\texttt{RecFer}(x, k)$ traces the path from x to y_0 in T_k, contributing $O(1)$ templates per recursion level. The path crosses at most k subtree boundaries, giving $\mathrm{len}(\texttt{RecFer}(x, k)) = O(k)$ for every x.

Number of distinct memoized calls. With a global cache shared across all transpositions, every call is identified by a pair (x, q) with $q \leq k$. For fixed q the number of possible values of x is $n(q) = 2F_q$. Since

$$\sum_{q=1}^{k} n(q) \;=\; 2\sum_{q=1}^{k} F_q \;=\; 2(F_{k+2} - 1) \;=\; O(F_k) = O(n),$$

the cache contains $O(n)$ distinct entries.

Total cost. Each of the $O(n)$ cached calls performs $O(k)$ work, giving total template cost $O(nk) = O(n \log n)$. Steps 1 and 3 (cycle decomposition and the final product) each take $O(n)$. Hence the total running time and space are both $O(n \log n)$.

Comparison with the published bound. The analysis in [EHMWV25, Theorem 11] establishes a $\Theta(n^2)$ bound by tracking the cost of $\texttt{RecFer}(x, k)$ for a single transposition with a cold cache and then multiplying by the $O(n)$ transpositions in the worst-case permutation. This is correct as stated, but it does not account for the savings from the global memoization shared across all transpositions in a single run of the algorithm, nor does it use the shallow transposition routine. The bound $O(n \log n)$ is therefore a genuine improvement in the algorithm. □

Remark 4.26. The space bound $O(n \log n)$ is consistent with the space complexity stated in Theorem 11 of [EHMWV25], which already observes $O(n \log n)$ space; the contribution here is that the same quantity also bounds the running time.

4.4 Amoeba trees

I present the properties of amoeba trees and recent advancements regarding their conjectures.

My Contribution: I independently conceptualized several key properties discussed in this section. Additionally, I authored abstractions and object-oriented frameworks that had a significant impact on the results presented in [GGHMW]. Apart from my immediate contributions, I focus on unpublished ideas that helped shaped the research.

4.5 Infinitary amoeba graphs

I synthesize ideas on infinite graphs, focusing on the interaction between automorphism groups, ends, and amenable groups, specifically placing my own results and ideas concerning amoeba graphs within this broader theoretical context.

Chapter 5

Closing Remarks and Open Problems

5.1 No countable basis for Borel directed graphs of dichromatic number at least three

This section collects several natural questions suggested by the results presented in Section 2.4, the work of Raghavan and Xiao [RX24], and suggestions of Thornton.

Question 5.1 (Minimality of the Raghavan–Xiao basis.)**.** Raghavan and Xiao [RX24] produce a continuum-size family of directed graphs that serves as a basis for Borel directed graphs with uncountable Borel dichromatic number. This family consists of pairwise Borel-incomparable directed graphs. *Does every basis for this class have size* $\mathfrak{c}$*?* Raghavan has noted (personal communication) that the directed graphs in his family are pairwise non-homomorphic, but it is unknown whether a smaller basis exists.

Question 5.2 (Structural restrictions.)**.** The result presented here applies to the full class of Borel directed graphs. *Does a countable basis exist when one restricts to structurally simpler classes, say locally finite acyclic Borel directed graphs, or directed graphs generated by a single Borel function?* In the undirected case, Miller [Mil08] proved that there is no countable $\leq_B$-basis for graphs of the form G_f (where f is a Borel function) with $\chi_B(G_f) \geq 3$. The analogous question for directed graphs generated by a single Borel function and dichromatic number remains open.

Question 5.3 (Borel dichromatic number and Borel order dimension.)**.** Raghavan and Xiao [RX24] established a close connection between Borel dichromatic number and a notion of Borel order dimension for Borel quasi-orders. *Does the complexity result of Proposition 2.15 yield new complexity results for Borel order dimension?*

Question 5.4 (Dichotomies for complete multipartite graphs.)**.** It is known that checking whether a Borel graph in which every connected component is complete k-partite admits a Borel k-coloring is Π^1_1, but no satisfying dichotomy theorem is known for this problem. Thornton has suggested (personal communication) that a finite basis may exist for each k, though its size appears to grow faster than $n!$. *Is there an explicit finite basis for each k? What is its growth rate?*

Question 5.5 (Bounded width CSPs.)**.** Among the constraint satisfaction problems studied in the Borel setting, the *bounded width* problems occupy a distinguished position. Thornton [Tho22a] showed that the structures whose every solvable Borel instance has a Borel solution are exactly the width 1 structures, and proved partial complexity results for certain bounded width structures. Recent work of Grebík and Vidnyánszky [GV25] shows that the split between easy and hard problems lies at a different place in the Borel setting than in the classical CSP dichotomy: in particular, there is no dichotomy for any Borel CSP that is not of bounded width. However, sharp complexity bounds within bounded width remain known only in very special cases. *Can the exact complexity dividing line within bounded width CSPs be determined?* This problem appears to be very difficult; even identifying good illustrative examples of bounded width CSPs beyond the well-studied special cases remains open.

Question 5.6 (Measurable arboricity.)**.** In the undirected setting, the *arboricity* of a graph (the minimum number of acyclic sets needed to cover the vertex set) is the undirected analogue of the dichromatic number. Several basic questions about arboricity in measurable combinatorics remain open. For instance: *Can the μ-measurable arboricity of a probability-measure-preserving graph differ from its (classical) arboricity by more than 1?* And: *What is the μ-measurable arboricity of the k-th power of the Bernoulli shift graph of F_2?*

5.2 Uncountable sets and an infinite linear order game

When does the converse of Proposition 2.32 hold? By the results of 2.5 such orderings must be ω_1-free and ω_1^*-free. Separable and even ccc linear orderings have this property. The separable case (the reals) was established in [BC23]. Then, the next step is to study dense linear orders that are ω_1-free, ω_1^*-free but not separable. A classical example of such ordering is the Aronszajn ordering.

Definition 5.1. An *Aronszajn line (A-line)* is an uncountable linearly ordered set that is ω_1-free, ω_1^*-free and contains no uncountable subset of the reals.

Following [Tod95], define a proper decomposition for A-lines. Let A be a dense Aronszajn line. Let $D_0 := \emptyset$. Given a countable $D_\alpha \subseteq A$, consider the equivalence relation on $A \backslash D_\alpha$: $x \sim_\alpha y$ if and only if there is no $z \in D_\alpha$ such that $x < z < y$. Each $\sim_\alpha$ class is convex. Let T_α be the set of all $\sim_\alpha$ classes and let D'_α be a selector of the $\sim_\alpha$ classes. Since D_α is countable and dense in $D_{\alpha+1} := D_\alpha \cup D'_\alpha$, then $D_{\alpha+1}$ must be countable because A does not contain uncountable separable subsets. Given D_α for $\alpha < \gamma$ and γ limit, let $D_\gamma = \bigcup_{\alpha<\gamma} D_\alpha$. Since A does not contain copies of ω_1 and ω_1^*, $A = \bigcup_{\alpha<\omega_1} D_\alpha$. Also, $(T, \supseteq)$ is an Aronszajn tree where $T = \bigcup_{\alpha<\omega_1} T_\alpha$.

Question 5.7. Is there a dense Aronszajn line A where Player II has a winning strategy in $BG(A, S)$ for all $S \subseteq A$?

Another class of lines where a similar decomposition can be performed is the class of nowhere separable Souslin continua.

Definition 5.2. A *Souslin continuum* is a non-separable dense, complete linearly ordered set that has the countable chain condition (ccc), i.e., every family of pairwise disjoint open intervals is countable. A Souslin continuum is *nowhere separable* if no non-empty open interval is separable.

Define a *proper decomposition* for the nowhere separable Souslin continuum L as follows. Let $D_0 := D \neq \emptyset$ be any countable set. Given a countable $D_\alpha \subseteq L$ consider the equivalence relation on $L \backslash \overline{D_\alpha}$: $x \sim_\alpha y$ if and only if there is no $z \in D_\alpha$ such that $x < z < y$. Again, each $\sim_\alpha$ class is convex. Let T_α be the set of all $\sim_\alpha$ classes and let D'_α be a selector of the $\sim_\alpha$ classes. Since L has the ccc, then T_α is countable. Let $D_{\alpha+1} = D_\alpha \cup D'_\alpha$. For $\gamma < \omega_1$ limit let $D_\gamma = \bigcup_{\alpha<\gamma} D_\alpha$.

Lemma 5.3. *If $\{D_\alpha : \alpha < \omega_1\}$ is a proper decomposition of a nowhere separable Souslin continuum L, then*

$$L = \bigcup_{\alpha<\omega_1} \overline{D_\alpha}.$$

Proof. Suppose that $x \in L$ and $x \notin \bigcup_{\alpha<\omega_1} \overline{D_\alpha}$. For each $\xi < \omega_1$ let $I^0_\xi := [x]_\xi$ (the $\sim_\xi$ class of x). Note that $I^0_\eta \subseteq I^0_\xi$ for all $\xi < \eta$.

Claim 5.3.1. *For all $\alpha < \omega_1$, $|T_\alpha| > 1$.*

Proof of Claim. Let $x_0 \in D_\alpha$. Since L is nowhere separable, $\overline{D_\alpha}$ is nowhere-dense so pick $y_1 \in (L \backslash \overline{D_\alpha}) \cap (-\infty, x_0)$ and $y_2 \in (L \backslash \overline{D_\alpha}) \cap (x_0, \infty)$. Then, $[y_1]_\alpha, [y_2]_\alpha \in T_\alpha$. $\dashv$

For each $\xi < \omega_1$ let $I^1_{\xi+1} \in T_{\xi+1}$ such that $I^1_{\xi+1} \cap I^0_{\xi+1} = \emptyset$. Then, $\{I^1_{\xi+1} : \xi < \omega_1\}$ is an uncountable pairwise disjoint family of open sets, contradicting ccc. $\square$

Using a proper decomposition to show the following remains an open question of great interest to the authors of [MWS25].

Conjecture 5.1. Let L be a nowhere separable Souslin continuum. Then Player II has a winning strategy for $BG(L, S)$ if and only if S is countable.

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